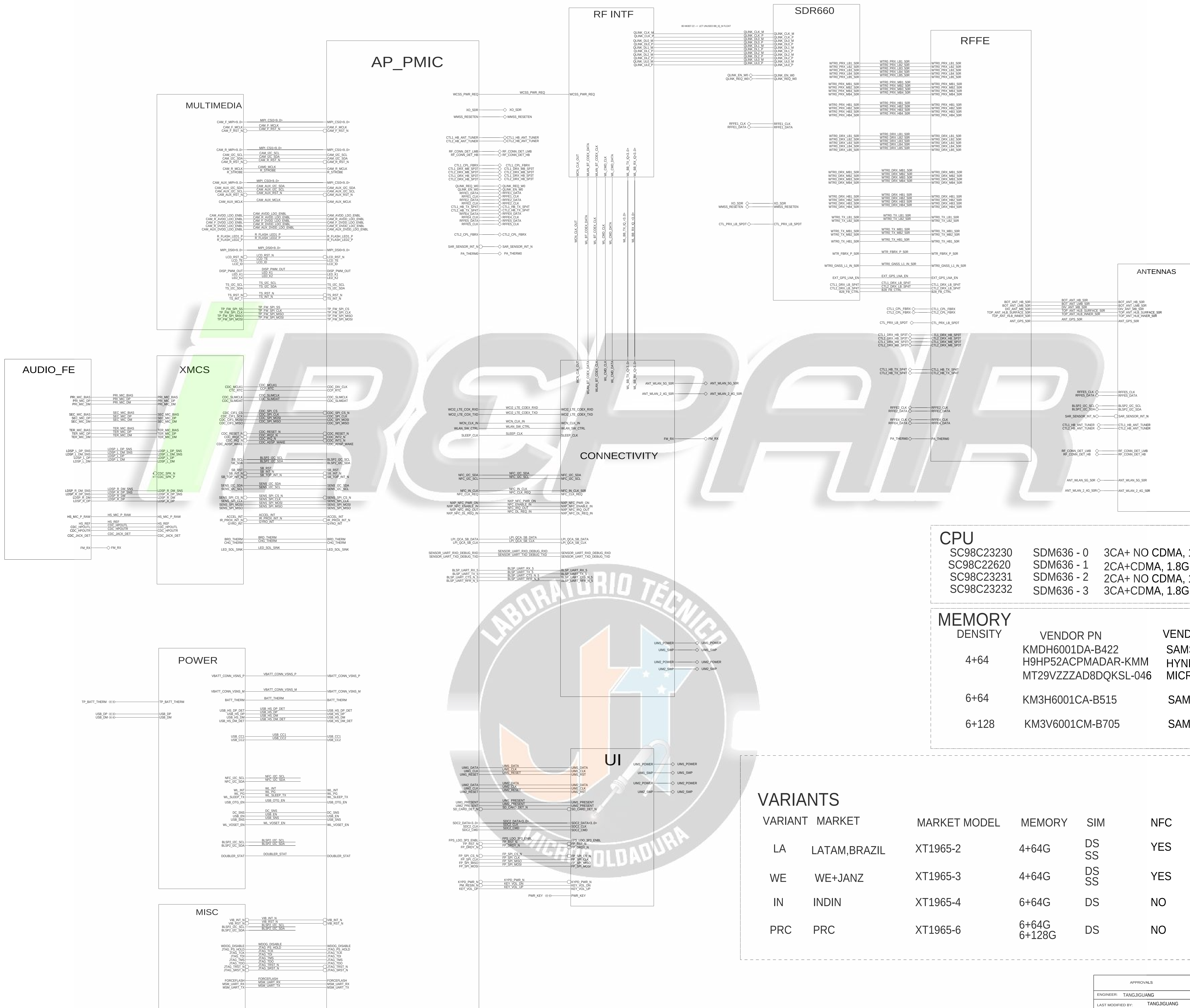


REVISIONS				
ZONE	LTN	DESCRIPTION	DATE	APPR.

**CPU**

SC98C23230	SDM636 - 0	3CA+ NO CDMA, 1.8G
SC98C22620	SDM636 - 1	2CA+CDMA, 1.8G
SC98C23231	SDM636 - 2	2CA+ NO CDMA, 1.8G
SC98C23232	SDM636 - 3	3CA+CDMA, 1.8G

MEMORY DENSITY

DENSITY	VENDOR PN	VENDOR	LENOVO PN
4+64	KMDH6001DA-B422 H9HP52ACPMADAR-KMM MT29VZZZAD8DQKSL-046	SAMSUNG HYNIX MICRON	SM38C28210 SM38C33882 SM38C35427
6+64	KM3H6001CA-B515	SAMSUNG	SM38C28211
6+128	KM3V6001CM-B705	SAMSUNG	SM38C20203

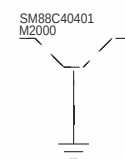
VARIANTS

VARIANT	MARKET	MARKET MODEL	MEMORY	SIM	NFC	CPU	PCBA
LA	LATAM,BRAZIL	XT1965-2	4+64G	DS SS	YES	SDM636-2	SP68C35674 SP68C40488
WE	WE+JANZ	XT1965-3	4+64G	DS SS	YES	SDM636-2	SP68C35675 SP68C40490
IN	INDIN	XT1965-4	6+64G	DS	NO	SDM636-2	SP68C35676
PRC	PRC	XT1965-6	6+64G 6+128G	DS	NO	SDM636-1	SP68C40451 SP68C40453

APPROVALS		BLOCK TITLE	
ENGINEER: TANGJIGUANG		MOBILE DEVICES	
LAST MODIFIED BY: TANGJIGUANG		rioh	
Wed Sep 12 08:23:54 2018		PROJECT NAME	
MOTOROLA CONFIDENTIAL RESTRICTED		E	rioh.cpm
The following is a list of the products and components that are covered by this document. It is not intended to be a complete list of all components and is not intended to be a list of all components that are covered by this document. It is intended to be a list of all components that are covered by this document.		REV	
© Copyright Motorola Inc. 2018		01	
PAGE 1 OF 85			

D

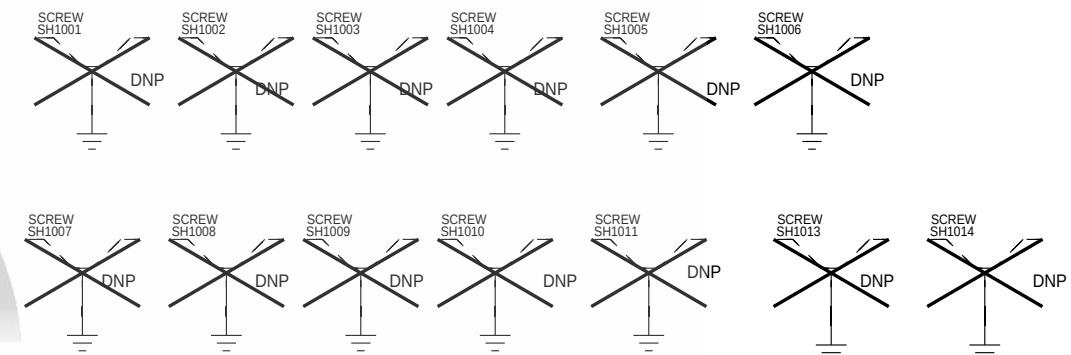
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D



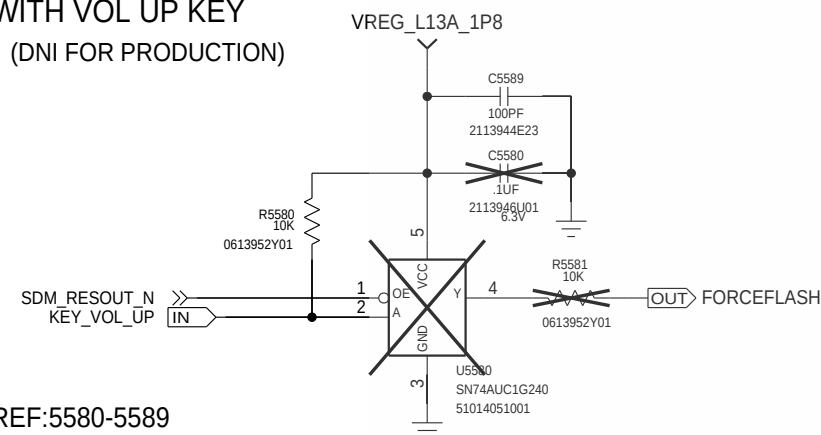
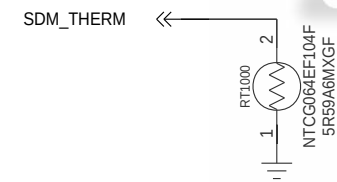
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MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

1

REF 1000-1099



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:24 2018			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		CAD FILE NAME:	
		PAGE 3 OF 66	

SDM: MIPI/DSI/CSI/EBI0

REF 1200-1299

D

C

B

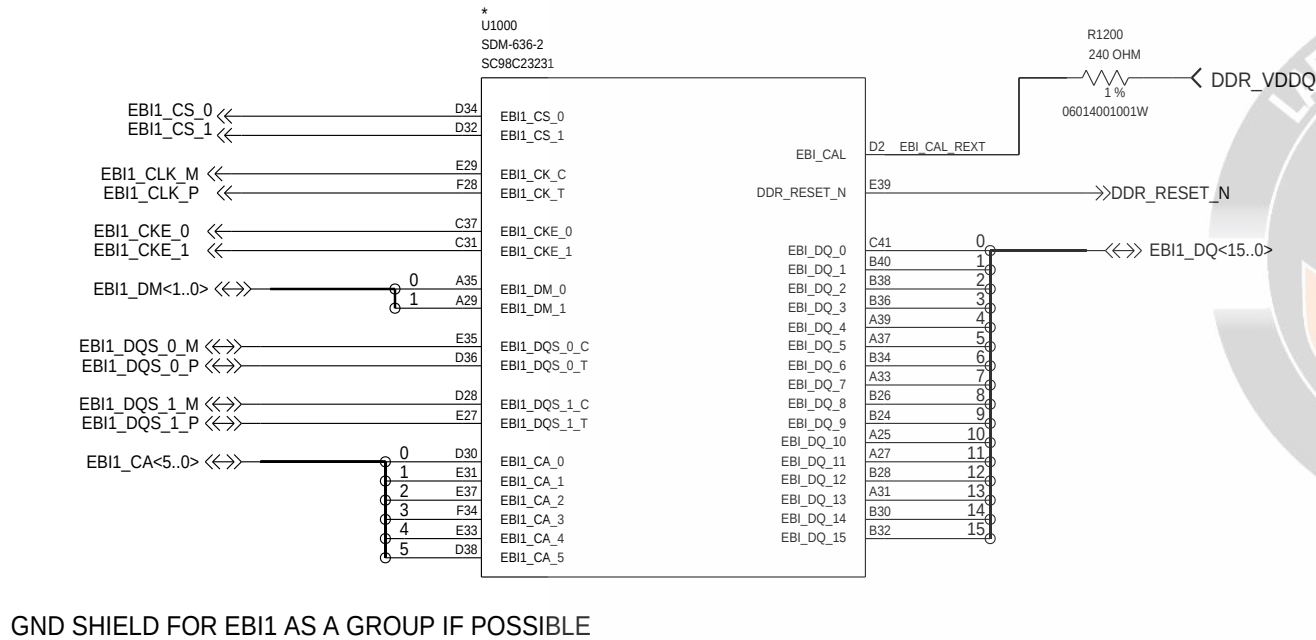
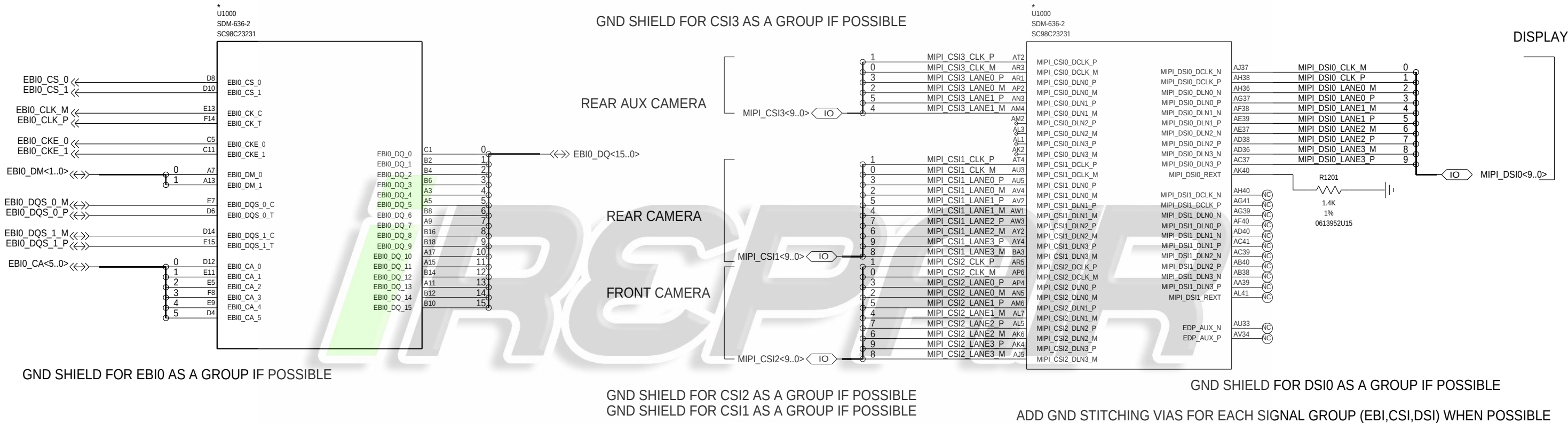
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D

C

B

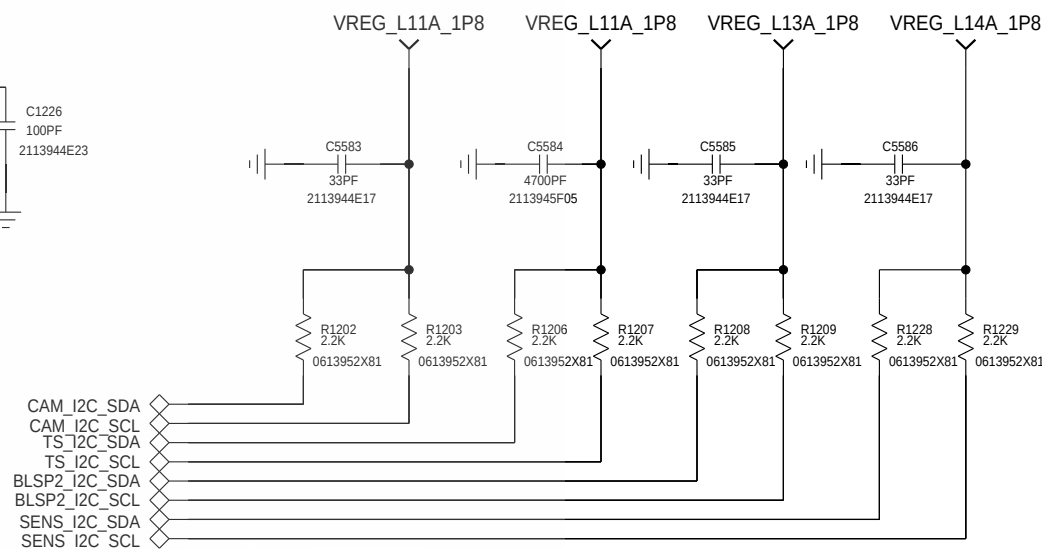
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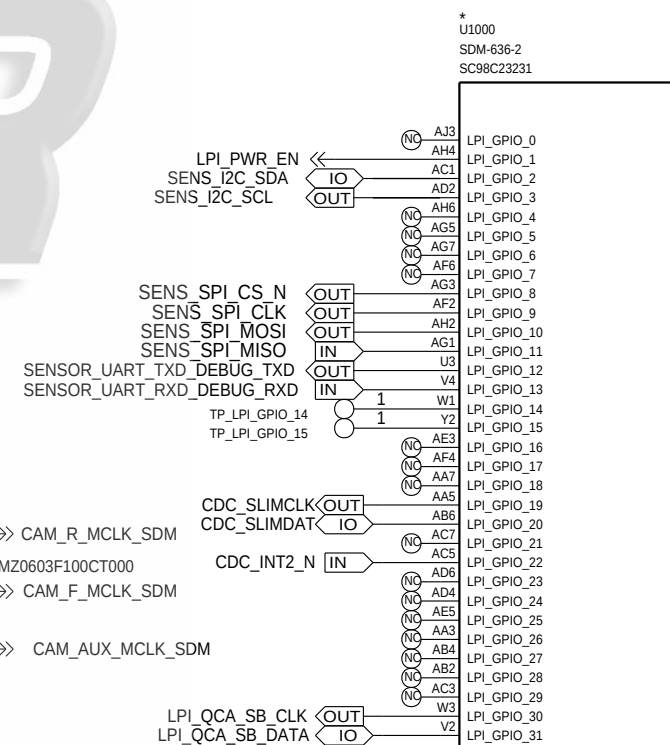
GND SHIELD FOR EBI1 AS A GROUP IF POSSIBLE

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:	ZHANG BEI	BLOCK TITLE: ap_pmic	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:25 2018		PROJECT NAME rio.h.cpm	
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		PAGE 4 OF 66	

FOR RUNTIME REVISION INFORMATION, REFER TO:
[HTTPS://DOCS.GOOGLE.COM/SPREADSHEETS/D/16MWPCNK0BTY06TPGAIPA2FM2YLOA15_B5TRXA0NFFXS/EDIT#GID=0](https://docs.google.com/spreadsheets/d/16MWPCNK0BTY06TPGAIPA2FM2YLOA15_B5TRXA0NFFXS/EDIT#GID=0)



GPI096,
GPI097,
GPI098,
GPI060,
GPI011,
FAST BOOT DEFAULT 0X00, MAKE SURE NO PULLUP.

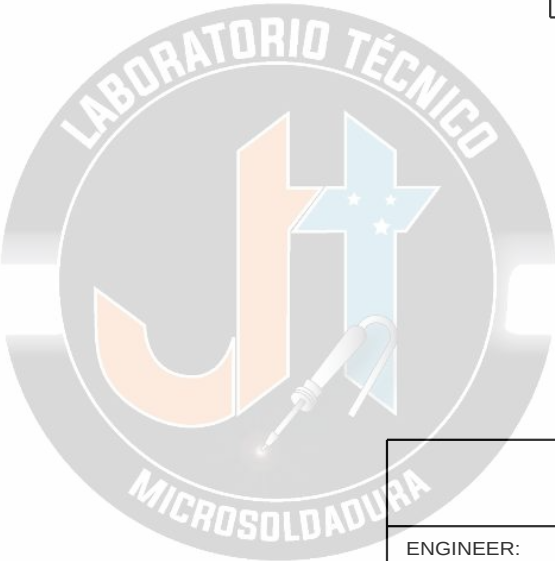
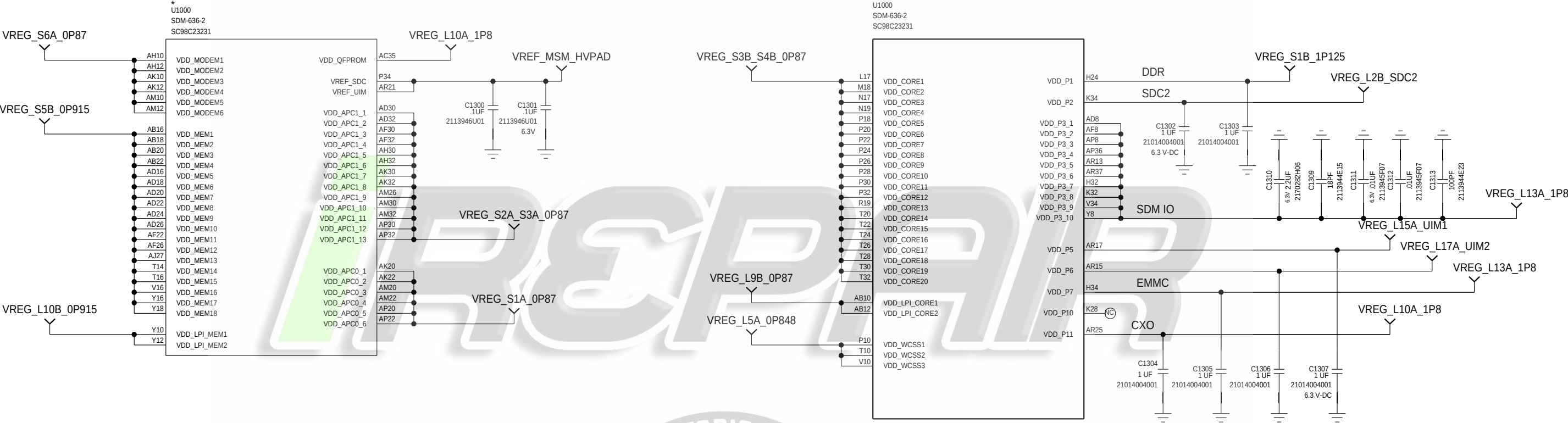


PAGE 5 OF 66

SDM: PWR1

REF 1300-1399

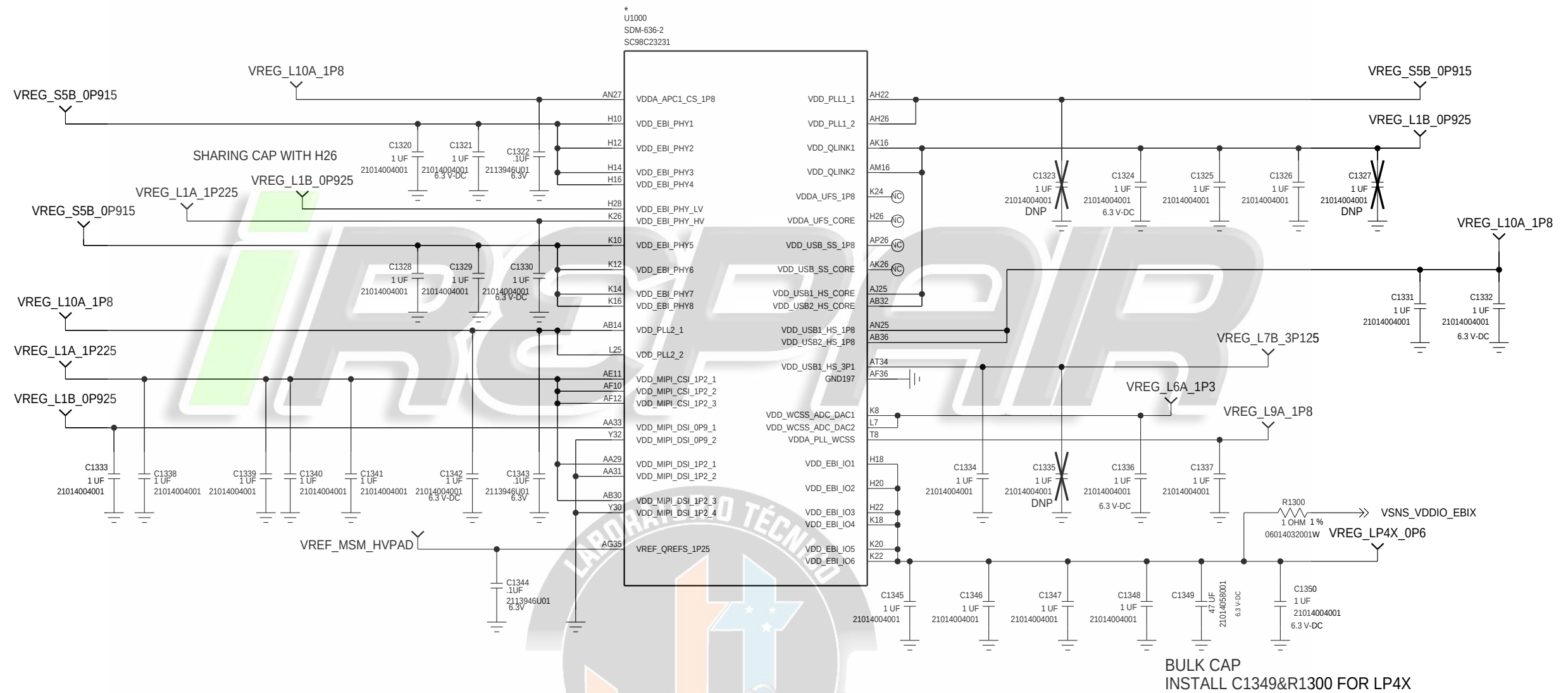
CAD NOTE: PLACE ONE DECAP NEAR PINS V5, V6 AND
THE OTHER DECAP NEAR PIN R5



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:26 2018		ap_pmic	
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		CAD FILE NAME:	
		REV 01	
		PAGE 6 OF 66	

SDM: PWR2

REF 1300-1399

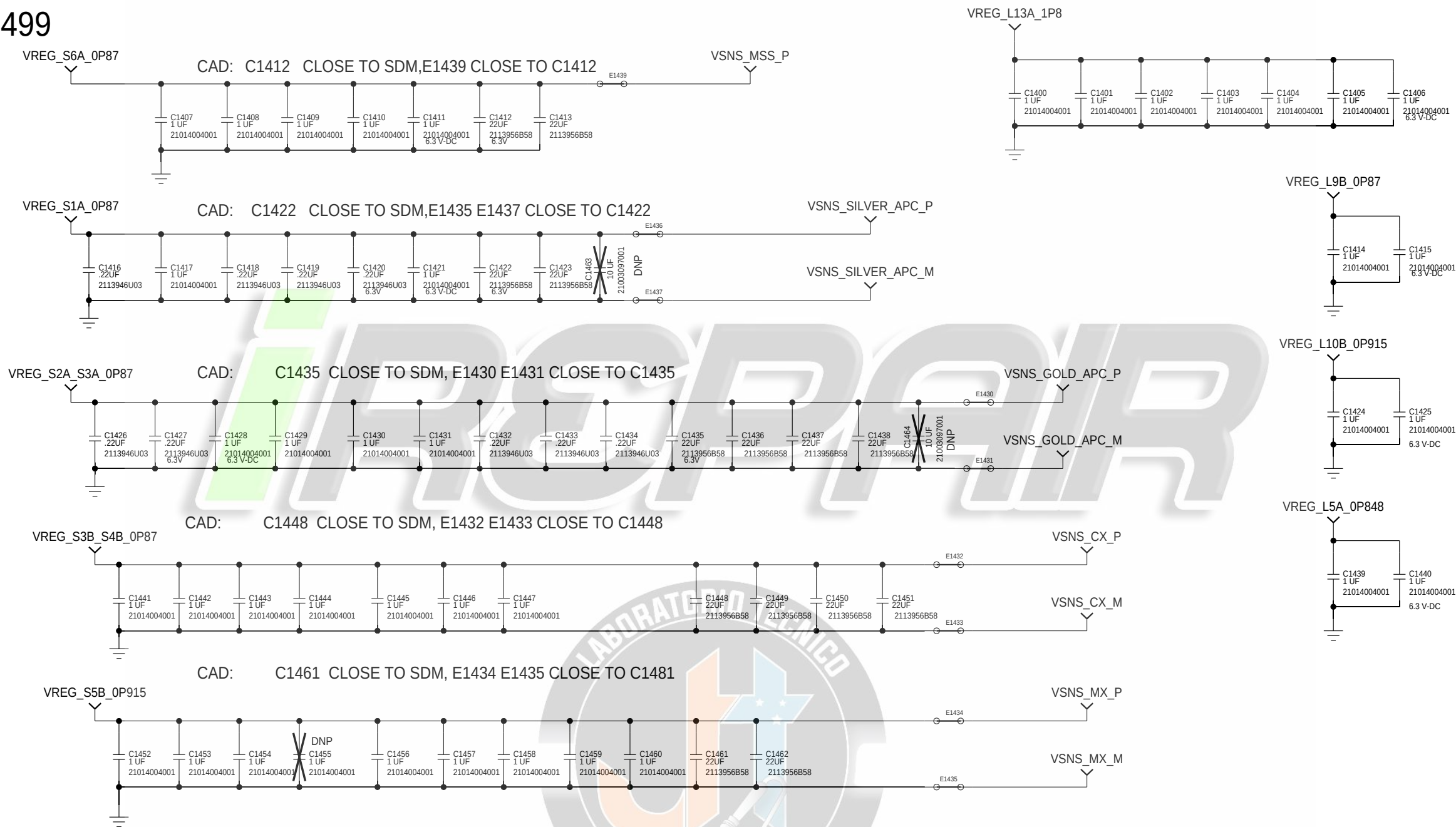


CAD: C1348 CLOSE TO SDM, R1300 CLOSE TO C1348

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:27 2018			
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		CAD FILE NAME:	PAGE 7 OF 66

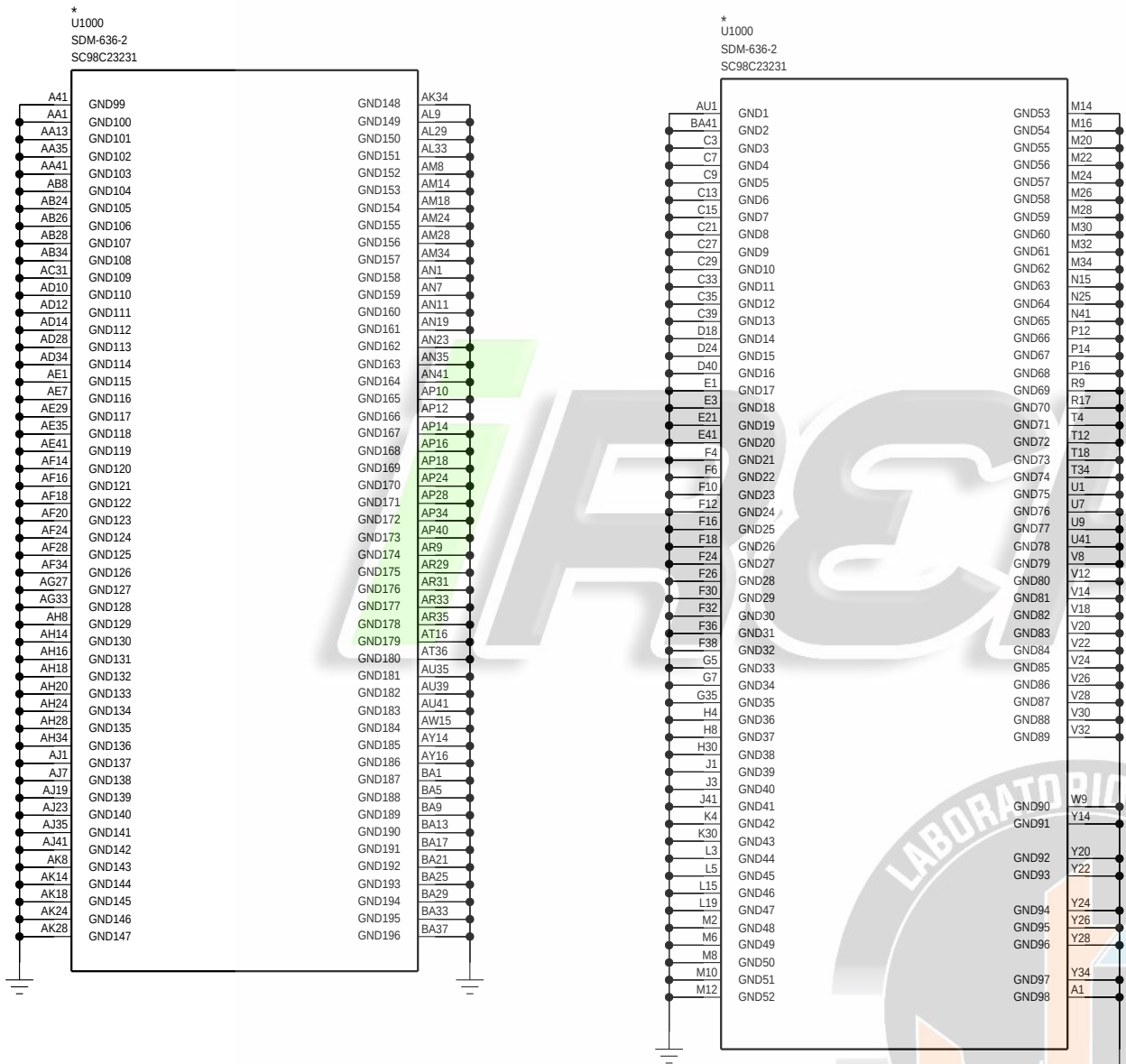
SDM: BYPASS CAPS

REF 1400-1499



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:26 2018		ap_pmic	
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		CAD FILE NAME: PAGE 8 OF 66	

SDM: GROUND/DNC



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:26 2018		ap_pmic	
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		B rioh.cpm	REV 01
		CAD FILE NAME:	
		PAGE 9 OF 66	

MEMORY: EMCP

REF: 1600-1699

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B

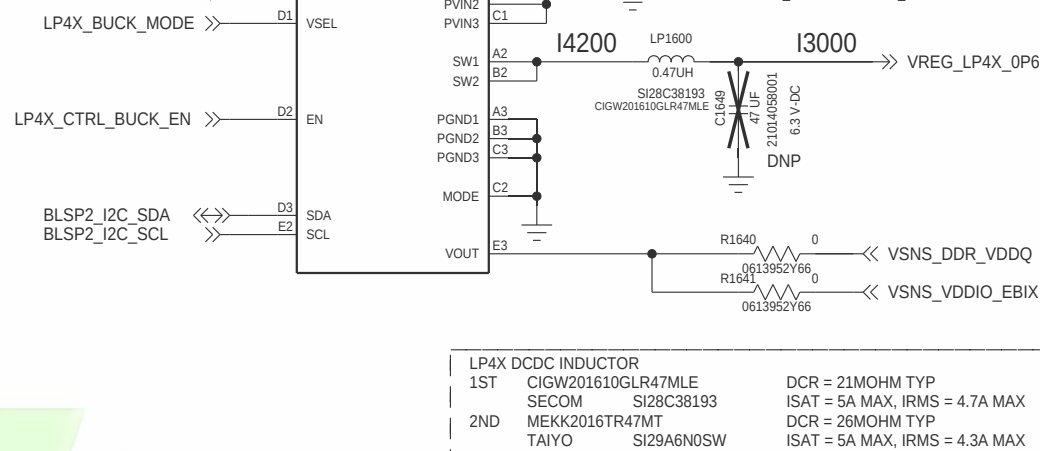
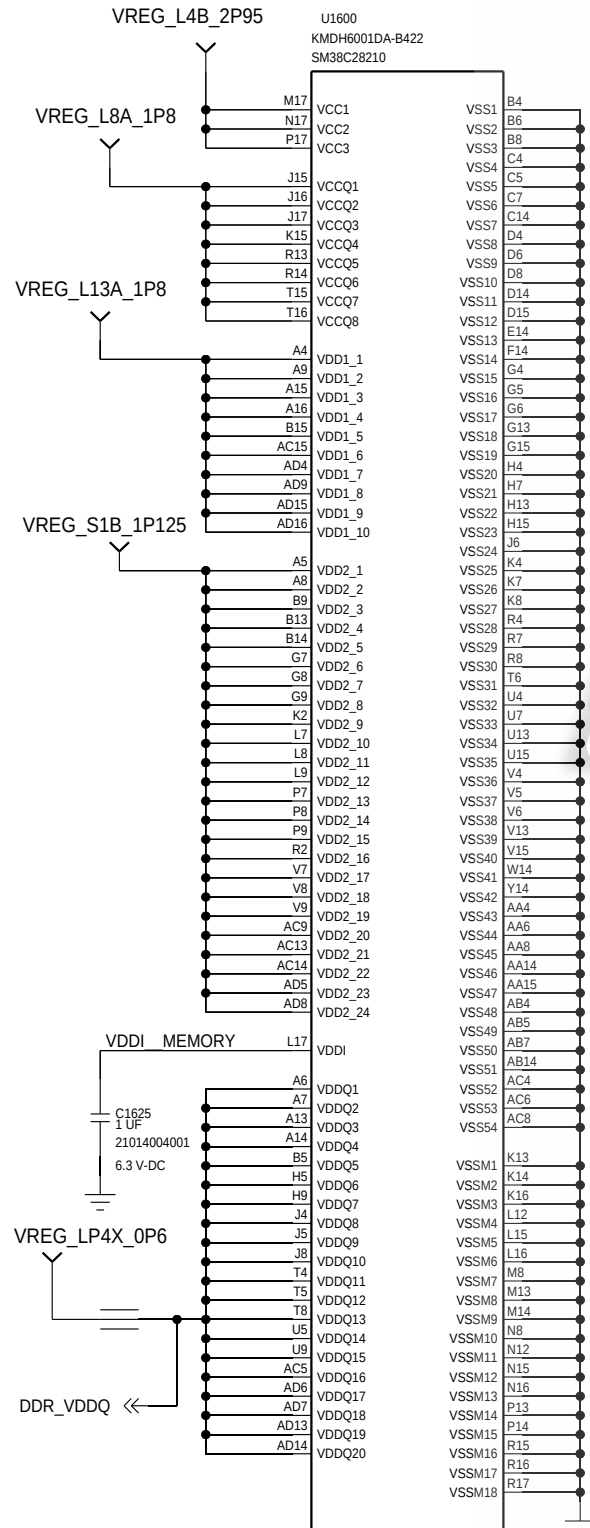
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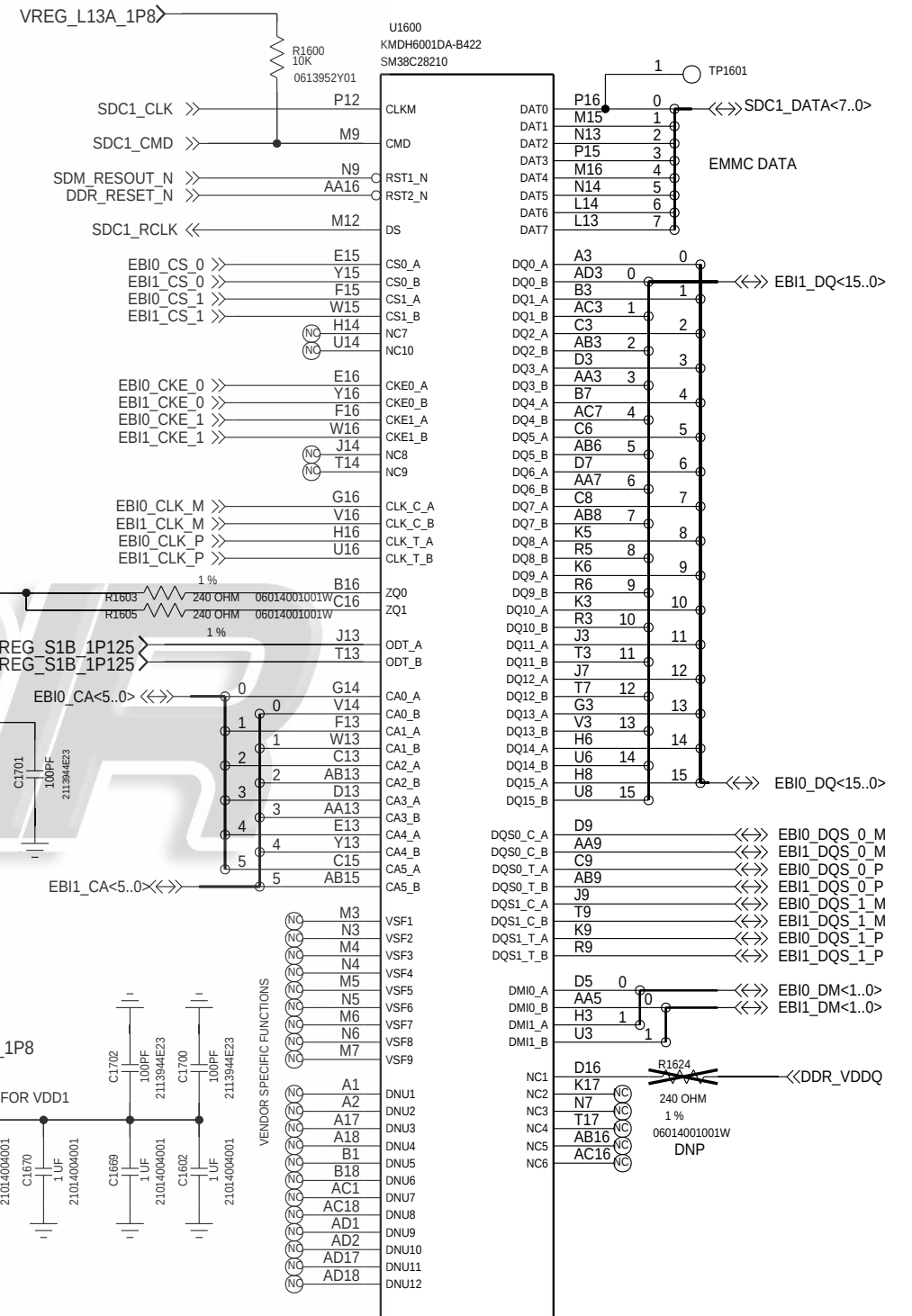
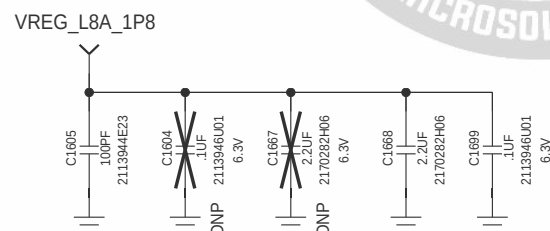
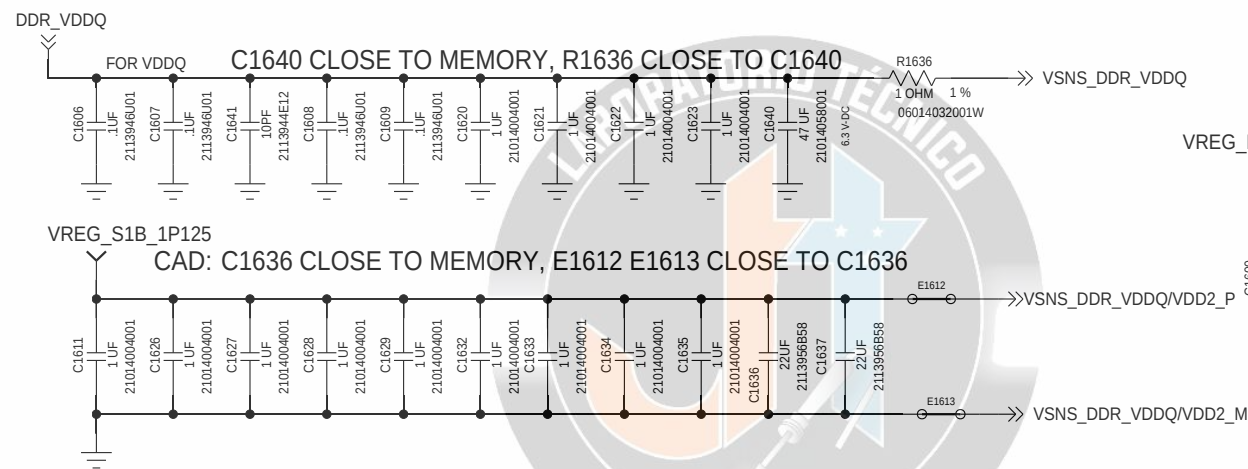
C

B

A



DENSITY	VENDOR PN	VENDOR	LENOVO PN
4+64	KMDH6001DA-B422 H9HP52ACPMADAR-KMM MT29VZZZAD8DQKSL-046	SAMSUNG HYNIX MICRON	SM38C28210 SM38C33882 SM38C35427
6+64	KM3H6001CA-B515	SAMSUNG	SM38C28211
6+128	KM3V6001CM-B705	SAMSUNG	SM38C20203



ADD STITCHING VIAS ALONG THE OUTLINE OF DDR AND SDC SIGNALS
THOSE PARALLEL INTERFACE CAN HAVE LARGE CURRENT, SSN ARE HIGH
MAY CAUSE RF DESNE IF RETURN CURRENT ARE NOT WELL LIMITED
AVOID ROUTING SENSITIVE SIGNAL CORSS THIS SECTION
SENSITIVE SIGNALS CAN BE CLOCKS, (CK, MCLK), ANALOG SIGNALS AND SO ON

APPROVALS

ENGINEER: ZHANG BEI

LAST MODIFIED BY: ZHANG BEI
Wed Sep 12 09:13:26 2018

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

ap_pmic

PROJECT NAME
rio.h.cpm

CAD FILE NAME:

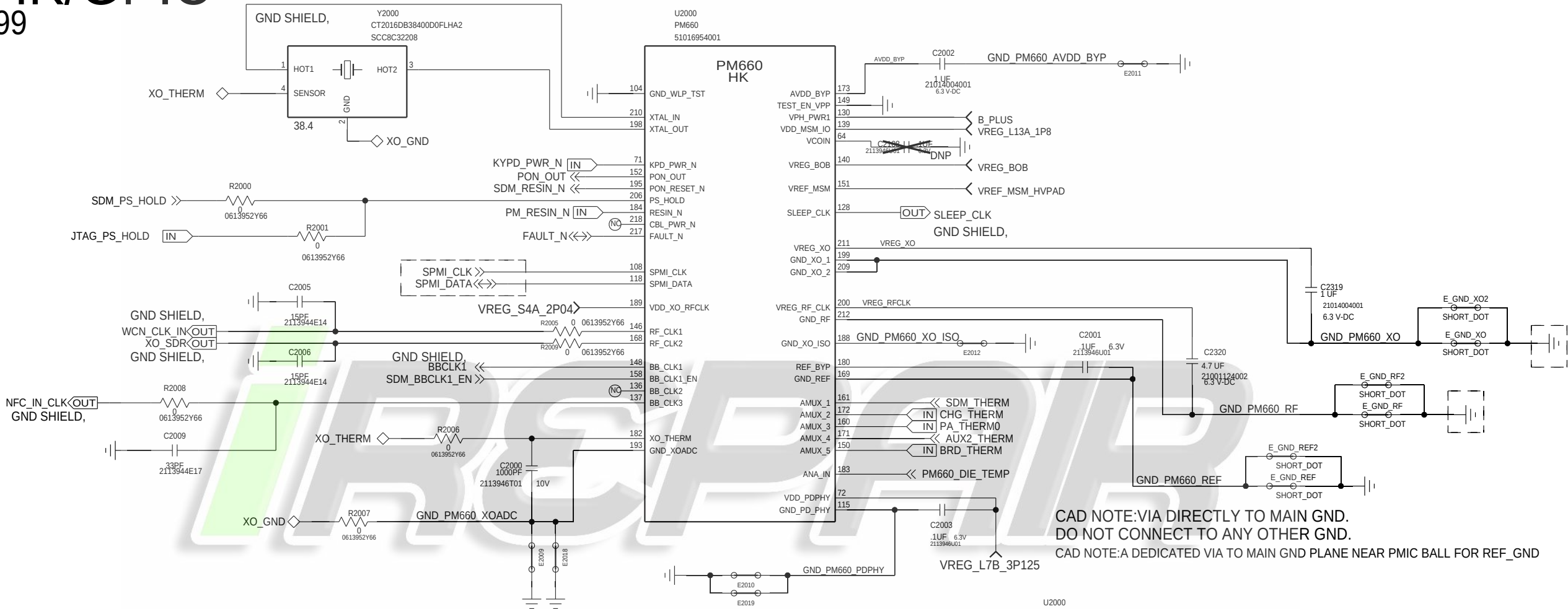
REV
01

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PM660: HK/GPIO

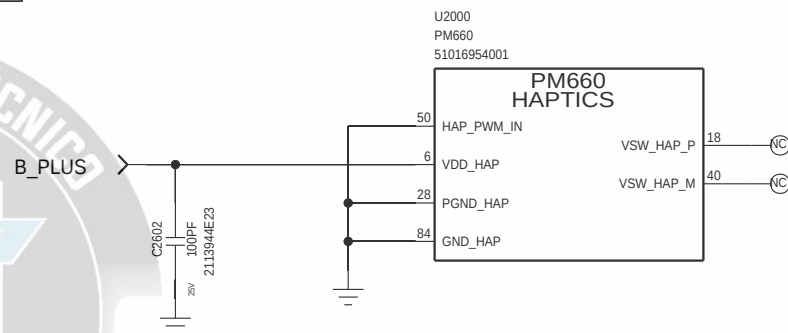
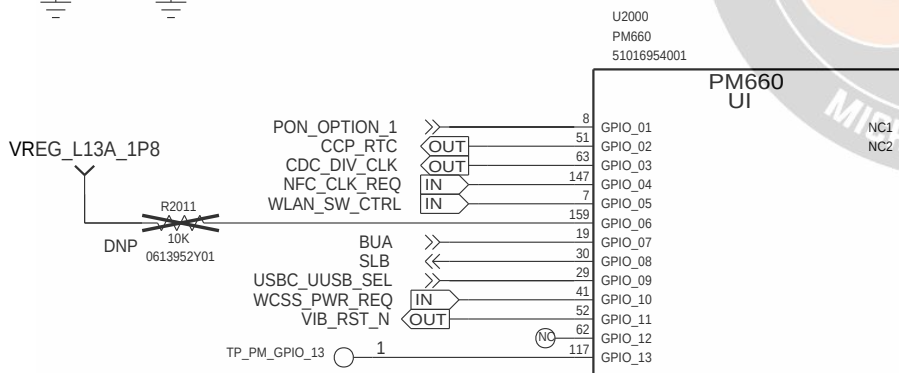
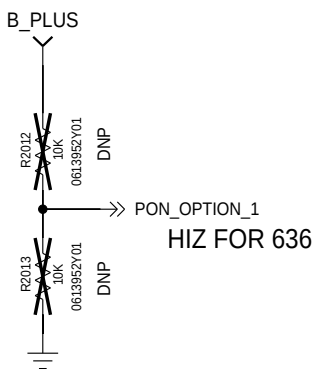
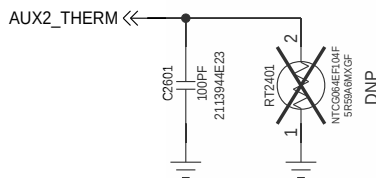
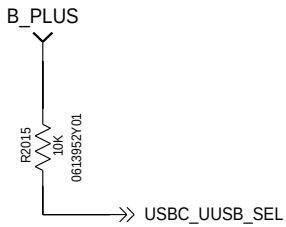
REF: 2000-2099

XTAL_IN, AND XTAL_OUT ROUTED ON 1ST LAYER, NOT AS DIFFERENT PAIR. REFER TO 80-P7747-3 REV. B FOR CRYSTAL LAYOUT GUIDE



CAD NOTE: VIA DIRECTLY TO MAIN GND.
DO NOT CONNECT TO ANY OTHER GND.
CAD NOTE: A DEDICATED VIA TO MAIN GND PLANE NEAR PMIC BALL FOR REF_GND

USBC_UUSB_SEL IS HIGH/HI-Z FOR TYPE-C

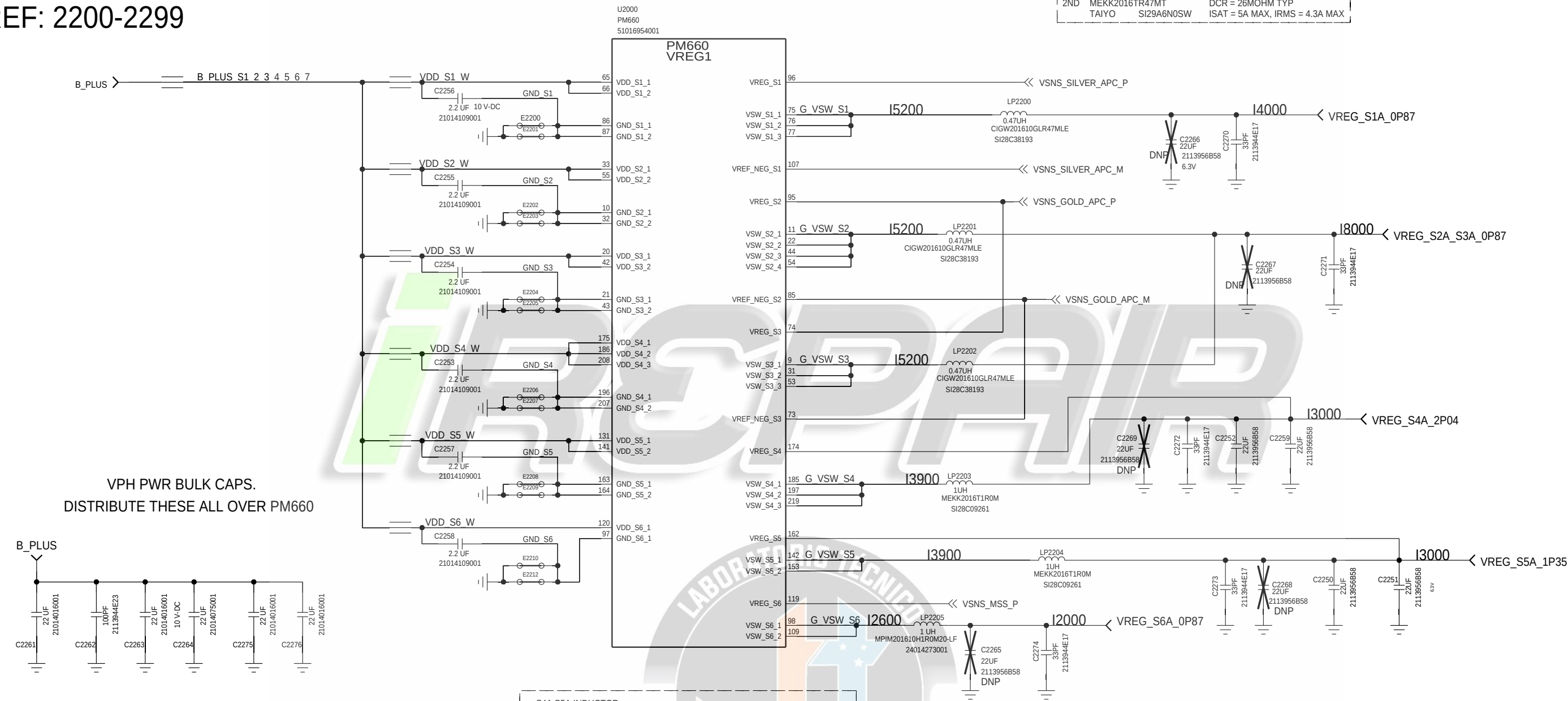


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:25 2018			
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		CAD FILE NAME:	PAGE 11 OF 66

PM660: SMPS

REF: 2200-2299

S1A,S2A,S3A INDUCTOR		
1ST	CIGW201610GLR47MLE	DCR = 21MOHM TYP
	SECOM SI28C38193	ISAT = 5A MAX, IRMS = 4.7A MAX
2ND	MEKK2016TR47MT	DCR = 26MOHM TYP
	TAIYO SI29A6N0SW	ISAT = 5A MAX, IRMS = 4.3A MAX

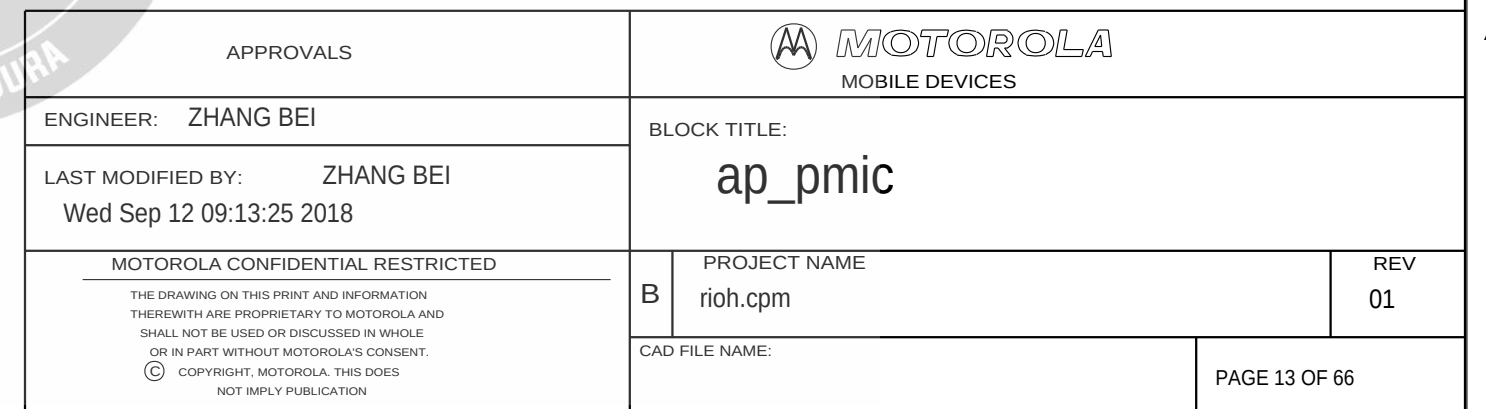


S4A,S5A INDUCTOR		
1ST	MEKK2016T1R0M	DCR = 260OHM
	TAIYO SI28C09261	ISAT = 3.6A MAX, IRMS = 3.1A MAX
2ND	MHCHL201610C-1R0M-Q8	DCR = 490OHM
	CHILISIN SI28XXXX	ISAT = 3.6A MAX, IRMS = 3.0A MAX
S6A INDUCTOR		
1ST	MPIM201610H1R0M20-LF	DCR = 58MOHM
	MICROGATE 24014273001	ISAT = 3.1A MAX, IRMS = 2.2A MAX
2ND	CIGT201610UM1R0MNE	DCR = 48MOHM
	SECOM SI28C38192	ISAT = 3.1A MAX, IRMS = 2.8A MAX

NOTE: LOCAL CAP PLACE HOLDER PROVIDED FOR LOAD CURRENT MEASUREMENTS

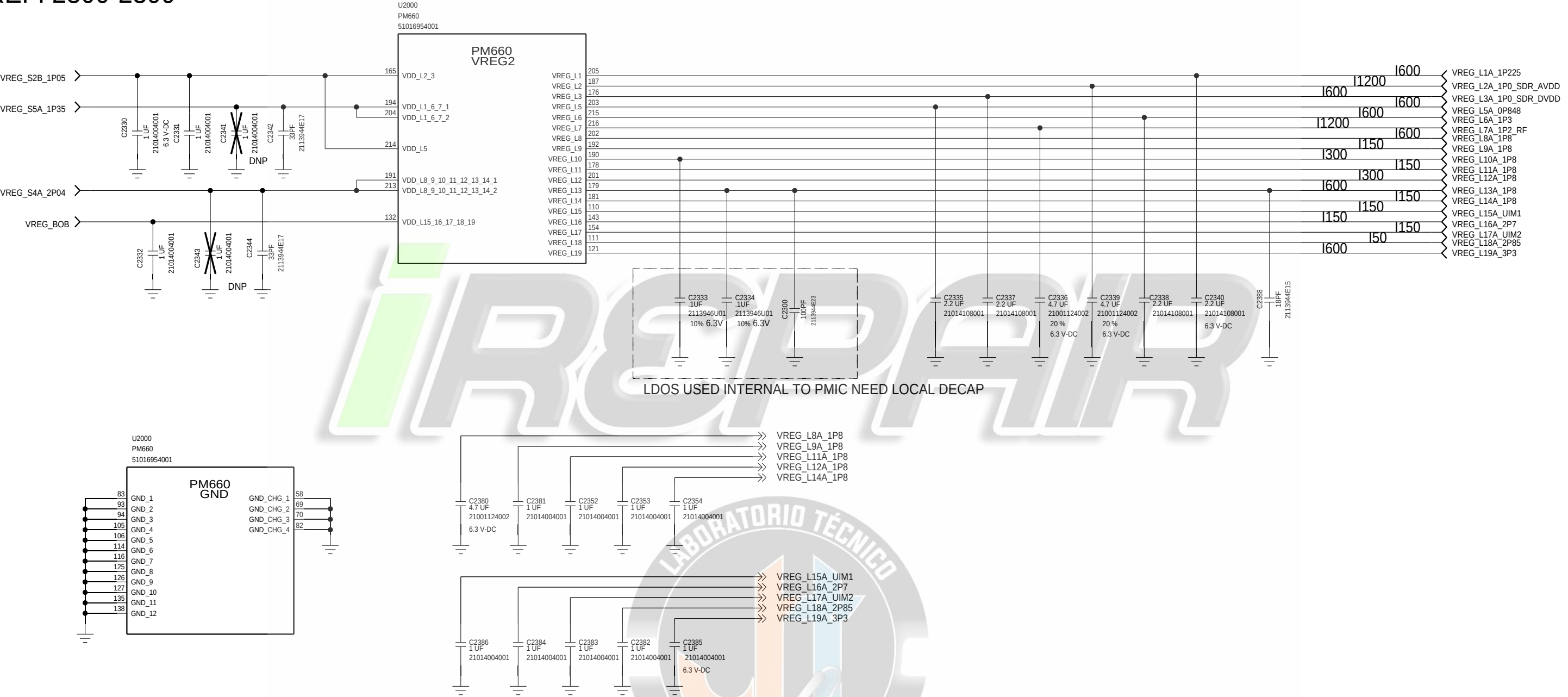
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:27 2018		ap_pmic	
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		CAD FILE NAME:	PAGE 12 OF 66

REF: 2400-2419



PM660: VREGS

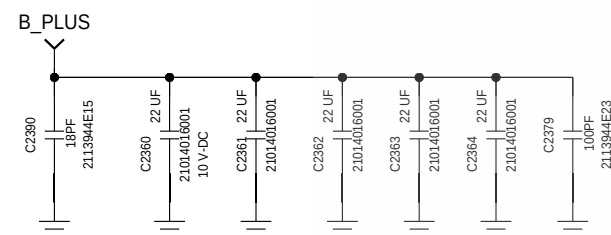
REF: 2300-2399



PSEUDO CAPLESS LODS
FOR CAPLESS LODS:IF DECAPS ON THE LOAD SIDE DO NOT ADD UP TO LDO SPEC, THEN INSTALL THE CAP CLOSE TO THE PMIC

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:27 2018		ap_pmic	
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		CAD FILE NAME:	
		PAGE 14 OF 66	
REV 01			

REF: 2300-2399

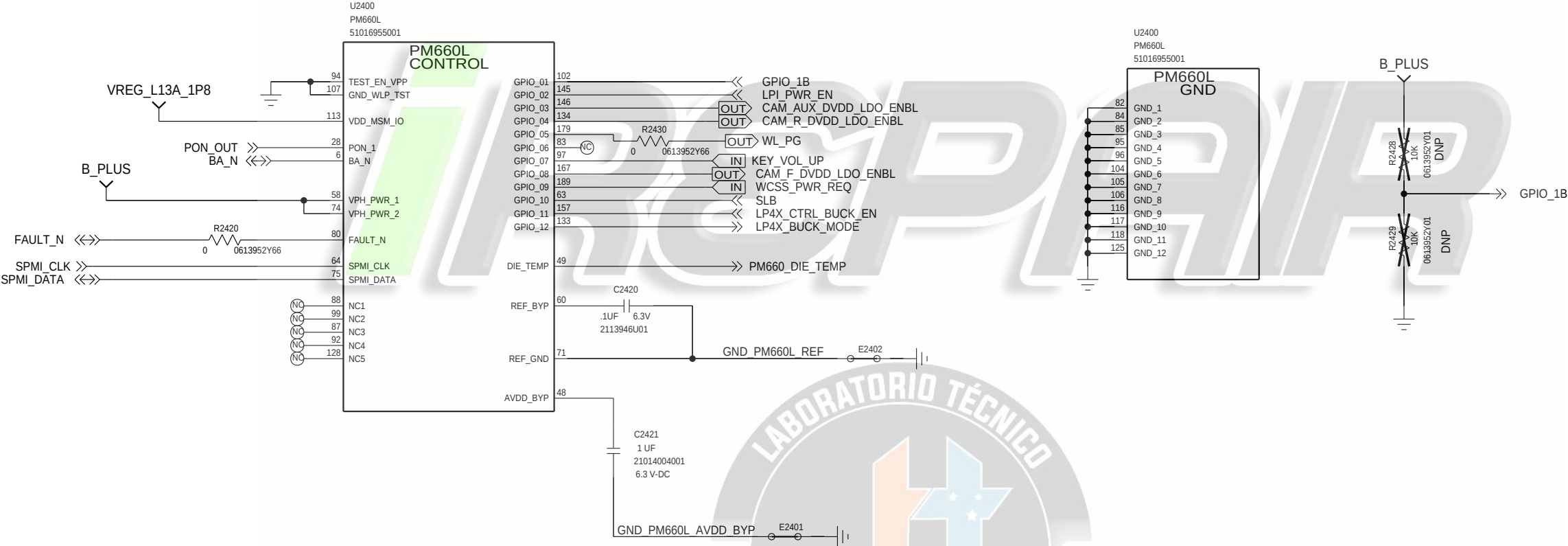


NOTE: LOCAL CAP PLACE HOLDER PROVIDED FOR LOAD CURRENT MEASUREMENTS

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:27 2018			
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		CAD FILE NAME:	PAGE 15 OF 66

PM660L: CONTROL

REF: 2420-2439



GND SHIELDED IF POSSIBLE, ROUTE DIFFERENTIALLY AND SPACING WITH 2X LW

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:26 2018		ap_pmic	
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		CAD FILE NAME:	
		REV 01	
		PAGE 16 OF 66	

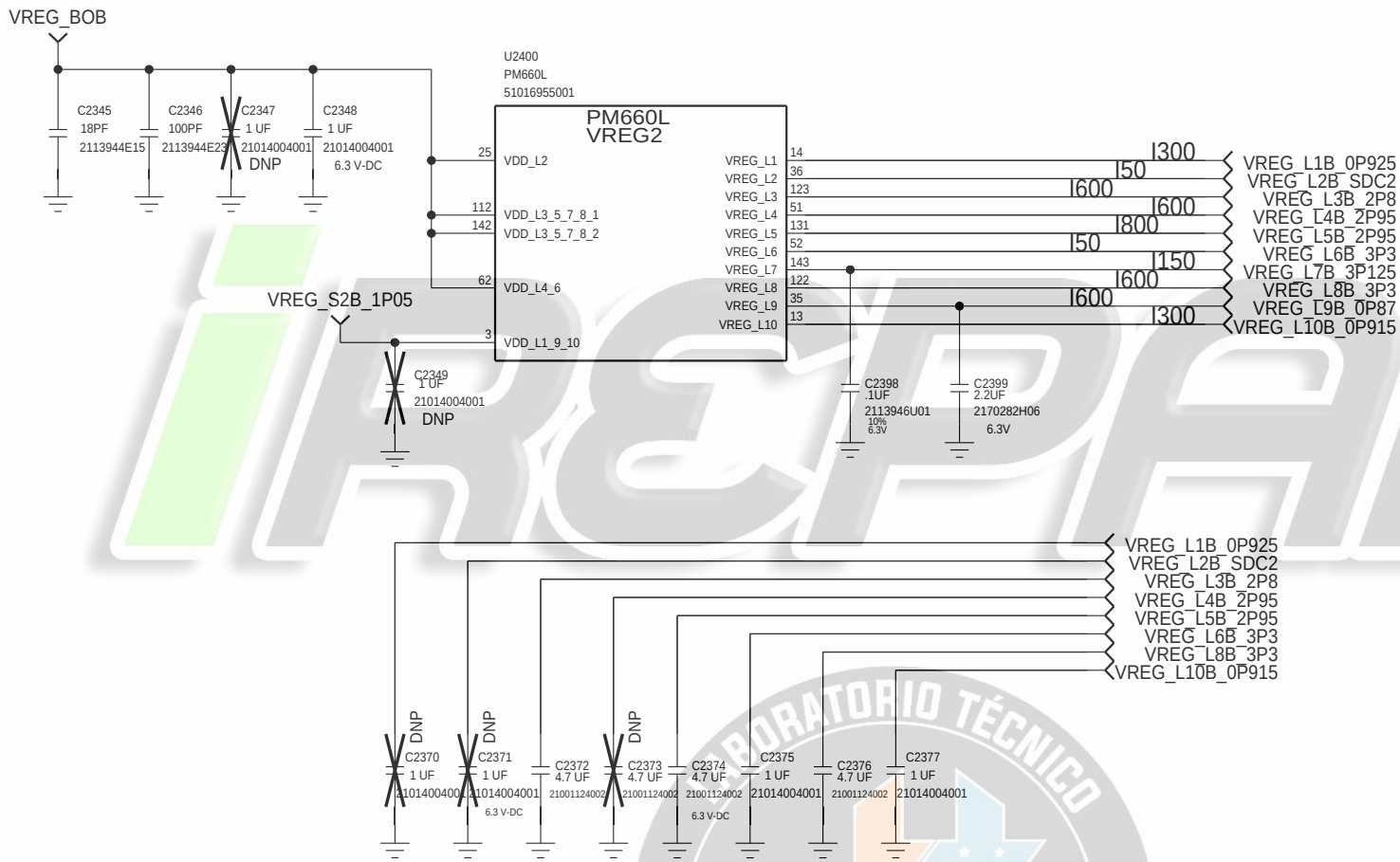
REF: 2440-2459



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: ZHANG BEI	BLOCK TITLE: ap_pmic			
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:25 2018				
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PM660L: VREGS

REF: 2300-2399

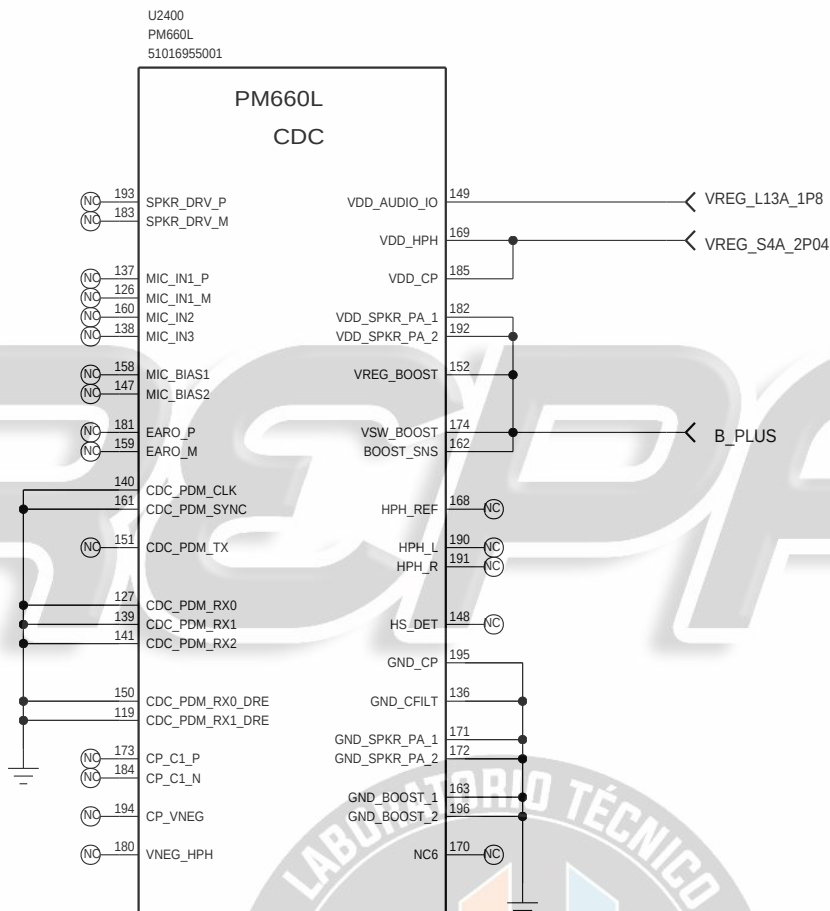


PSEUDO CAPLESS LODS
FOR CAPLESS LODS:IF DECAPS ON THE LOAD SIDE DO NOT ADD UP TO LDO SPEC, THEN INSTALL THE CAP CLOSE TO THE PMIC

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:27 2018		ap_pmic	
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		CAD FILE NAME:	
		PAGE 18 OF 66	

PM660L: CODEC

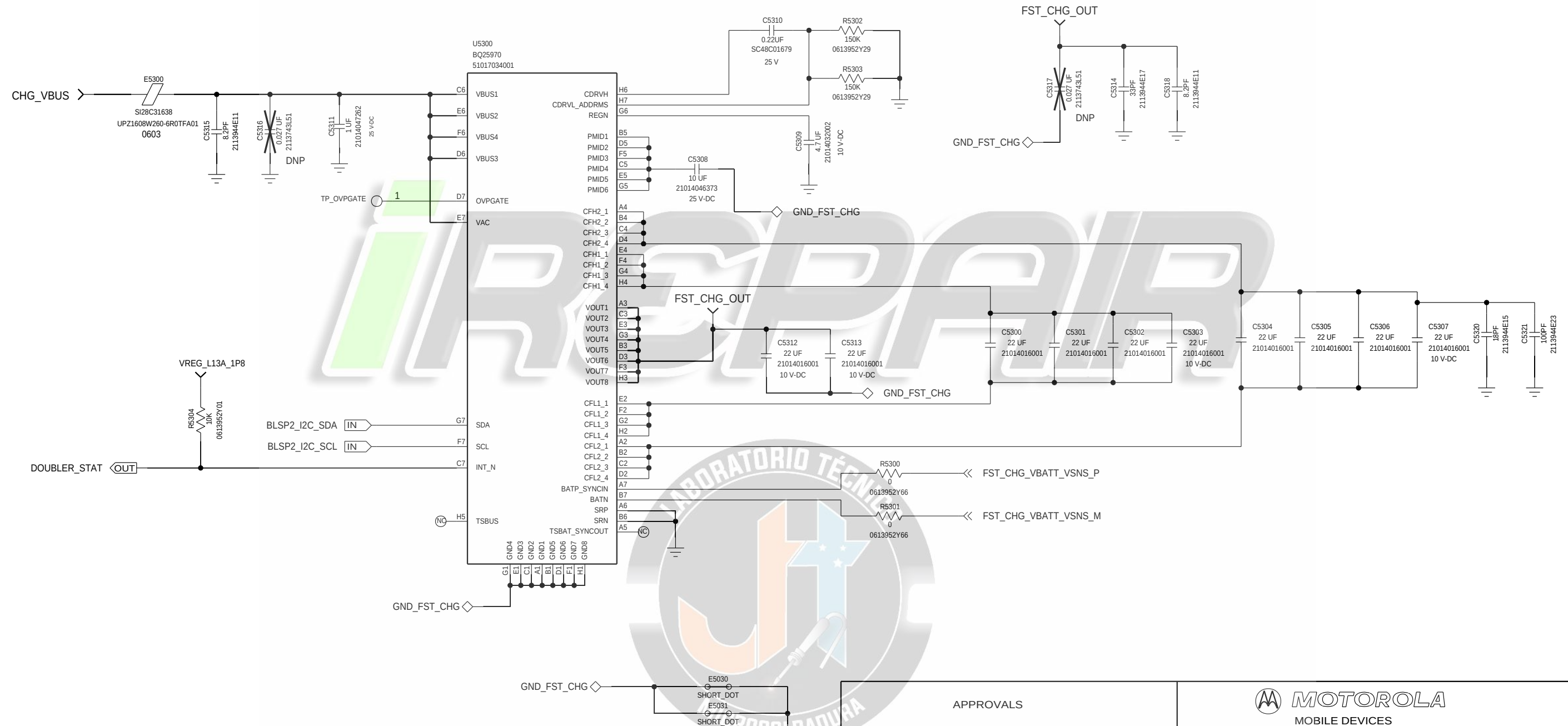
REF 2500-2579



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE: ap_pmic	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:27 2018			
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		CAD FILE NAME: PAGE 19 OF 66	

PWR: FAST CHARGER

REF:5300 - 5399



APPROVALS

ENGINEER: ZHANG BEI

LAST MODIFIED BY: ZHANG BEI
Wed Sep 12 09:13:23 2018

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BLOCK TITLE:

power

PROJECT NAME
rio.h.cpm

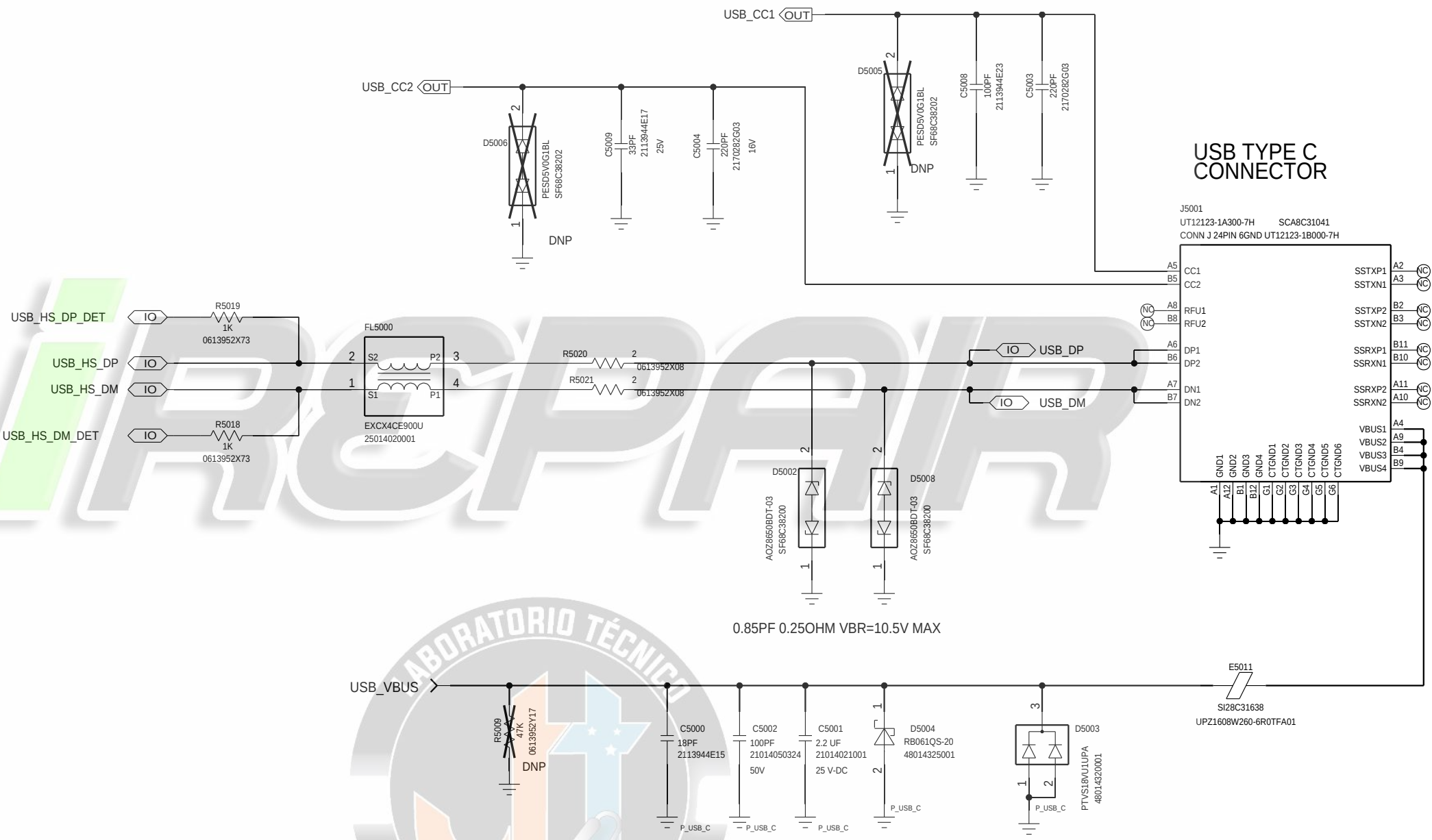
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CAD FILE NAME:

PAGE 20 OF 66

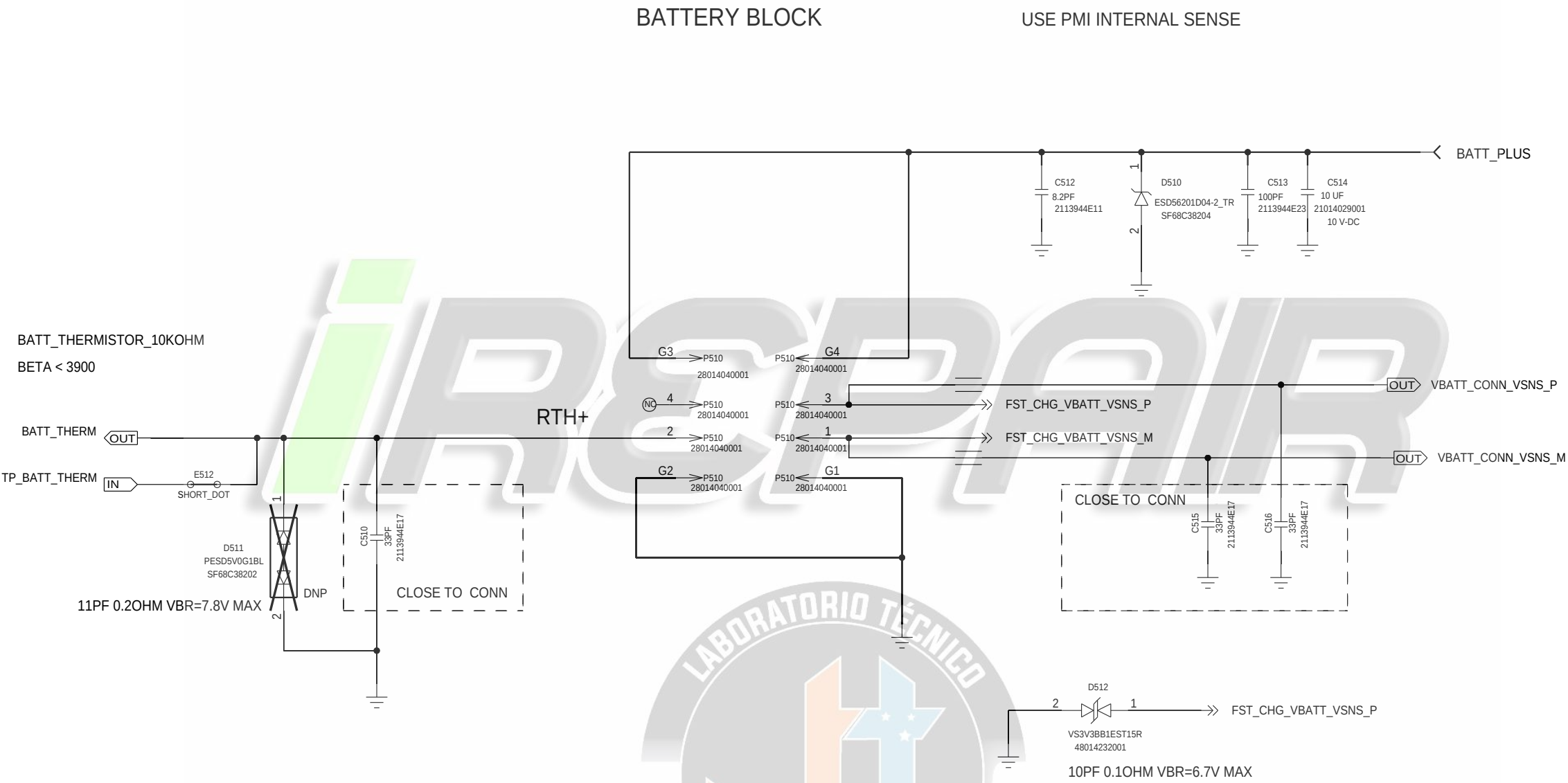
PWR: USB-C

REF:5300 - 5399



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:23 2018		power	
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		CAD FILE NAME:	
		REV 01	
		PAGE 21 OF 66	

PWR: BATT
REF:510 - 550

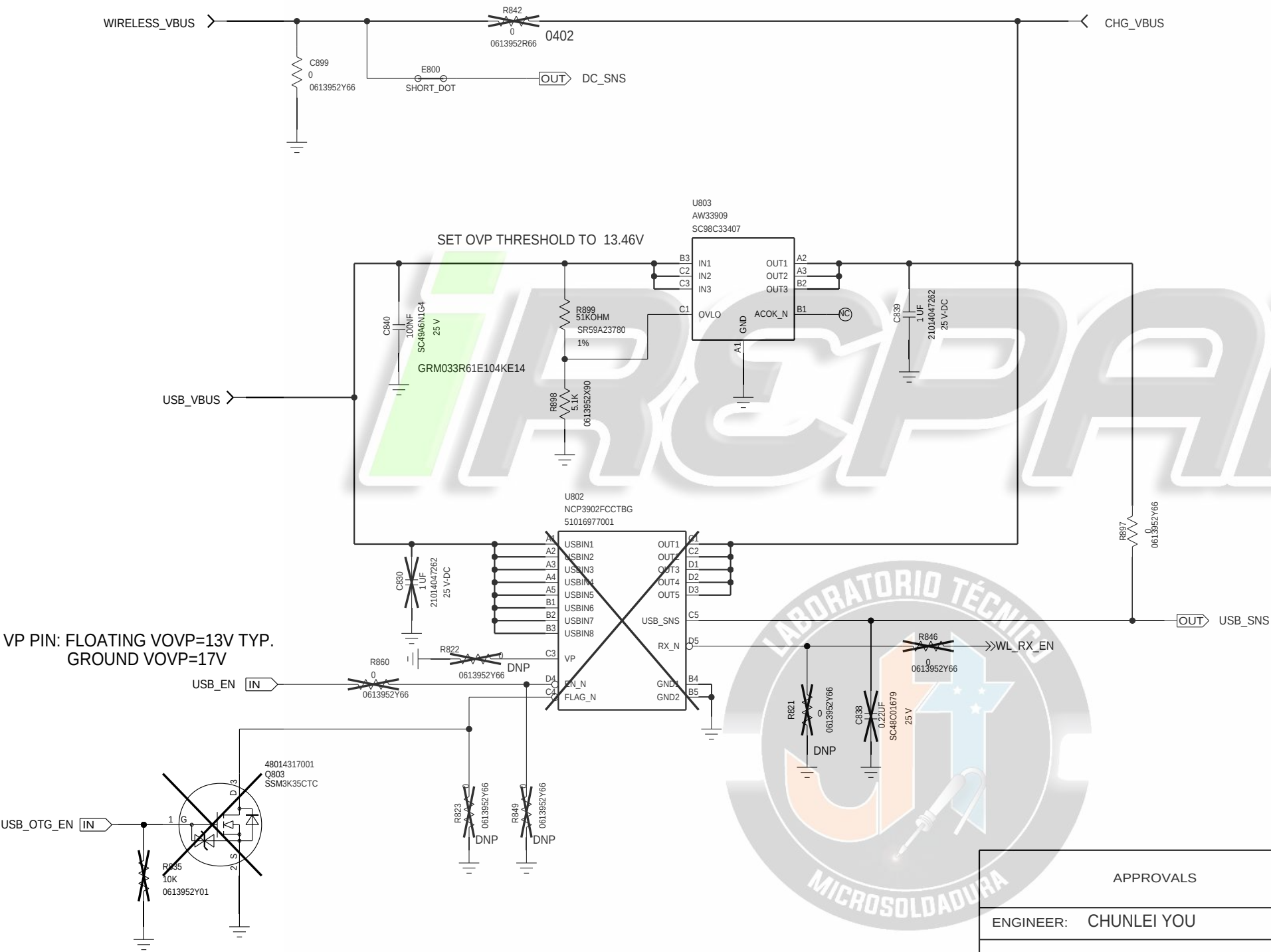


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:23 2018		power	
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		REV 01	
		CAD FILE NAME:	
		PAGE 22 OF 66	

PWR: OVP

REF:800 - 899

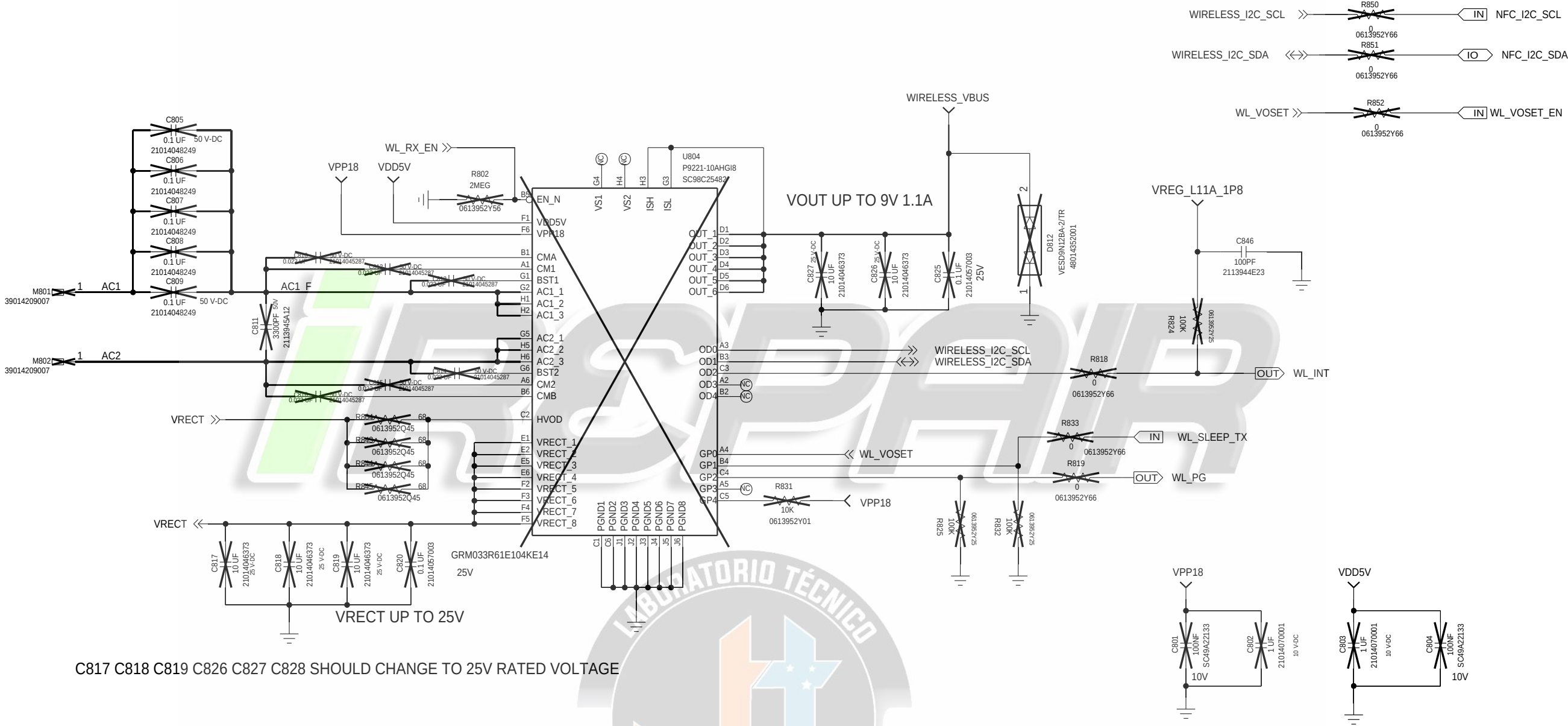
0402 RES 00HM RATED CURRENT 1A.



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: CHUNLEI YOU		BLOCK TITLE: power	
LAST MODIFIED BY: Wed Sep 12 09:13:23 2018		PROJECT NAME rioh.cpm	
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WIRELESS CHARGER

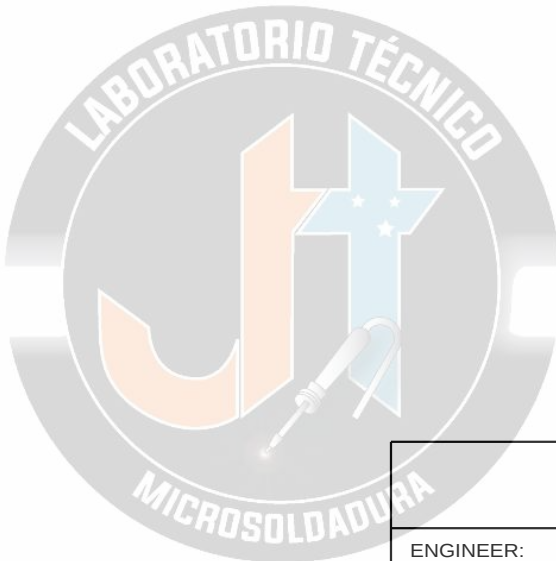
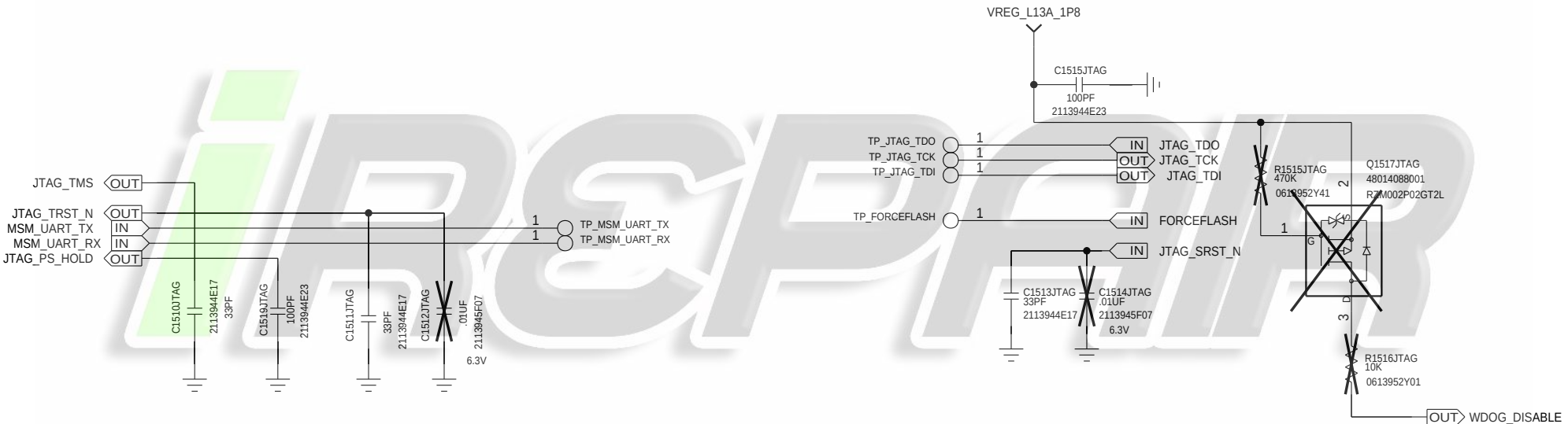
REF:800 - 899



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: CHUNLEI YOU		BLOCK TITLE: power	
LAST MODIFIED BY: Wed Sep 12 09:13:23 2018		PROJECT NAME rio.h.cpm	
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DEBUG PORT

REF 1000-1099



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:32 2018		misc	
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		CAD FILE NAME:	
		PAGE 25 OF 66	

D

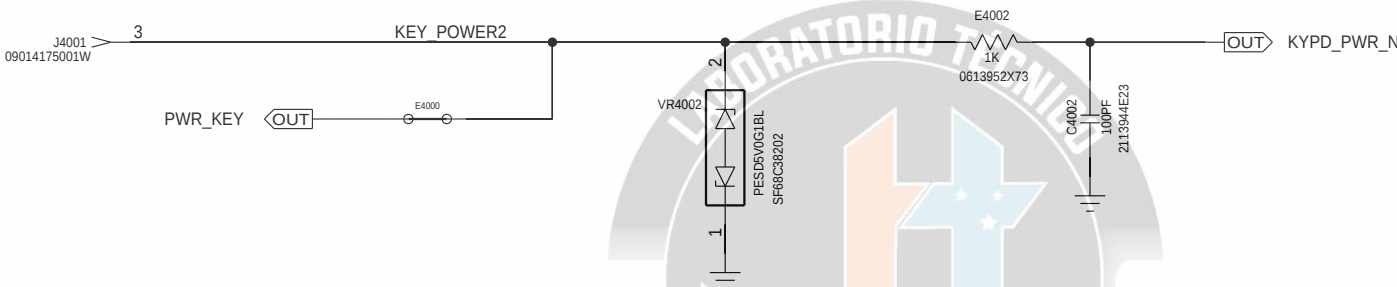
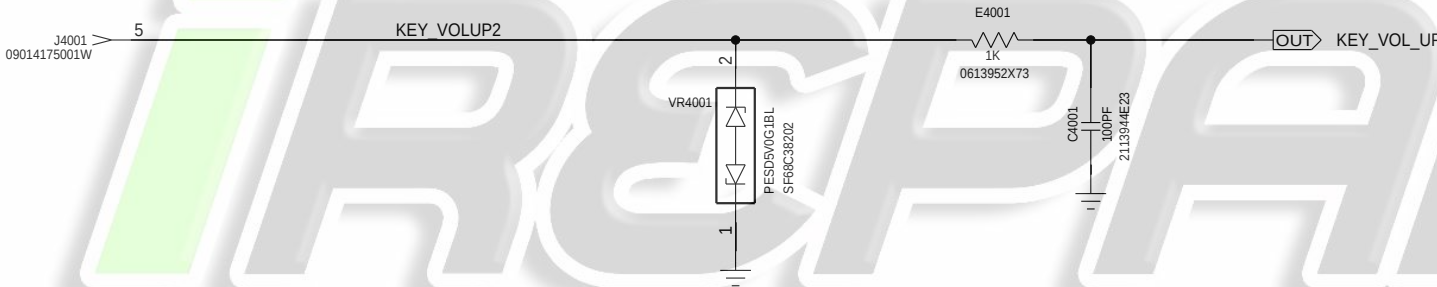
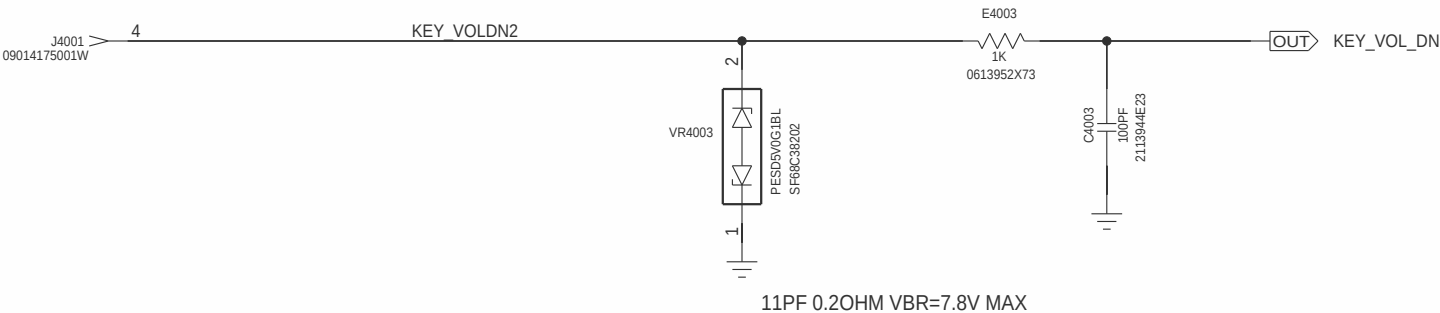
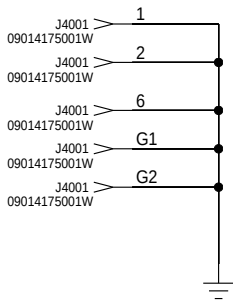
A

UI: SIDE KEY

REF:4000-4199

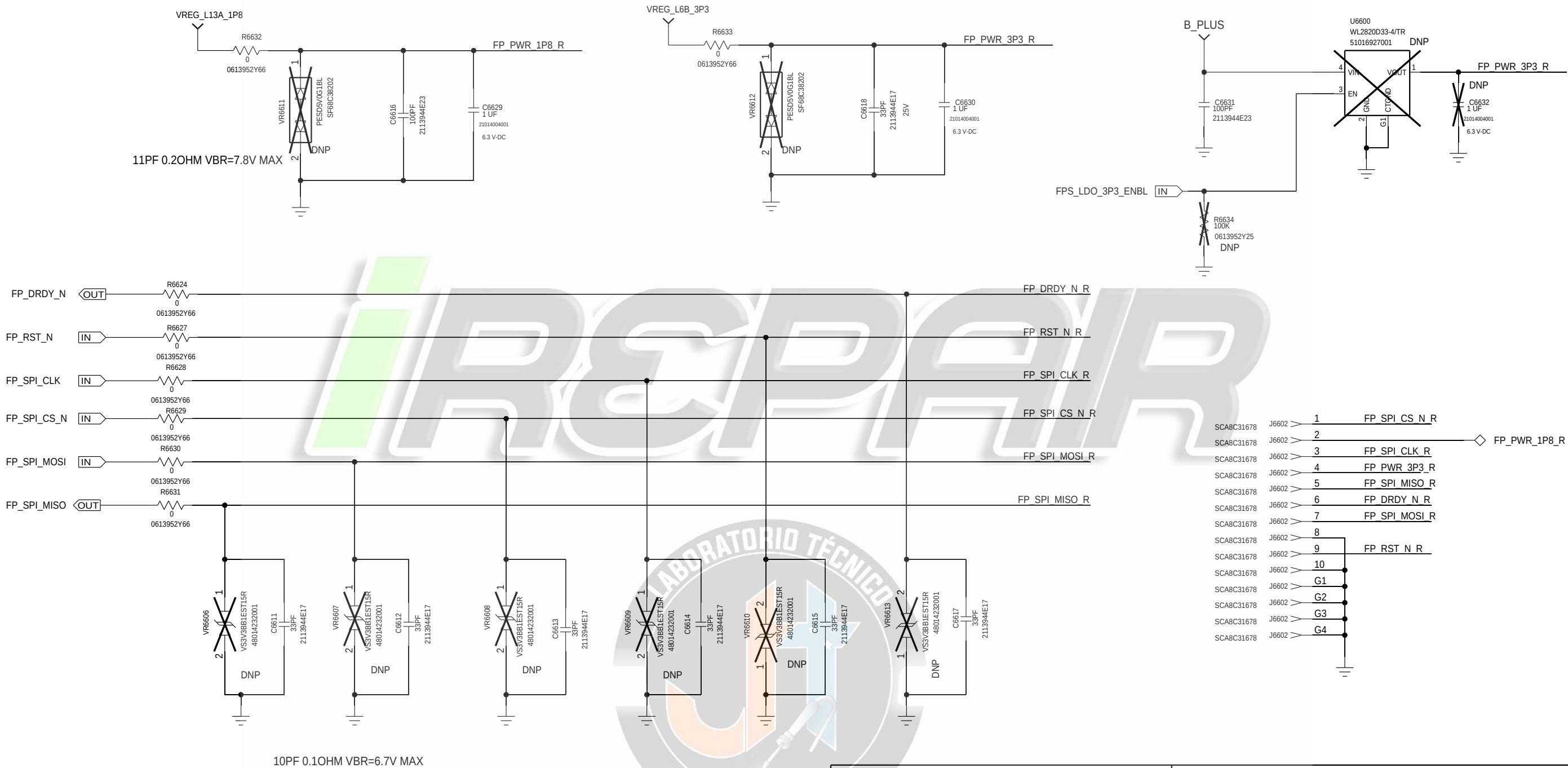
MECHANICAL KEYS

VOL_UP/DOWN/GND TPS NEED P0 BRINGUP ONLY



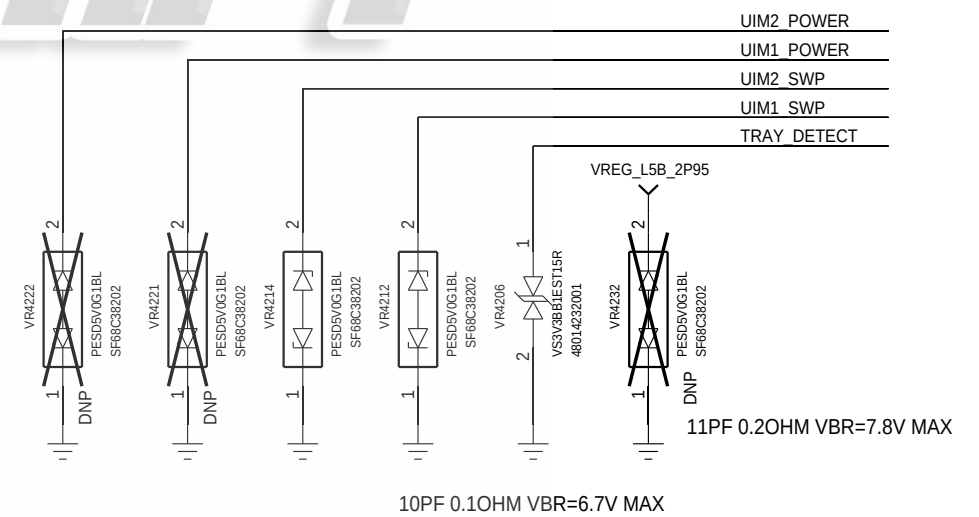
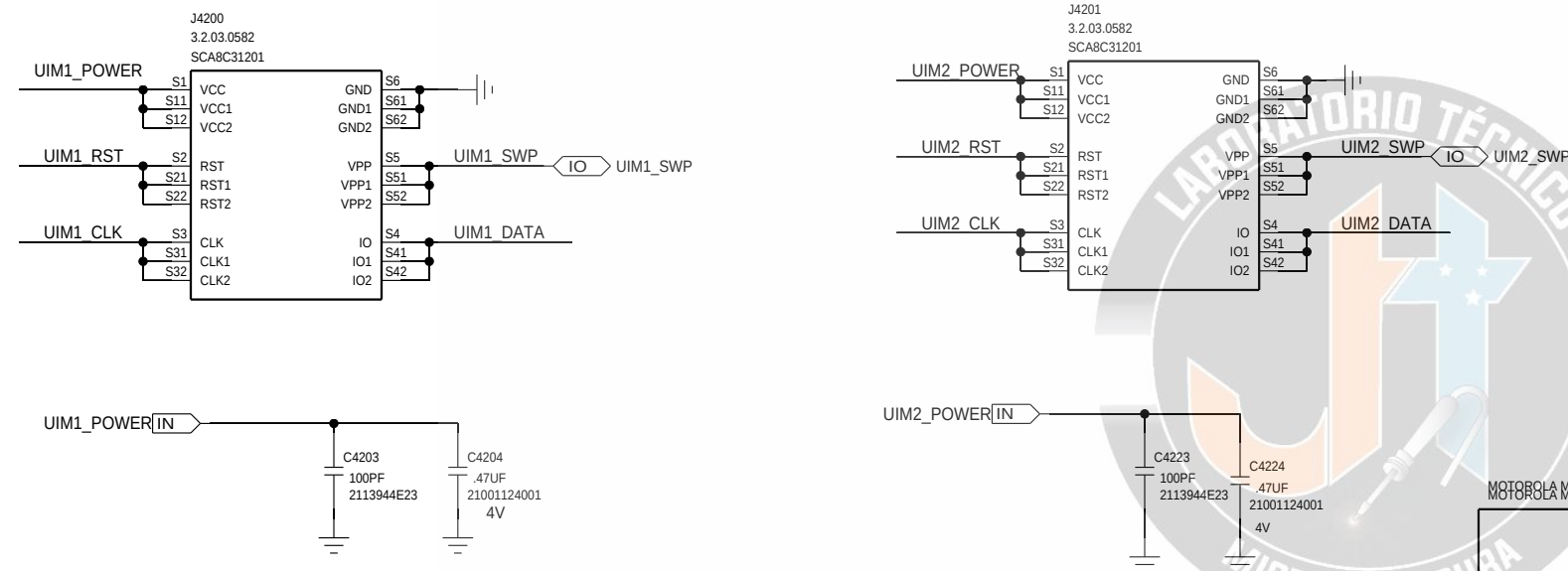
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Wed Sep 12 09:13:24 2018		ui	
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		PROJECT NAME rio.h.cpm	
		REV 01	
		CAD FILE NAME:	
		PAGE 27 OF 66	

UI: FPS
REF: 6600 - 6699



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: NATE CONNELL		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Wed Sep 12 09:13:24 2018		ui	
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		REV 01	
		PAGE 28 OF 66	

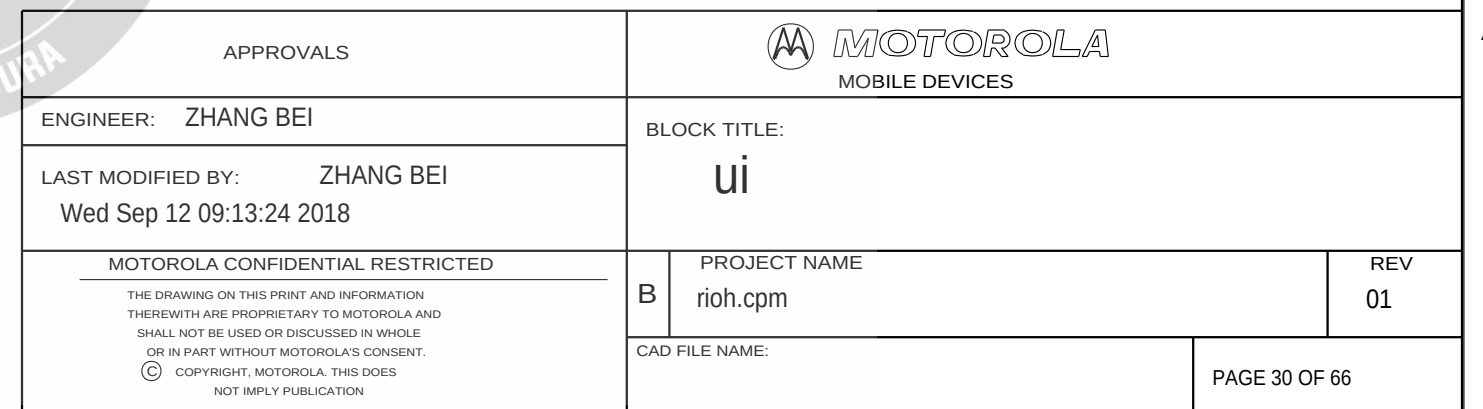
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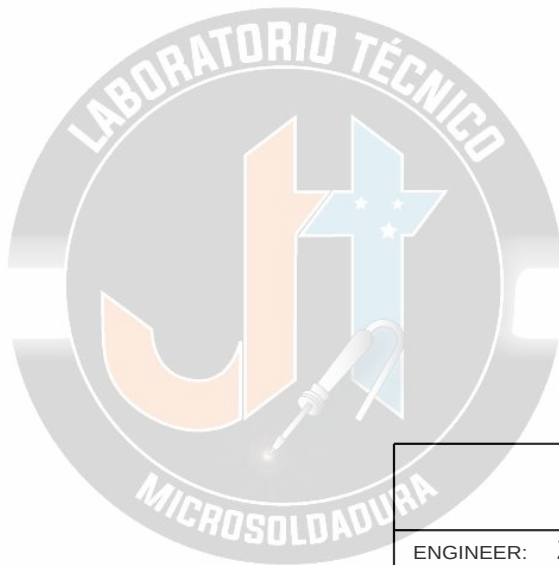
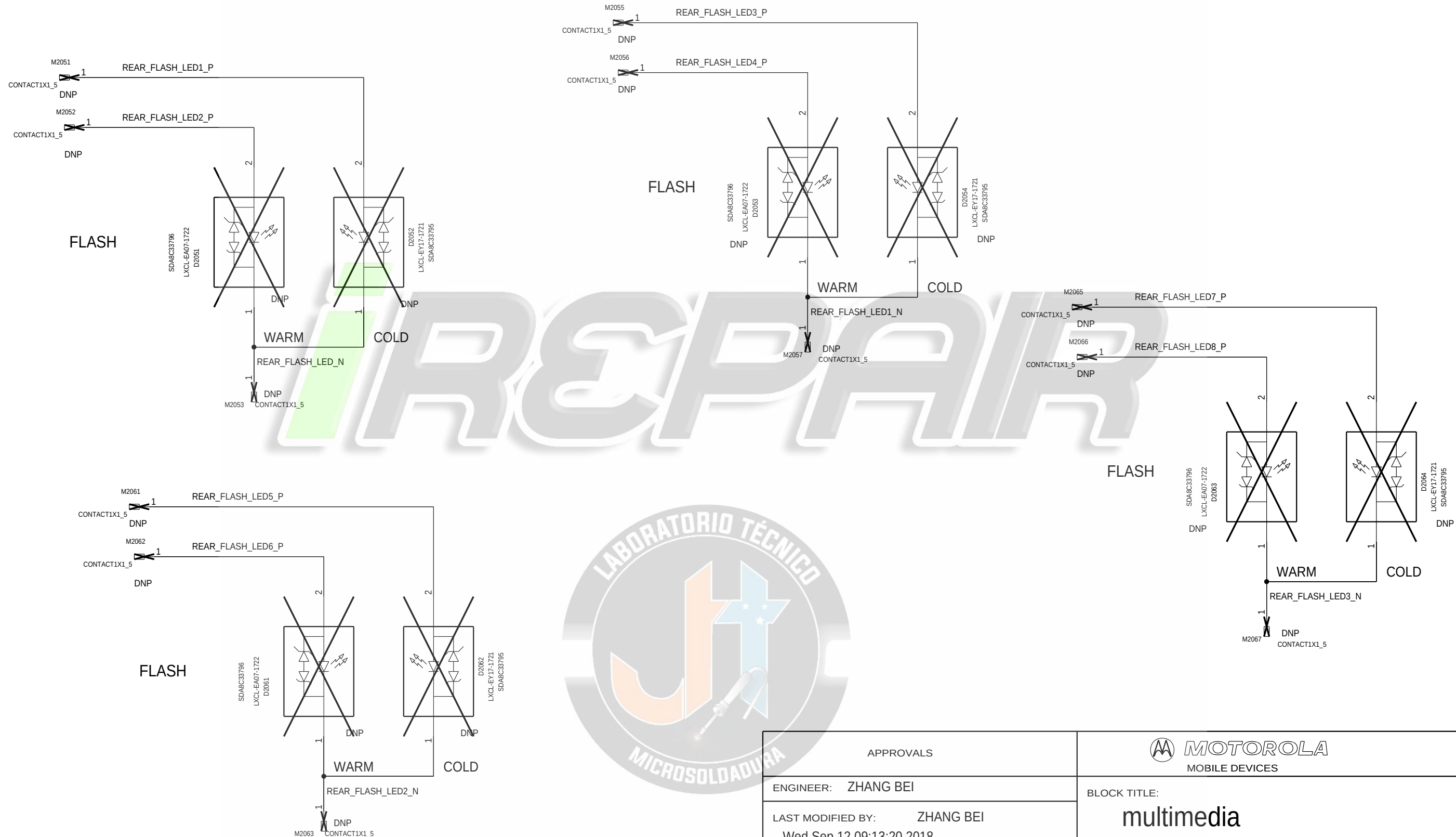
MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS	 MOTOROLA MOBILE DEVICES		
ENGINEER: ZHANG BEI	BLOCK TITLE: ui		
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:24 2018			
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CAD FILE NAME:		PAGE 29 OF 66	

REF:4200-4499



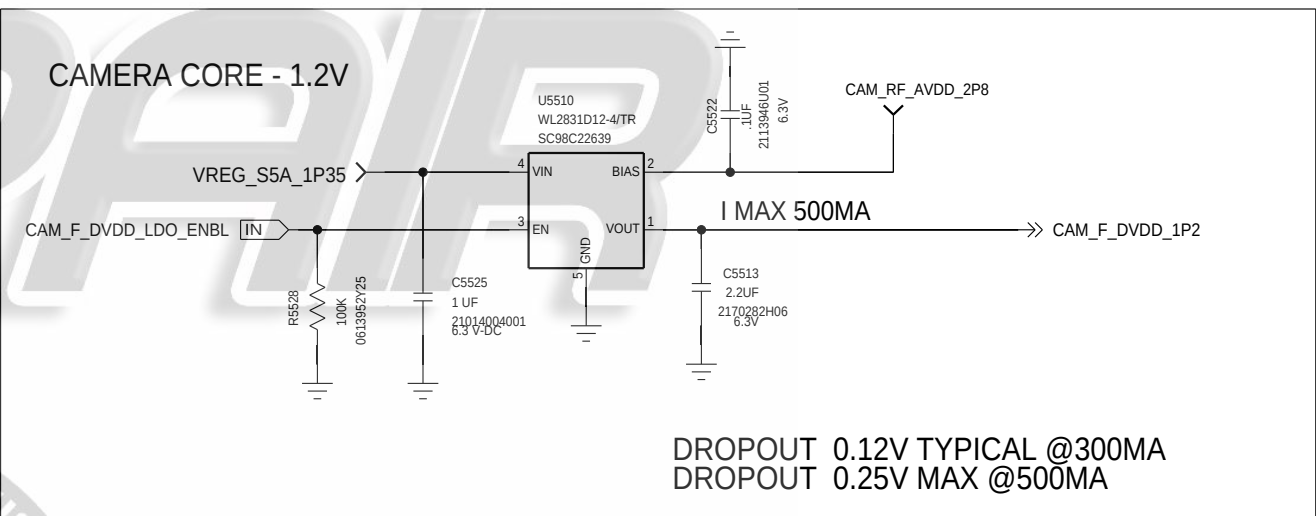
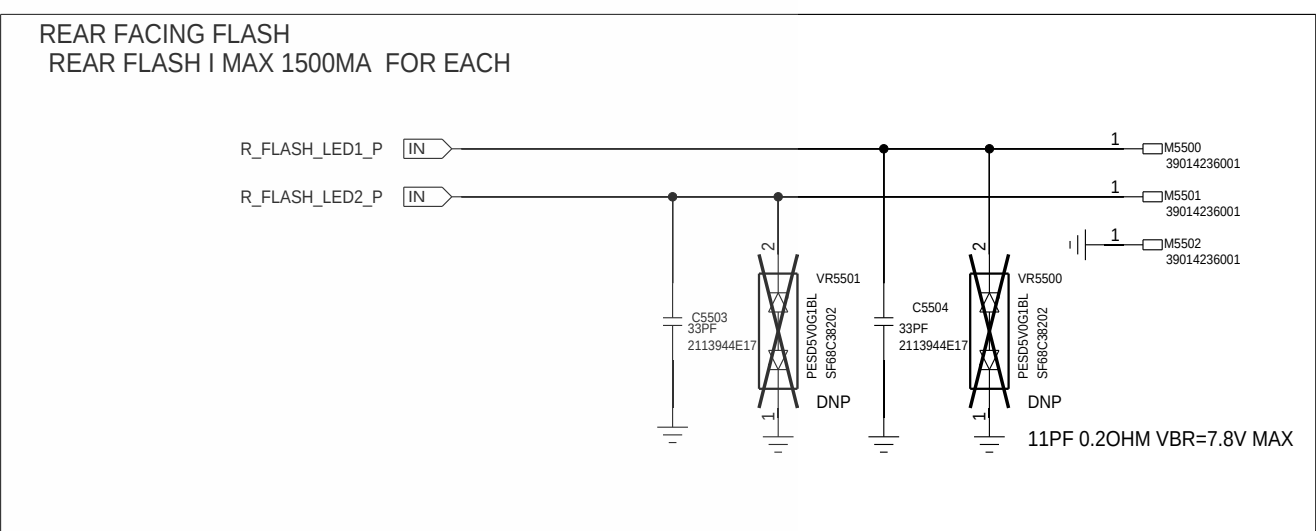
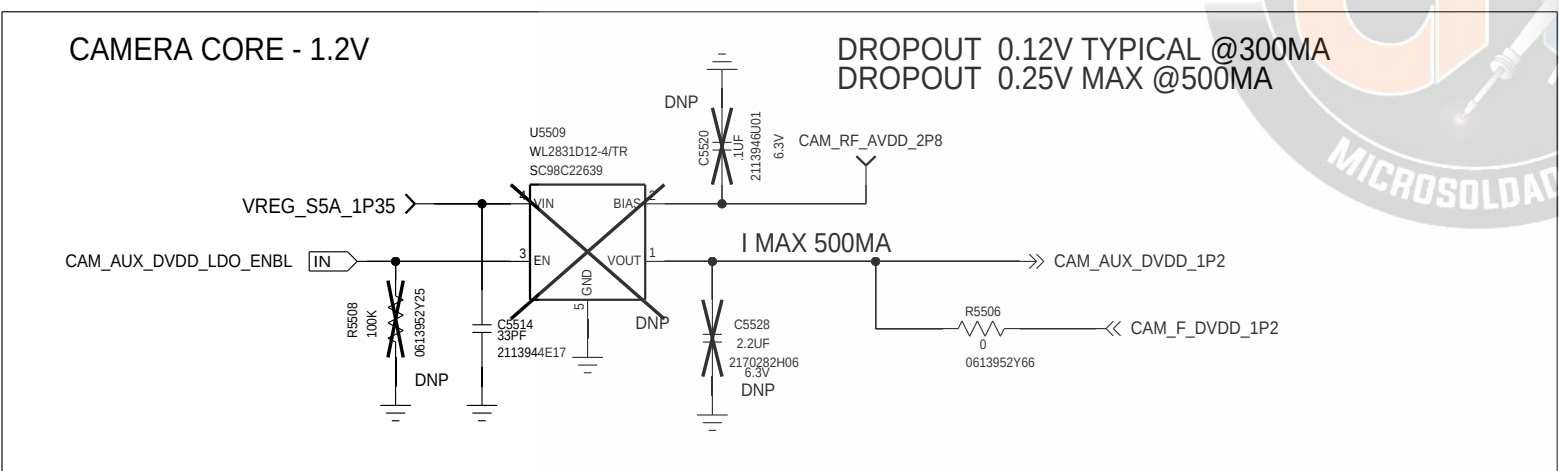
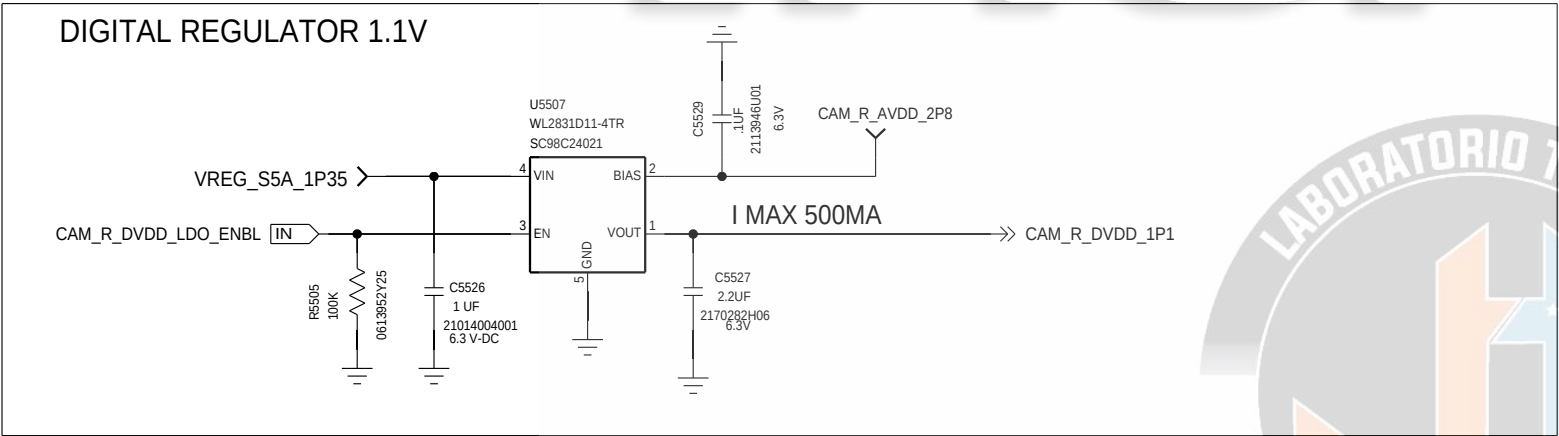
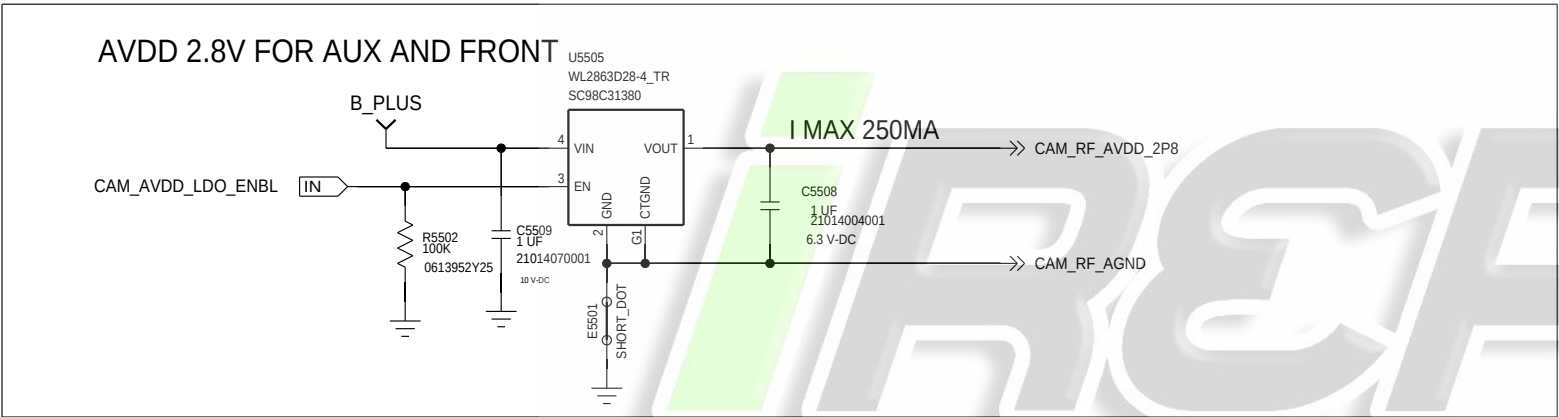
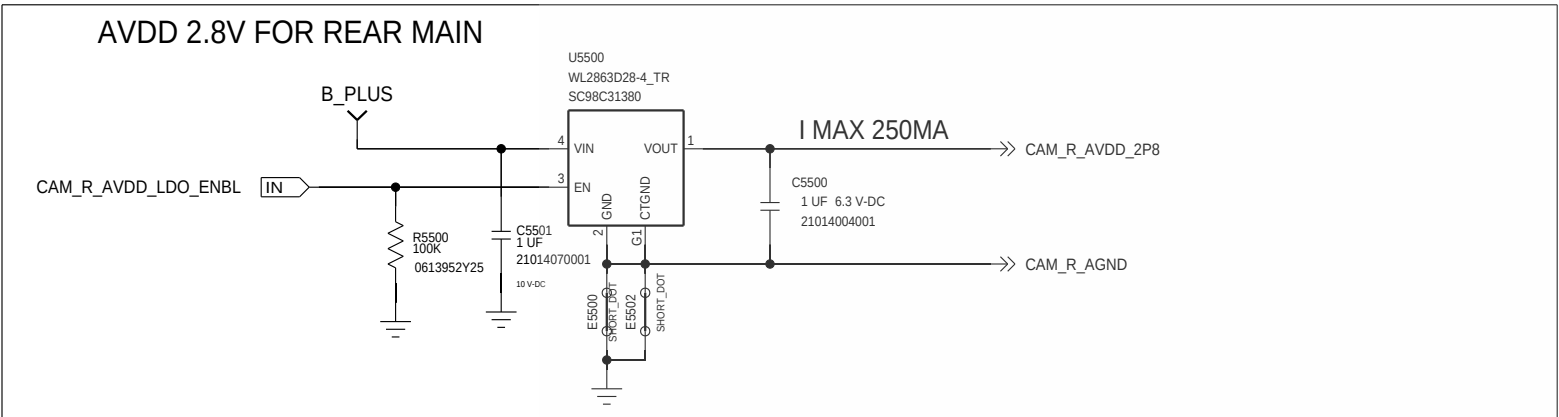
CAM: REAR FLASH



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:20 2018		multimedia	
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		REV 01	
		PAGE 31 OF 66	

CAM: POWER

REF:5500-5529

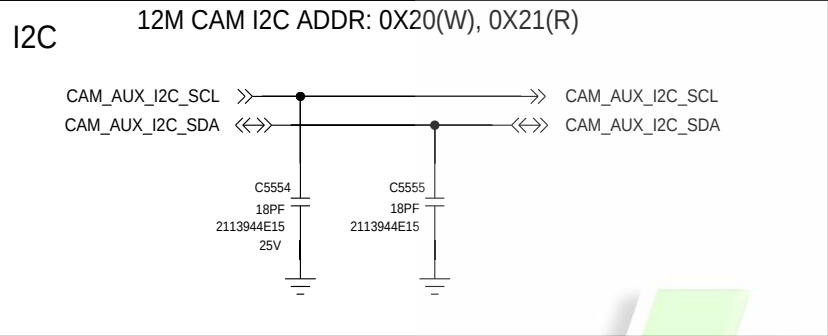


APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE: multimedia	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:21 2018		PROJECT NAME rio.h.cpm	
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CAM: FRONT CONNECTOR
REF 5550-5569

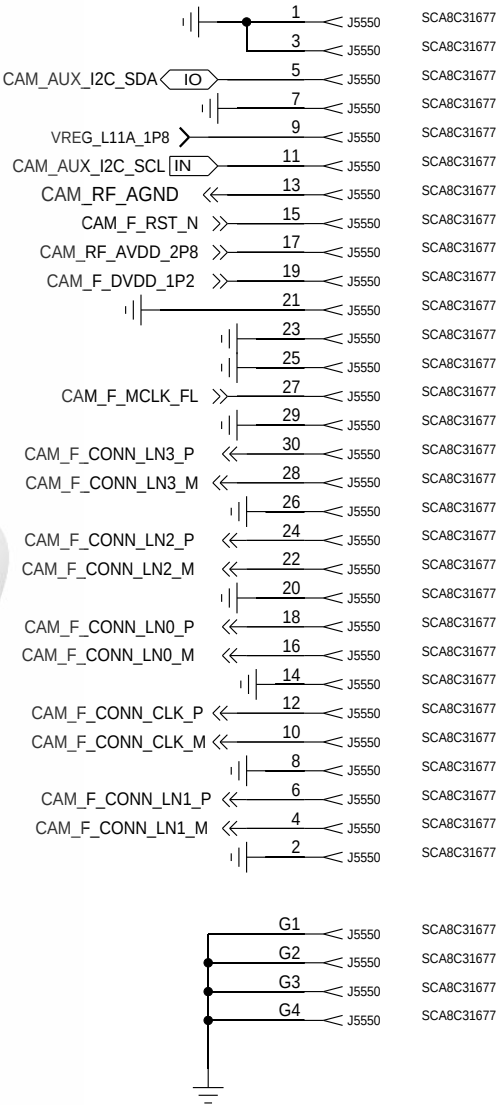
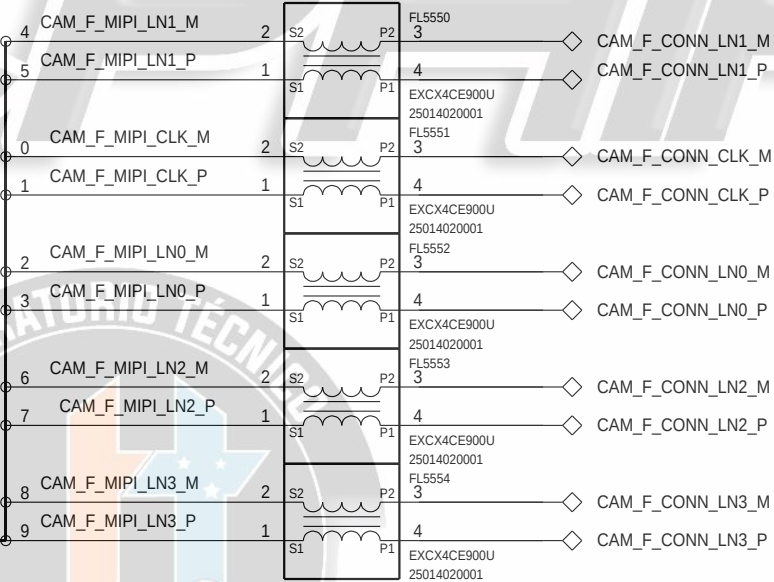
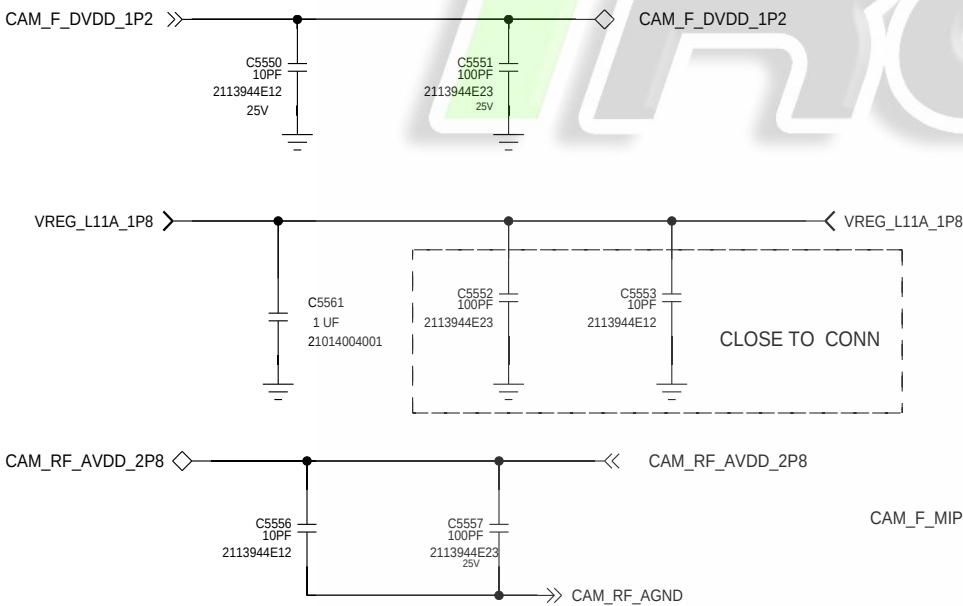
SENSOR: OV12A10
12M, 1.25UM, 1/2.8", 4096X3072

FF CAMERA CONNECTOR

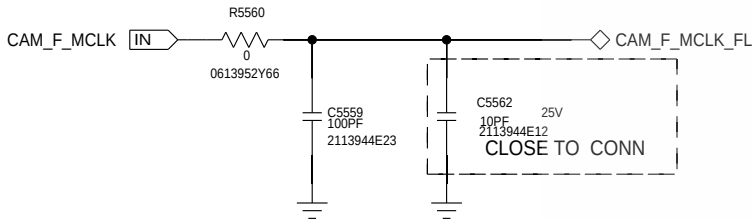


CAM I2C ADDRESS
FRONT OV12A10 CAM_AUX_I2C
SENSOR: 0X20,0X21
EEPROM: 0XA0,0XA1
AUX S5K5E9YX04 CAM_AUX_I2C
SENSOR: 0X20,0X21
REAR MAIN IMX519 CAM_I2C
SENSOR: 0X34,0X35
OIS: 0X48,0X49
EEPROM: 0XA0,0XA1

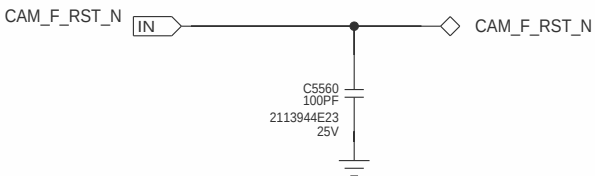
CAPACIORS BEADS ON THIS PAGE NEED BE PLACED CLOSE TO FRONT CAMERA CONNECTOR PIN.



EMI/ESD FILTER



XSHUTDOWN



APPROVALS

ENGINEER: ZHANG BEI

LAST MODIFIED BY: ZHANG BEI
Wed Sep 12 09:13:21 2018

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BLOCK TITLE:

multimedia

PROJECT NAME
rio.h.cpm

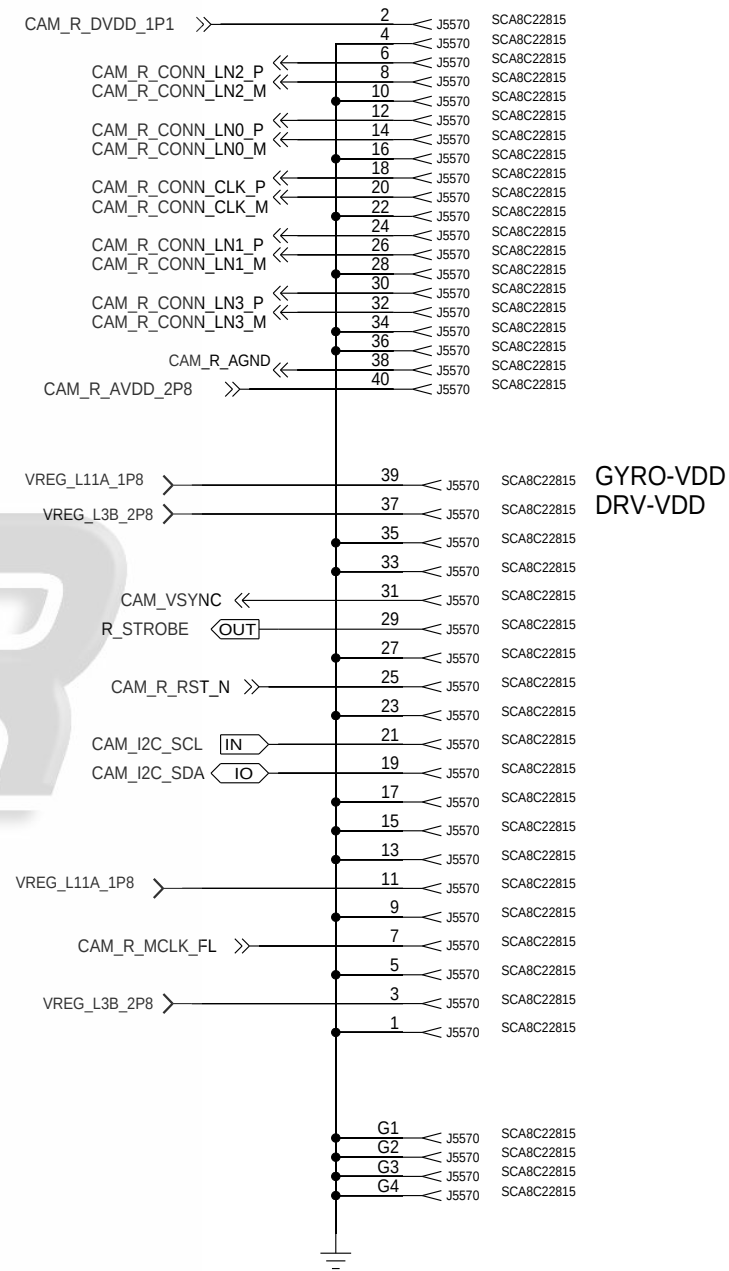
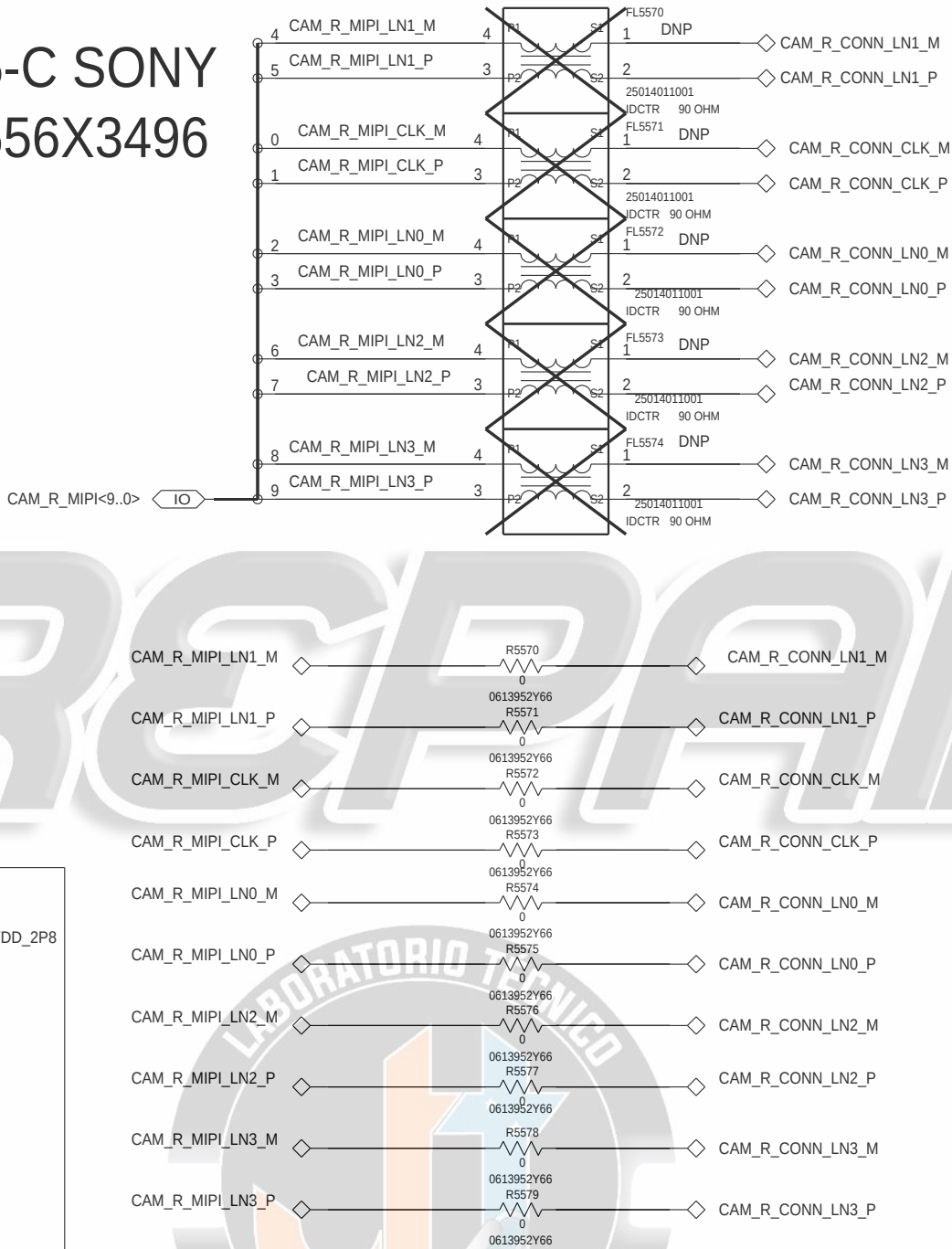
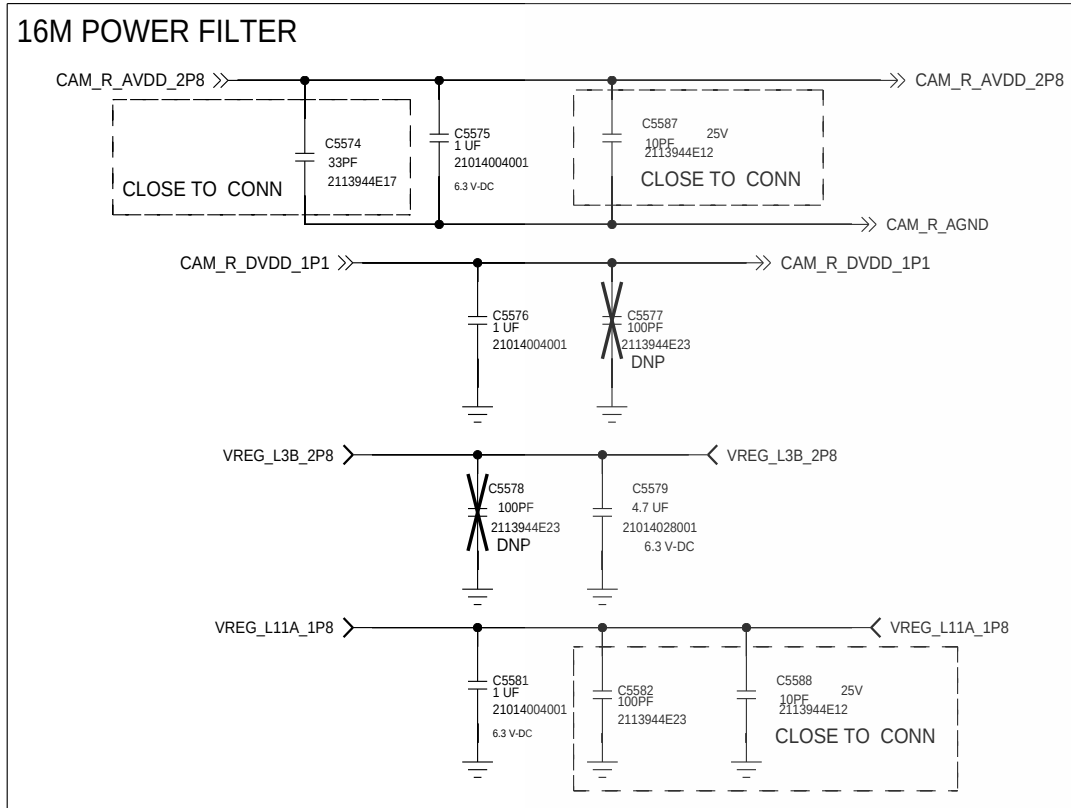
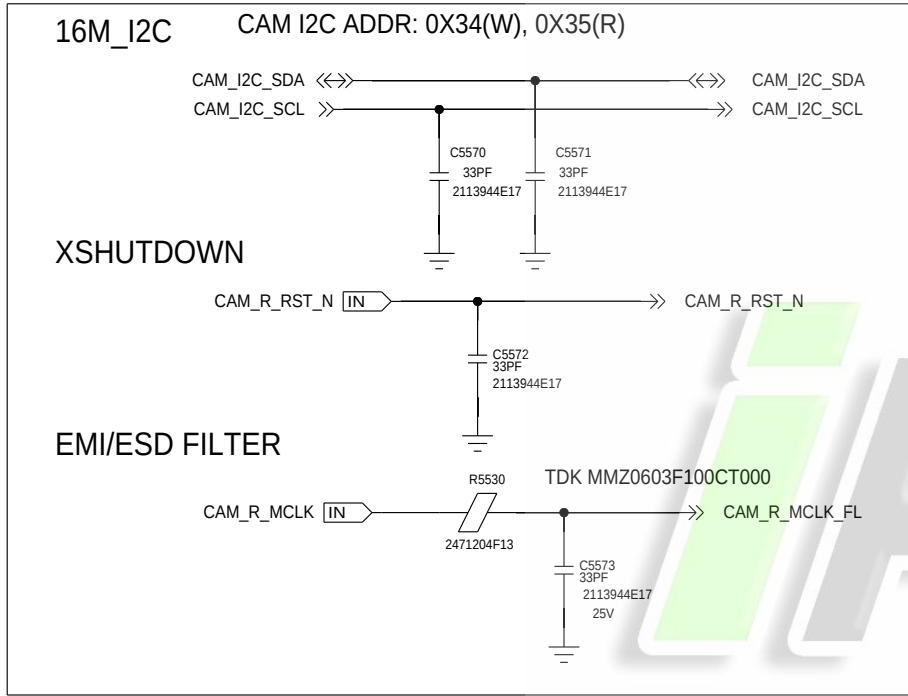
CAD FILE NAME:

REV
01

cam: rear main connector

REF:5570-5589

SENSOR: IMX519-AAQH5-C SONY
16M, 1.22UM, 1/2.534", 4656X3496



CAPACIORS BEADS ON THIS PAGE NEED BE PLACED CLOSE TO MAIN REAR CAMERA CONNECTOR PIN.

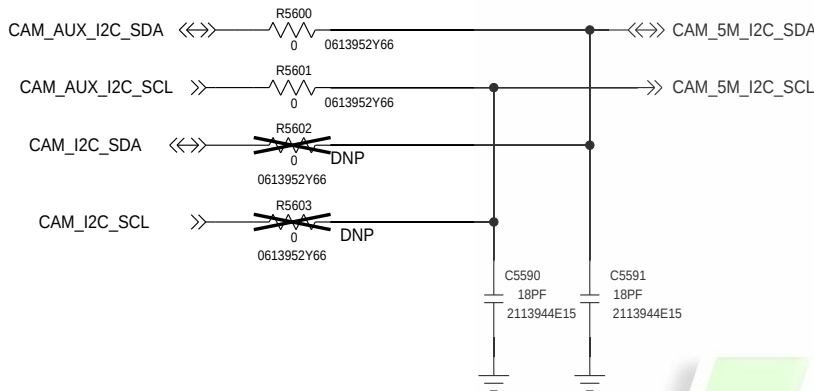
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:21 2018		multimedia	
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CAD FILE NAME:		PAGE 34 OF 66	

CAM: REAR AUX CONNECTOR

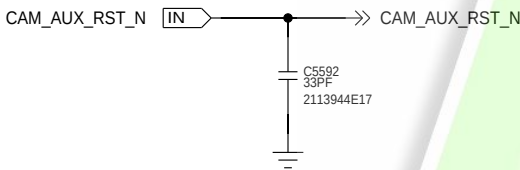
REF:5590-5610 5M CAM I2C ADDR: 0X20(W), 0X21(R)

SENSOR: S5K5E9YX04 SUMSANG
5M, 1.12UM, 1/5", 2592X1944

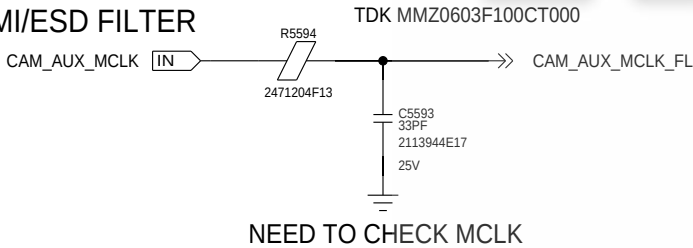
AUX_I2C



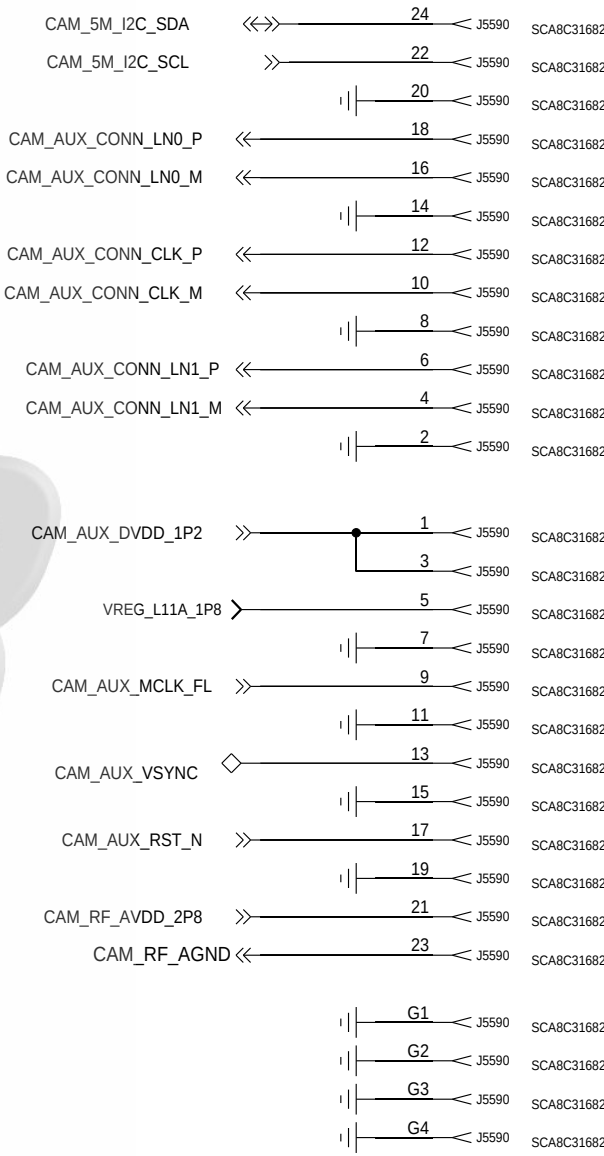
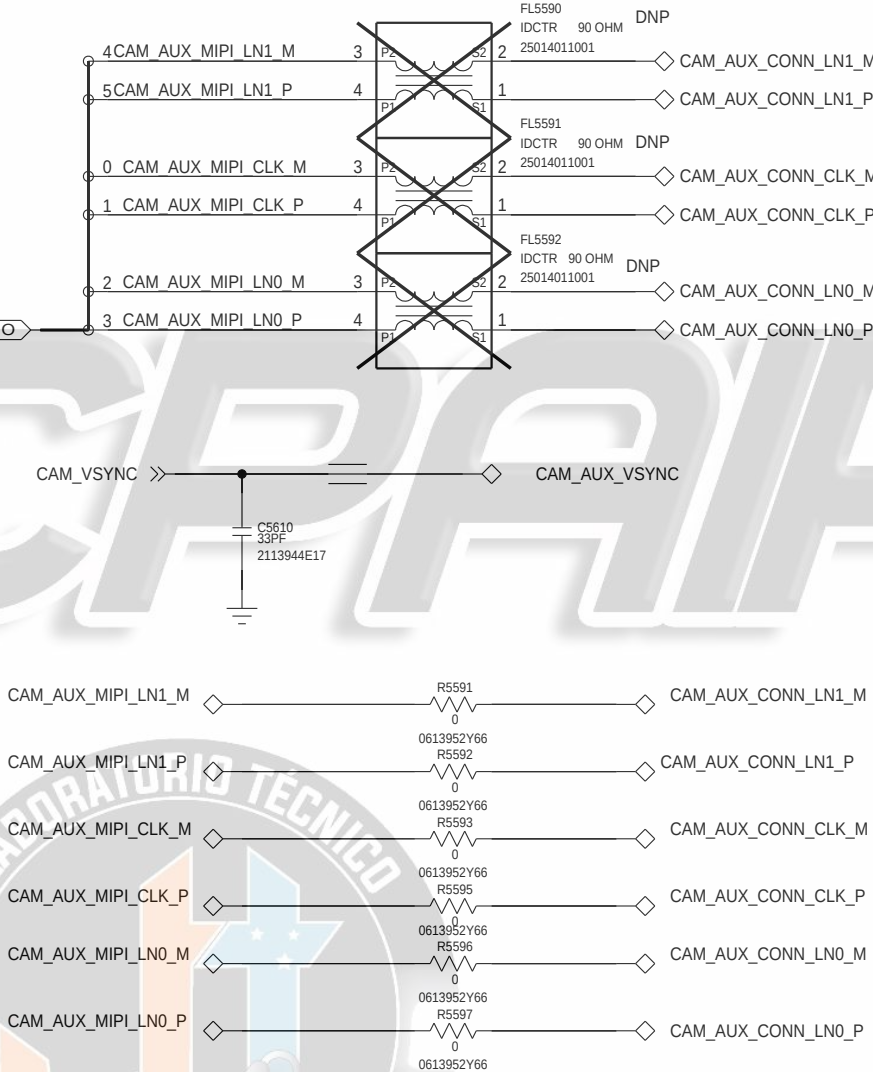
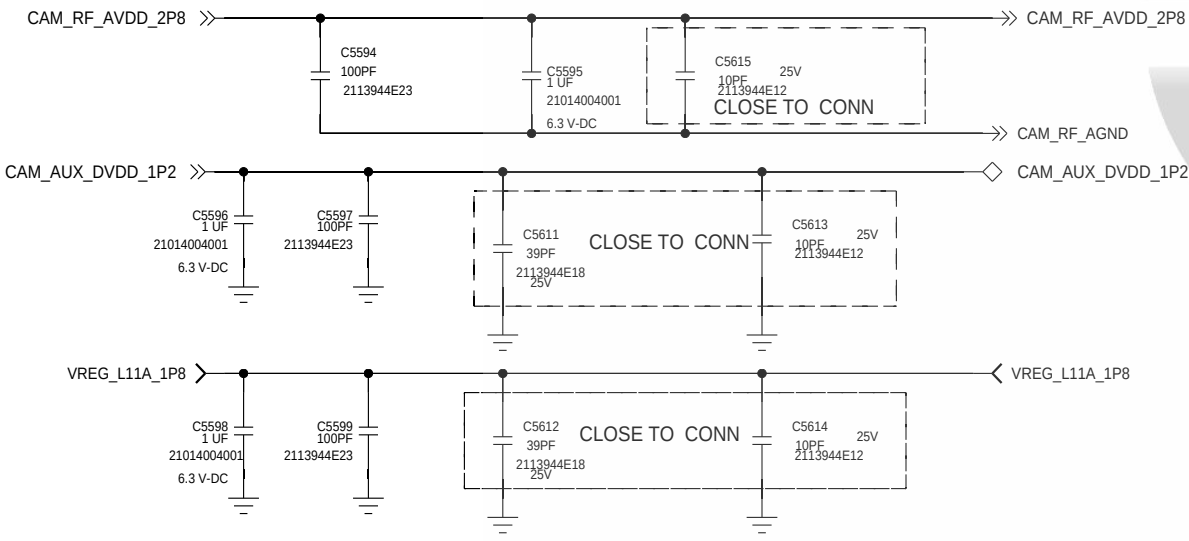
XSHUTDOWN



EMI/ESD FILTER



AUX POWER FILTER



CAPACIORS BEADS ON THIS PAGE NEED BE PLACED CLOSE TO AUX REAR CAMERA CONNECTOR PIN.

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE: multimedia	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:21 2018		PROJECT NAME rio.h.cpm	
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DISP: CONNECTOR

REF:5620--5650

ATTENTION TO POWER SUPPLY

LCD BACKUP ID

NO

1

J5620 SCA8C31679

LCD ID 1

3

J5620 SCA8C31679

5

J5620 SCA8C31679

DATA 3N

7

J5620 SCA8C31679

DATA 3P

9

J5620 SCA8C31679

11

J5620 SCA8C31679

DATA 0N

13

J5620 SCA8C31679

DATA 0P

15

J5620 SCA8C31679

17

J5620 SCA8C31679

CLK N

19

J5620 SCA8C31679

CLK P

21

J5620 SCA8C31679

23

J5620 SCA8C31679

DATA 1N

25

J5620 SCA8C31679

DATA 1P

27

J5620 SCA8C31679

29

J5620 SCA8C31679

DATA 2N

31

J5620 SCA8C31679

DATA 2P

33

J5620 SCA8C31679

35

J5620 SCA8C31679

LED PWM

37

J5620 SCA8C31679

LCD TE 1

39

J5620 SCA8C31679

LCD RST N 1

40

J5620 SCA8C31679

IOVCC

38

J5620 SCA8C31679

G4

J5620 SCA8C31679

G3

J5620 SCA8C31679

G2

J5620 SCA8C31679

G1

J5620 SCA8C31679

LCD_VSP

36

J5620 SCA8C31679

LCD_VSN

32

J5620 SCA8C31679

30

J5620 SCA8C31679

BLM03HG102SN1D

28

J5620 SCA8C31679

26

J5620 SCA8C31679

24

J5620 SCA8C31679

22

J5620 SCA8C31679

20

J5620 SCA8C31679

TS_INT_T

18

J5620 SCA8C31679

TS_I2C_SCL

16

J5620 SCA8C31679

TS_I2C_SDA

14

J5620 SCA8C31679

TS_RST_N_CONN

12

J5620 SCA8C31679

TP_FW_SPI_SS

10

J5620 SCA8C31679

TP_FW_SPI_MISO

8

J5620 SCA8C31679

TP_FW_SPI_MOSI

6

J5620 SCA8C31679

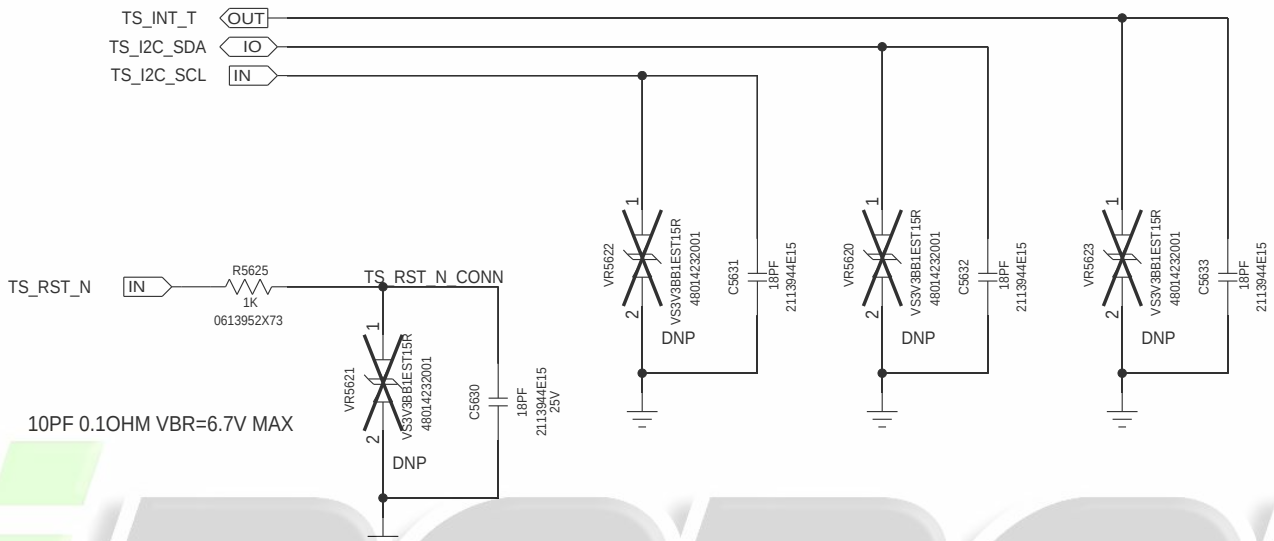
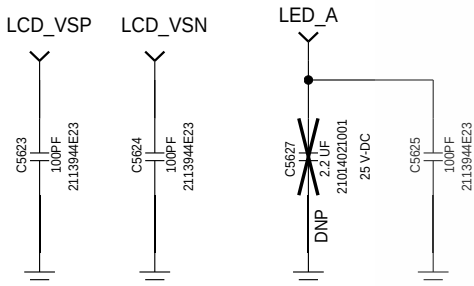
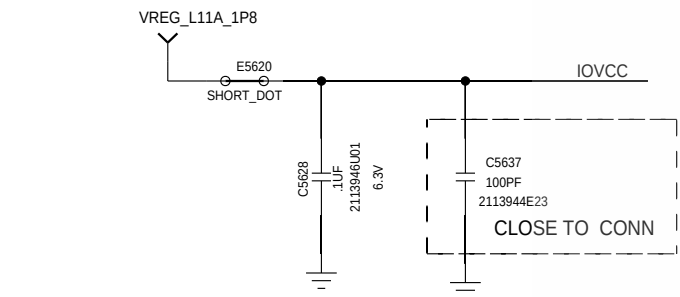
TP_FW_SPI_CLK

4

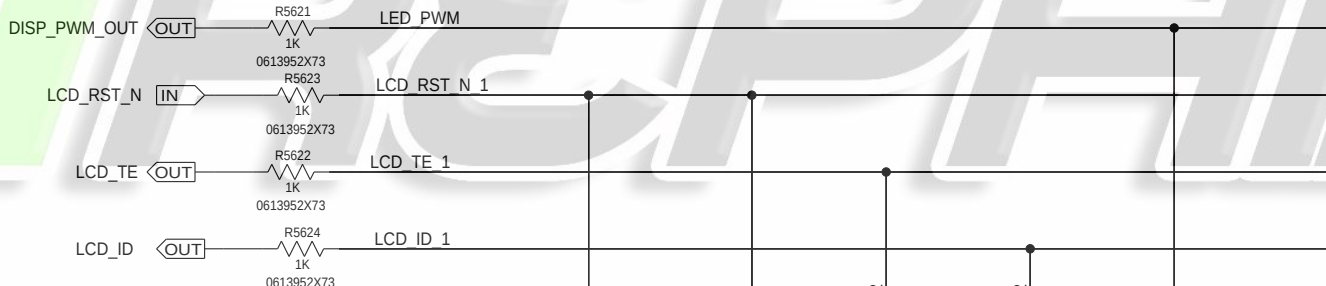
J5620 SCA8C31679

2

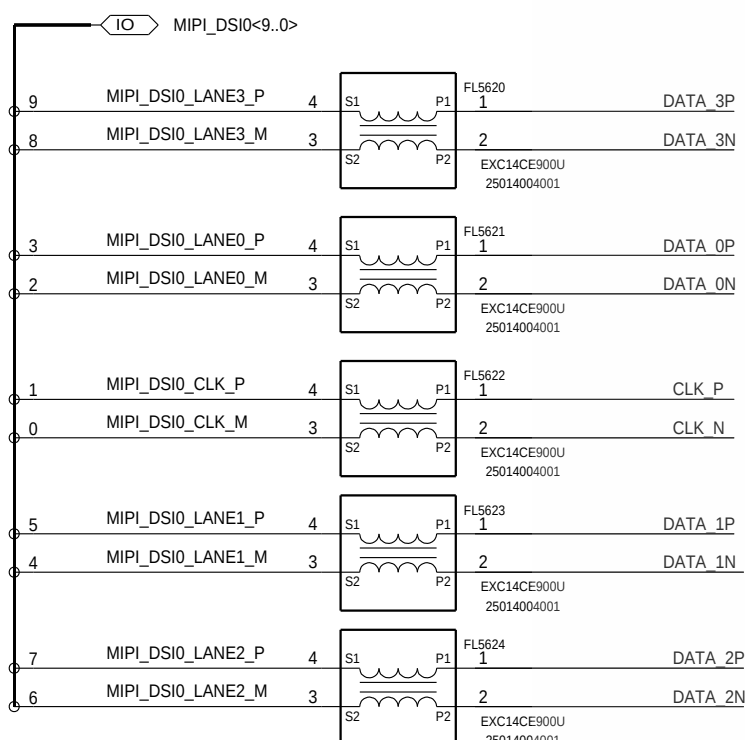
J5620 SCA8C31679



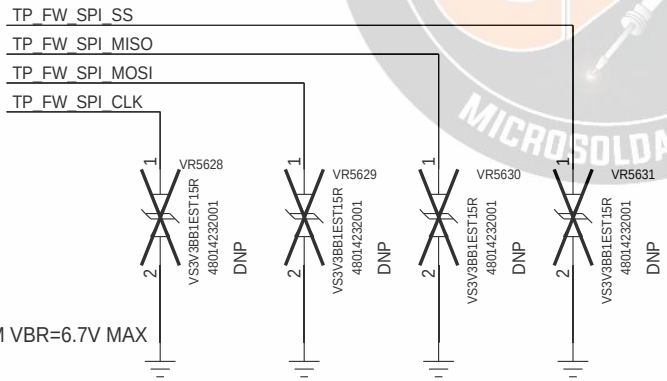
10PF 0.10HM VBR=6.7V MAX



10PF 0.10HM VBR=6.7V MAX



10PF 0.10HM VBR=6.7V MAX



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:20 2018		multimedia	
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		PAGE 36 OF 66	

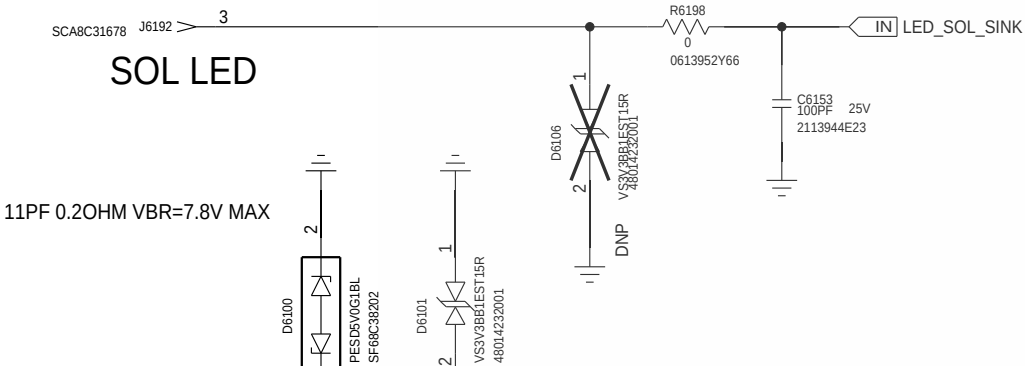
GND SHIELDED FOR MIPI DSI SIGNALS AS A GROUP
ADD GND STITCHING VIAS FOR EACH SIGNAL GROUP (EBI,CSI,DSI) WHEN POSSIBLE TO MINIMIZE RETURN CURRENT LOOP

XMCS: SENSORS

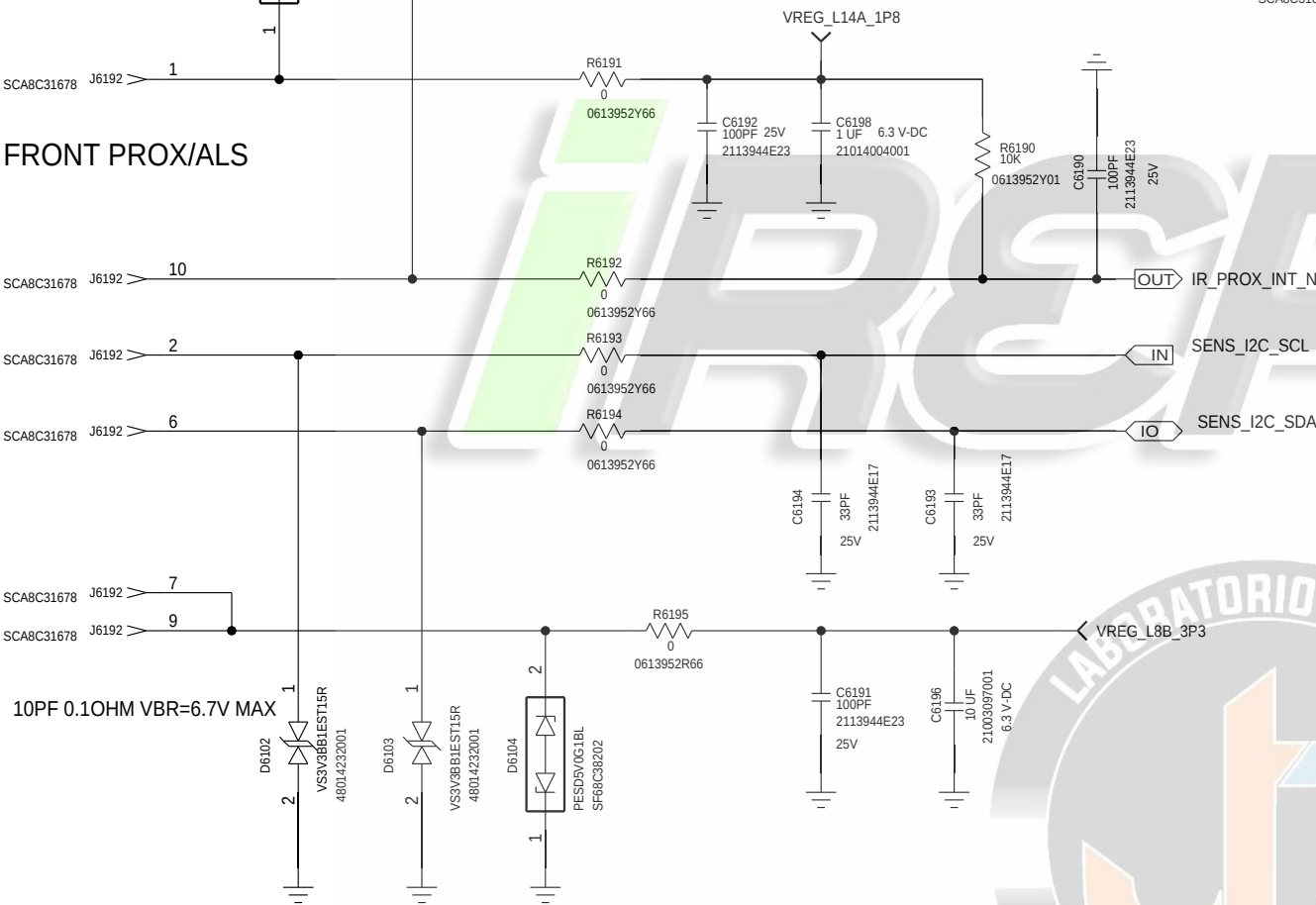
REF: 6100-6199

AWAY FORM NOISY SIGNALS AND HIGH POWER SOURCES

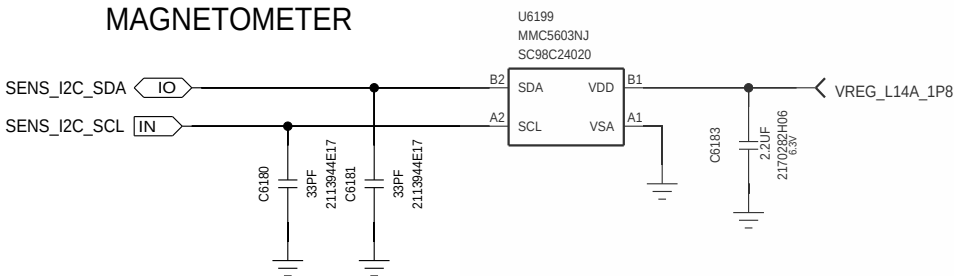
SOL LED



FRONT PROX/ALS



MAGNETOMETER

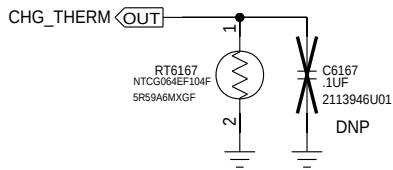


I2C ADDRESS
MMC5603NJ 0X60,0X61
CS47L35 0X34,0X35
TMD3702V 0X49,0X4A

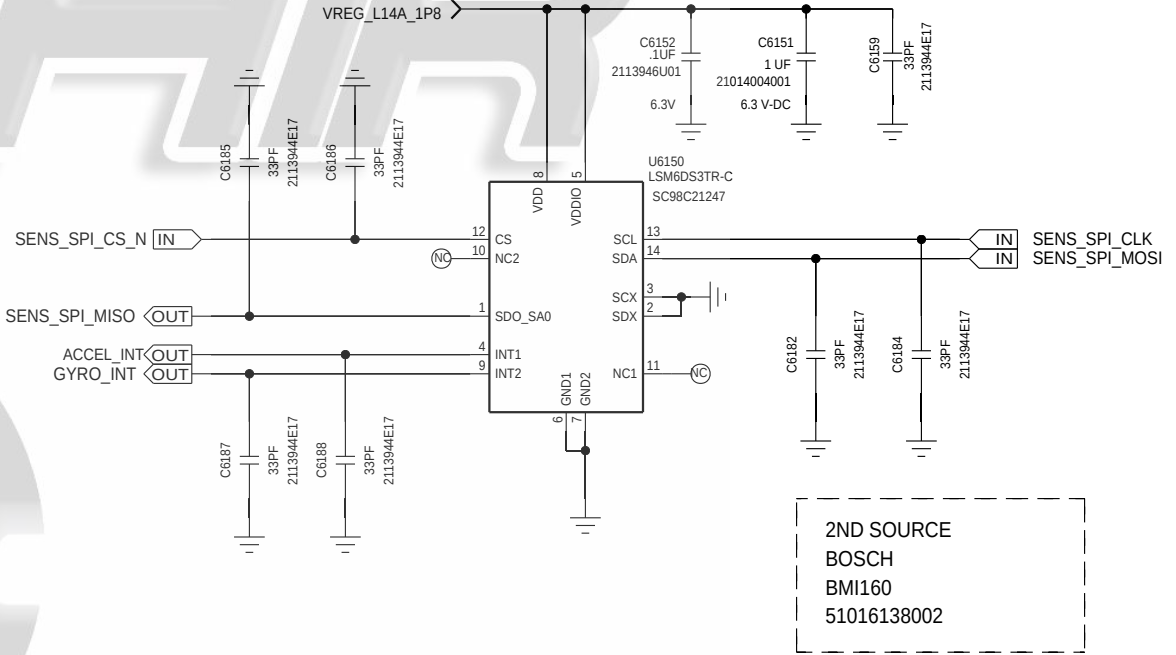
BOARD TEMP SENSOR



CHRG TEMP SENSOR



ACCELEROMETER+GYRO



2ND SOURCE
BOSCH
BMI160
51016138002

JOHN GORSICA/NATE CONNELL
APPROVALS

ENGINEER: ZHANG BEI
LAST MODIFIED BY: ZHANG BEI
Wed Sep 12 09:13:22 2018

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

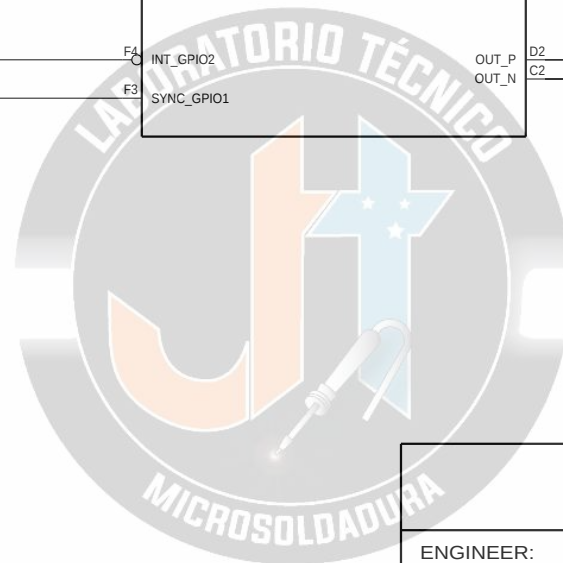
xmcs

PROJECT NAME
rio.h.cpm

CAD FILE NAME:

REV
01

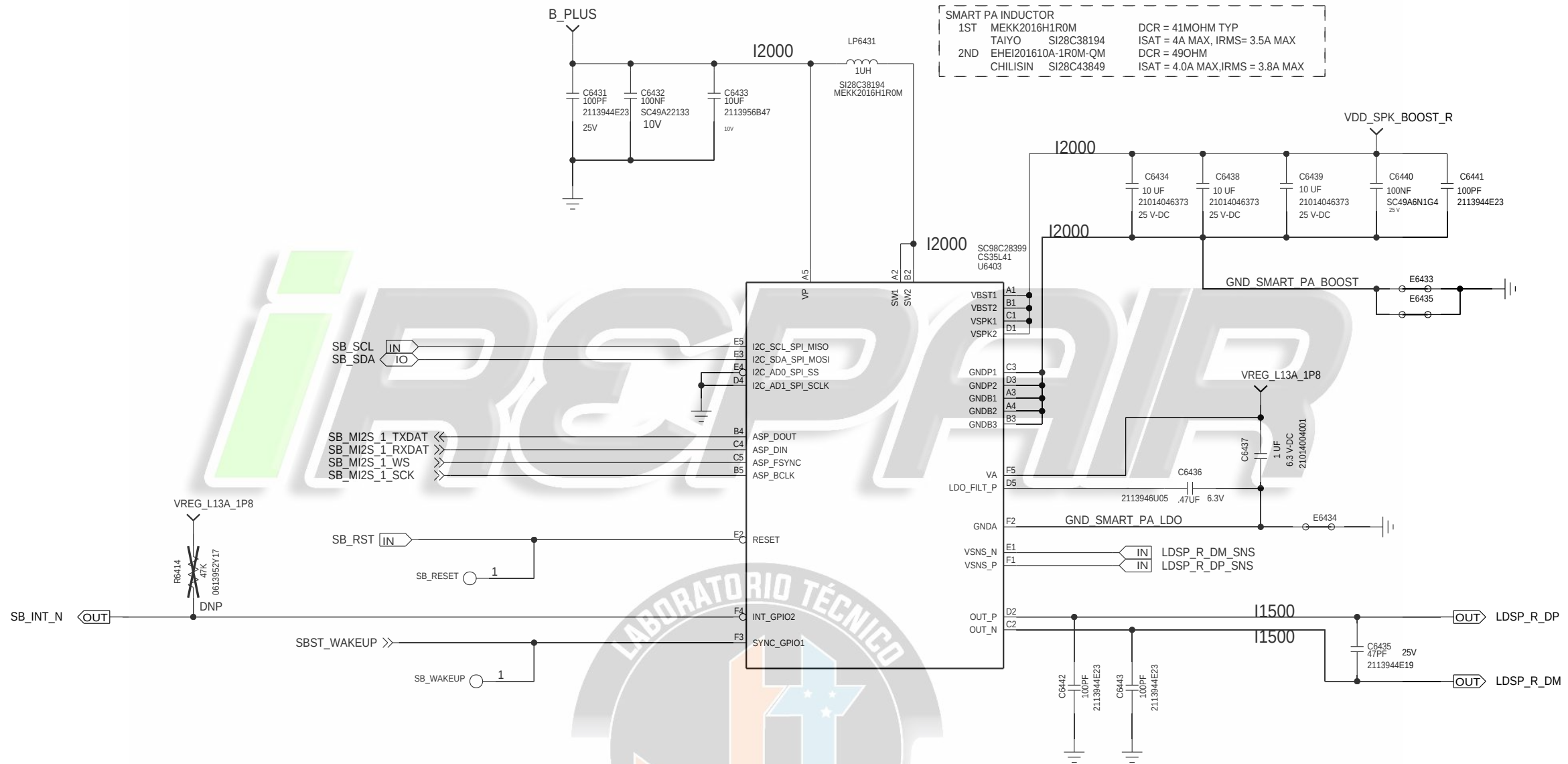
*** BOOST IC I2C ADD. 0X??

PAGE 38 OF 66

XMCS: SMART BOOST

REF 6400-6499

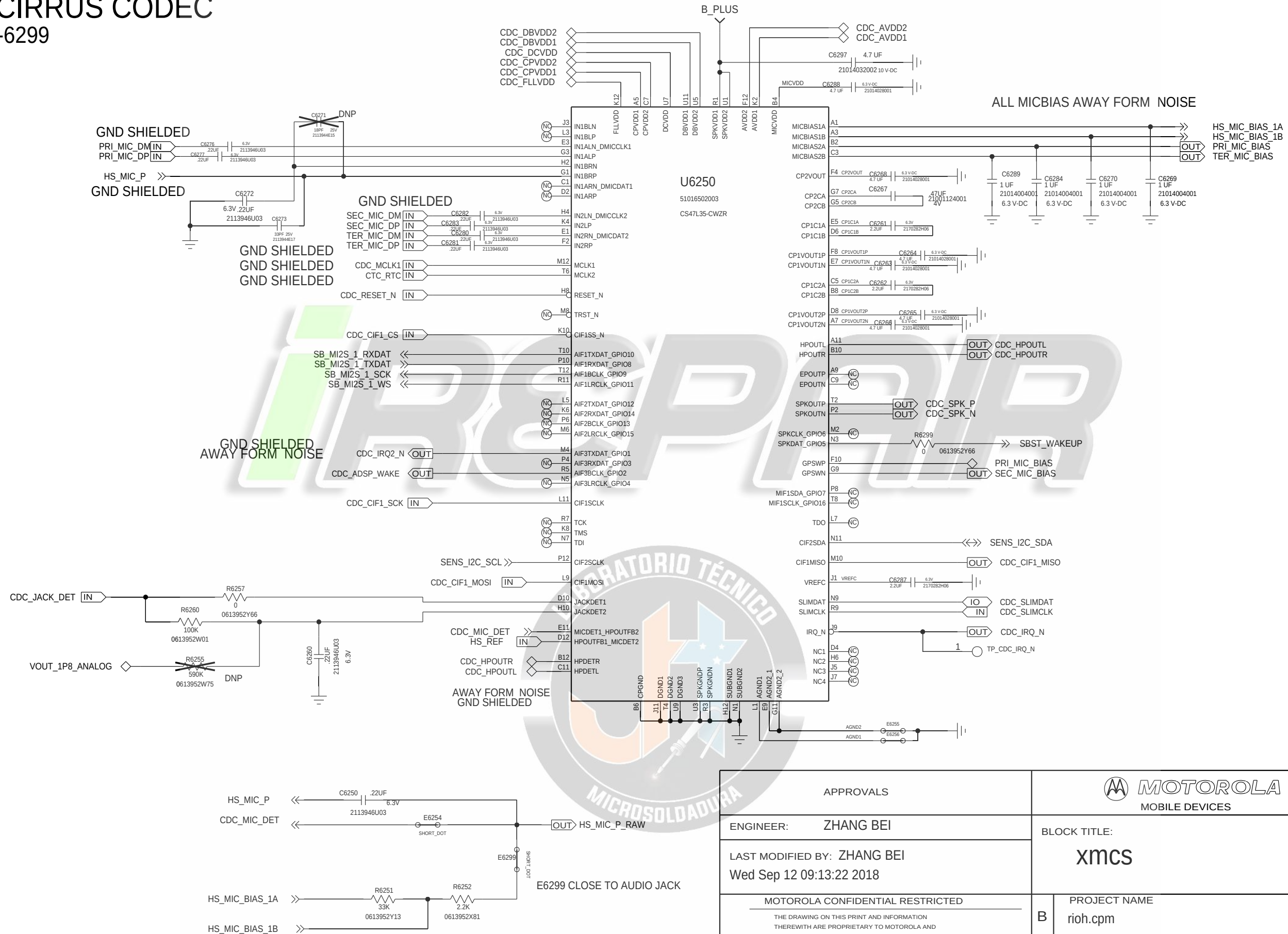
*** BOOST IC I2C ADD. 0X??



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:22 2018		xmcs	
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		REV 01	
		CAD FILE NAME:	
		PAGE 39 OF 66	

XMCS: CIRRUS CODEC

REF 6250-6299



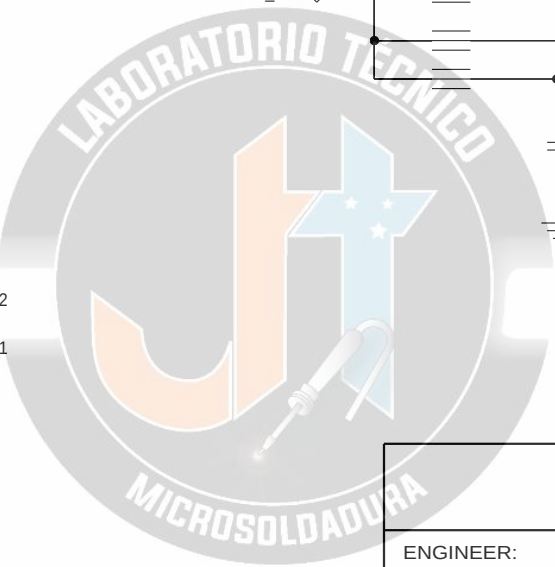
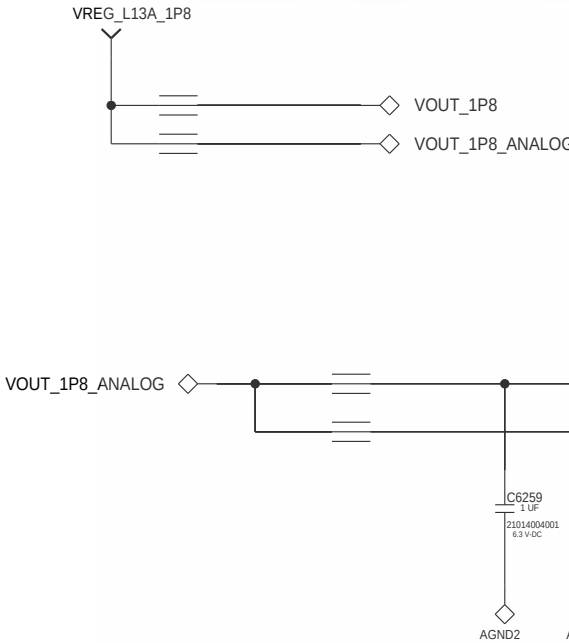
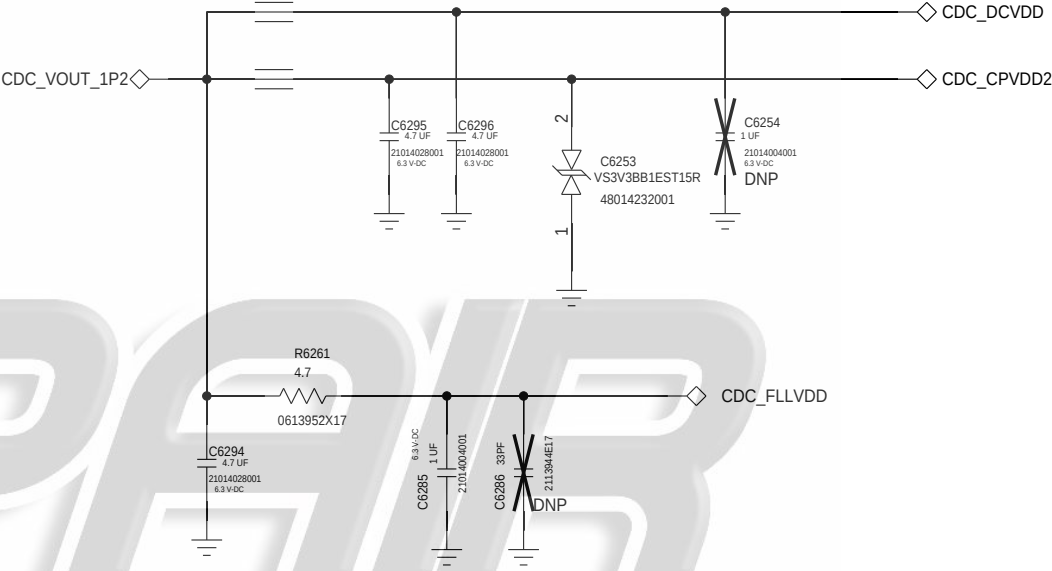
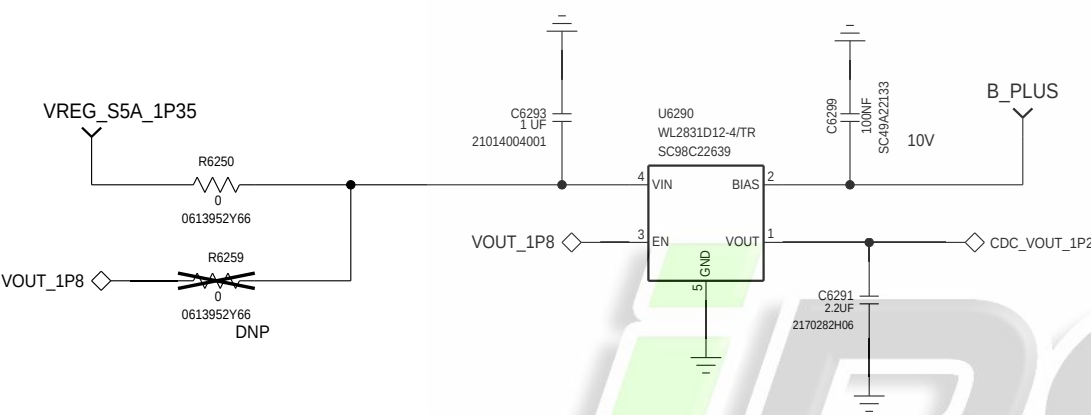
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI	BLOCK TITLE: xmcs		
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:22 2018			
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		CAD FILE NAME:	PAGE 40 OF 66

XMCS: CODEC POWER

REF 6250-6299

1.2V POWER SUPPLY

DROPOUT 0.12V TYPICAL @300MA
DROPOUT 0.25V MAX @500MA



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE:	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:22 2018		xmcs	
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		REV 01	
		PAGE 41 OF 66	

REF: 5700-5789



REF:5750-5759

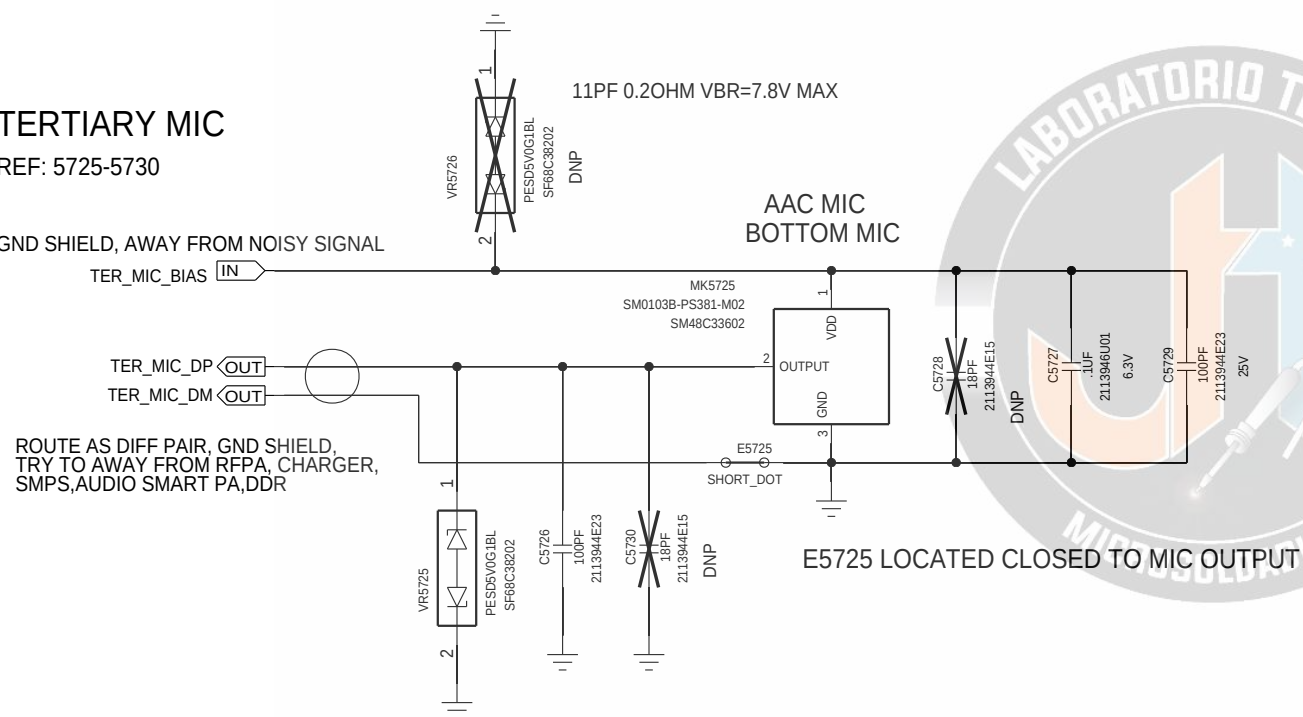
GND SHIELD, AWAY FROM NOISY SIGNAL



REF: 5725-5730

GND SHIELD, AWAY FROM NOISY SIGNAL

TER_MIC_BIAS



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI	BLOCK TITLE: audio_fe		
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:20 2018			
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		CAD FILE NAME:	PAGE 42 OF 66

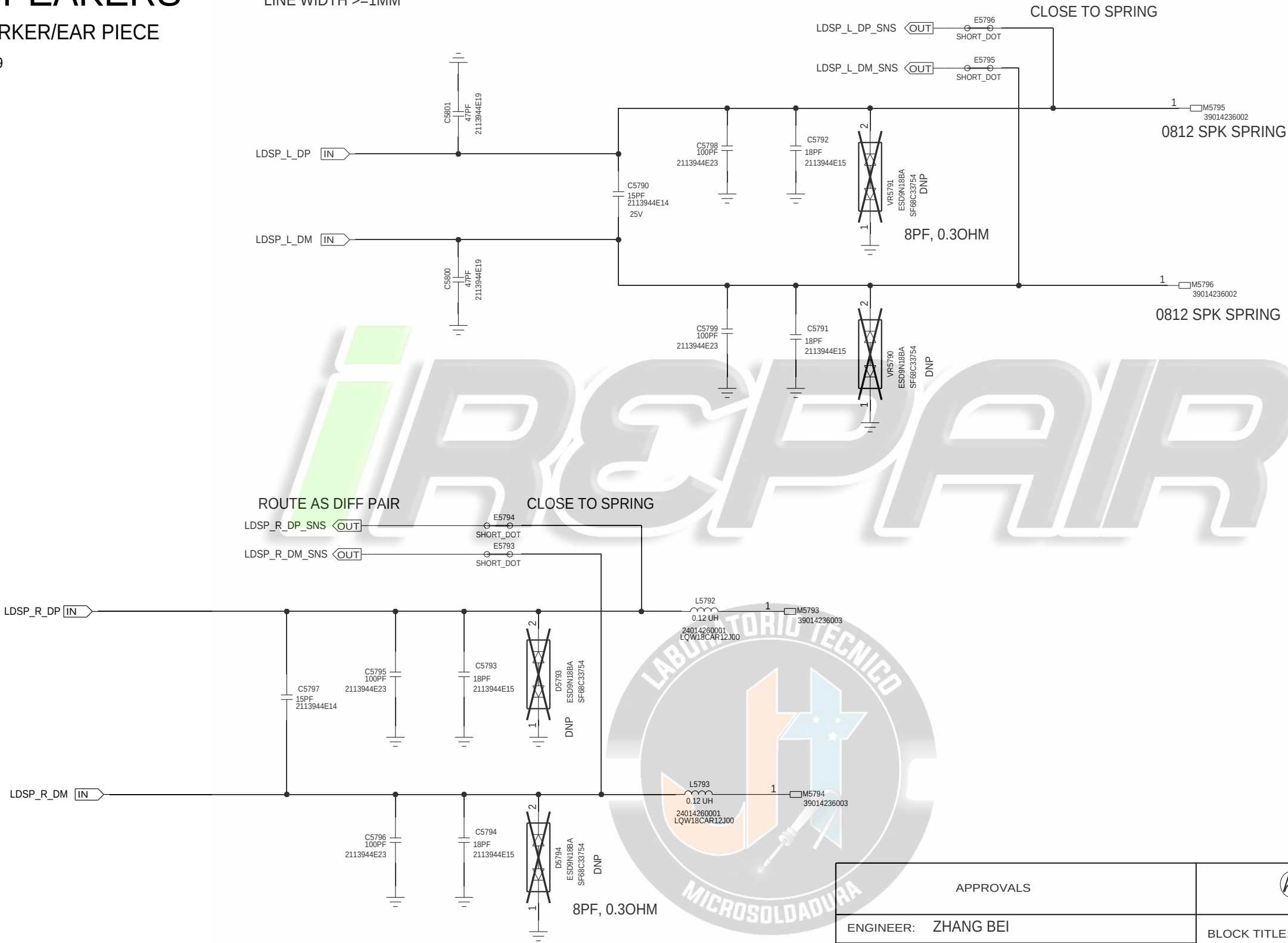
AFE: SPEAKERS

LOUDSPEARKER/EAR PIECE

REF:5790-5799

ROUTE AS DIFF PAIR
LINE WIDTH >=1MM

ADD STITCHING VIAS ALONGSIDE IF POSSIBLE

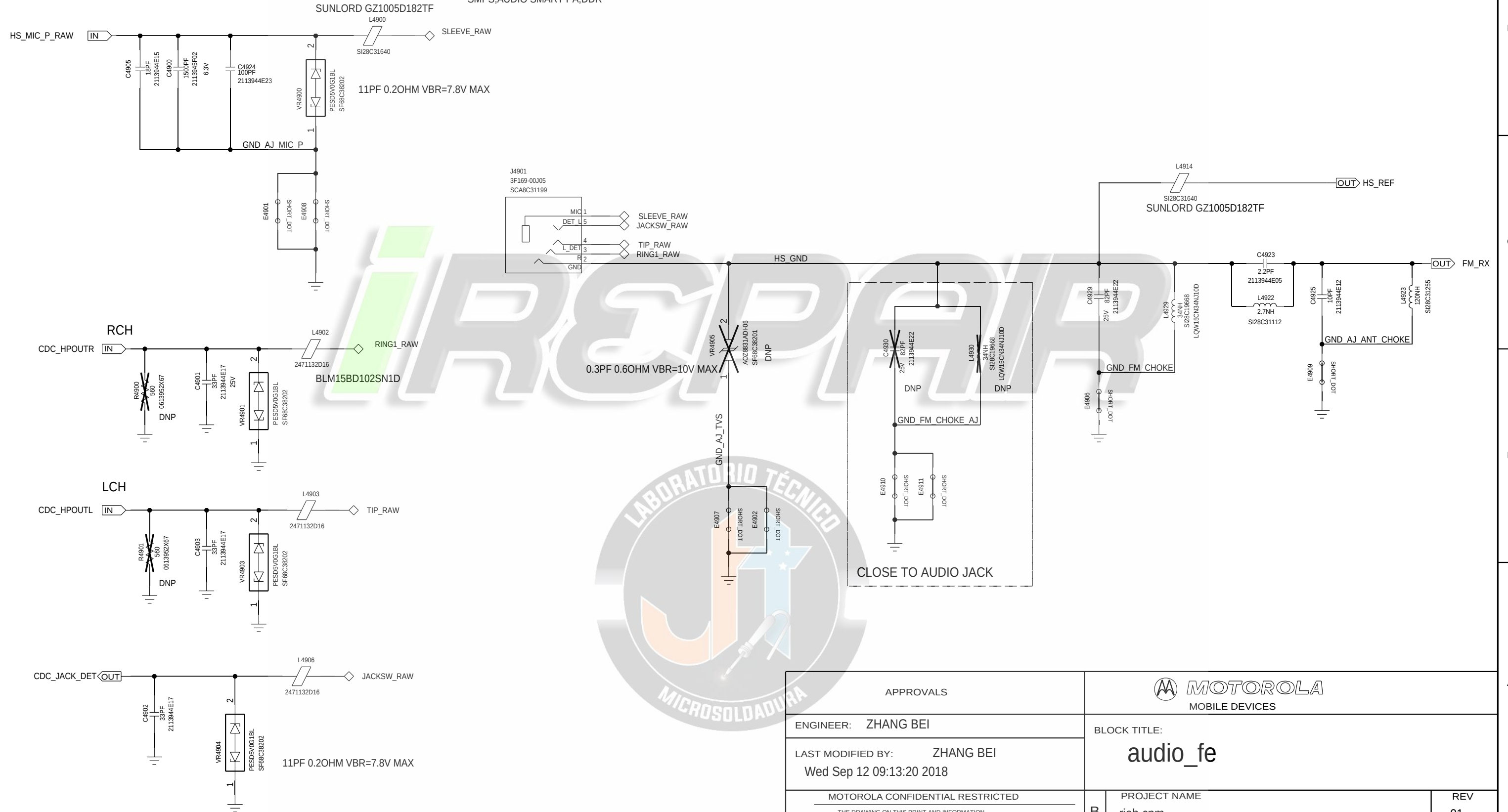


APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: ZHANG BEI		BLOCK TITLE: audio_fe	
LAST MODIFIED BY: ZHANG BEI Wed Sep 12 09:13:20 2018		PROJECT NAME B rioh.cpm	
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		REV 01	

AFE: 3.5MM HEADSET

REF:4900-4949

GND SHIELD,
TRY TO AWAY FROM RFPA, CHARGER,
SMPS,AUDIO SMART PA,DDR



APPROVALS

ENGINEER: ZHANG BE

LAST MODIFIED BY: ZHANG BE
Wed Sep 12 09:13:20 2018

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BLOCK TITLE

audio fe

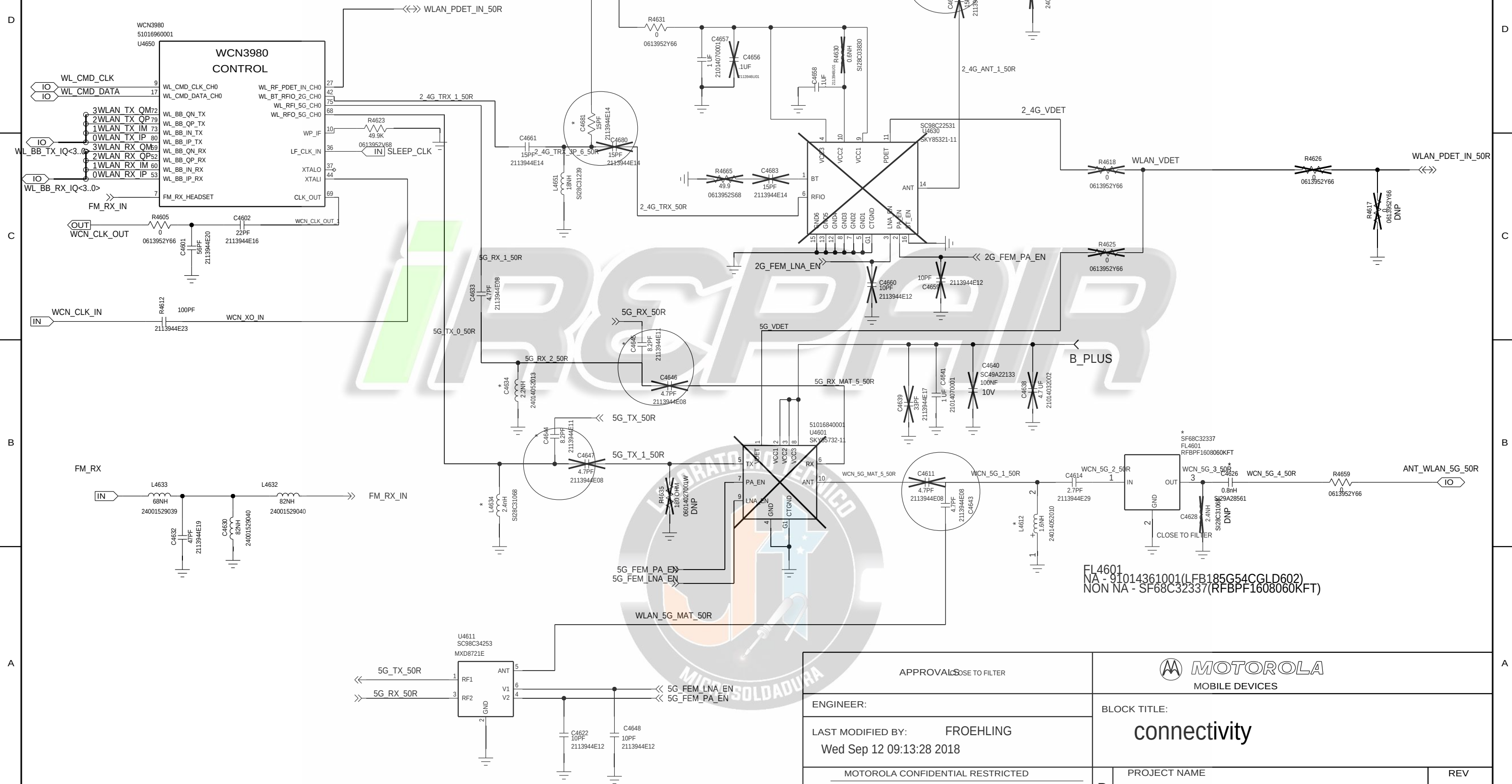
B	PROJECT NAME rio.h.cpm
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CAD FILE NAME

REV
01

PAGE 44 OF 66

WCN3980 RF/CTRL
REF 4600-4699

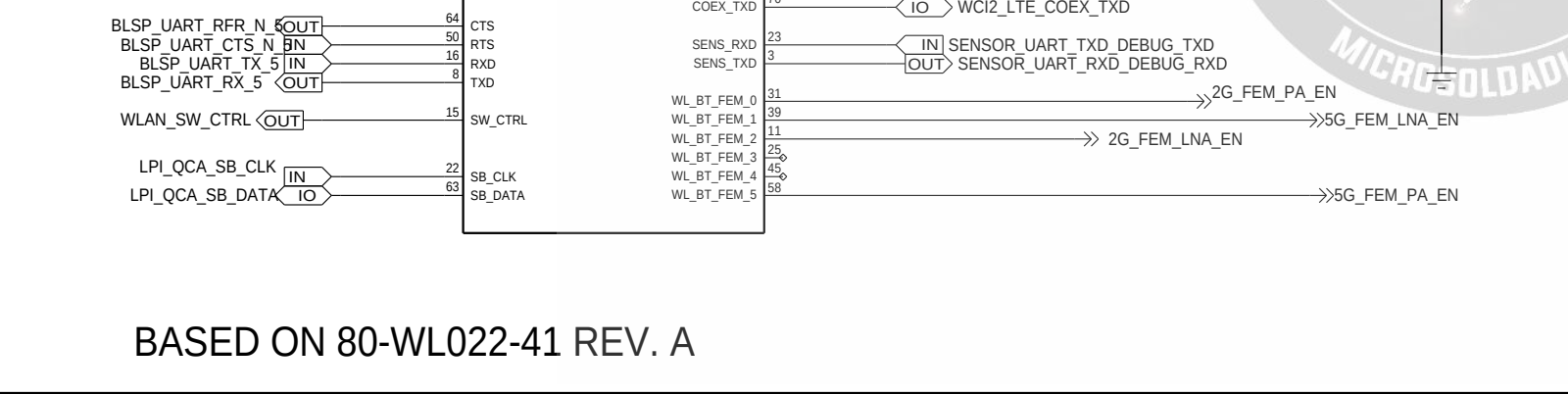
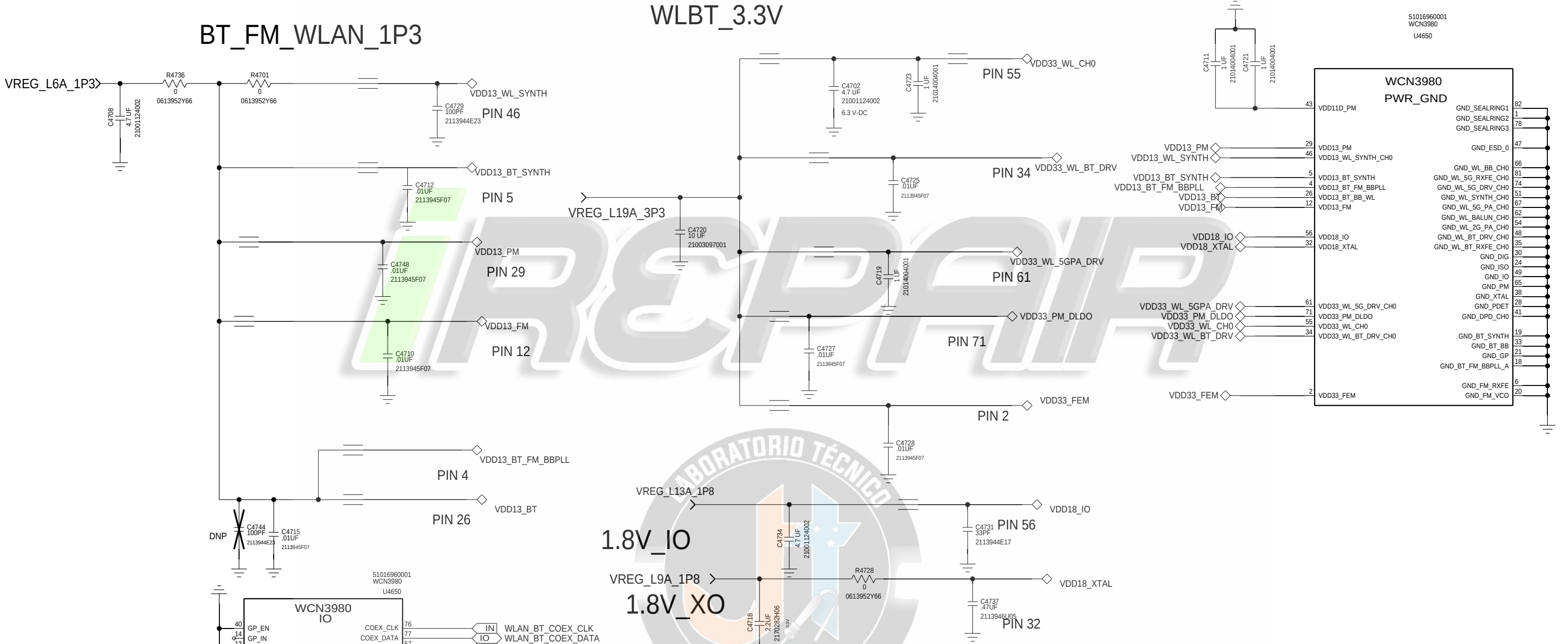


BASED ON 80-WL022-41 REV. A

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: connectivity	
LAST MODIFIED BY: FROEHLING Wed Sep 12 09:13:28 2018			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME B rioh.cpm	REV 01
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		PAGE 45 OF 66	

WCN3980 POWER AND GND,CONTROL

REF 4700-4799



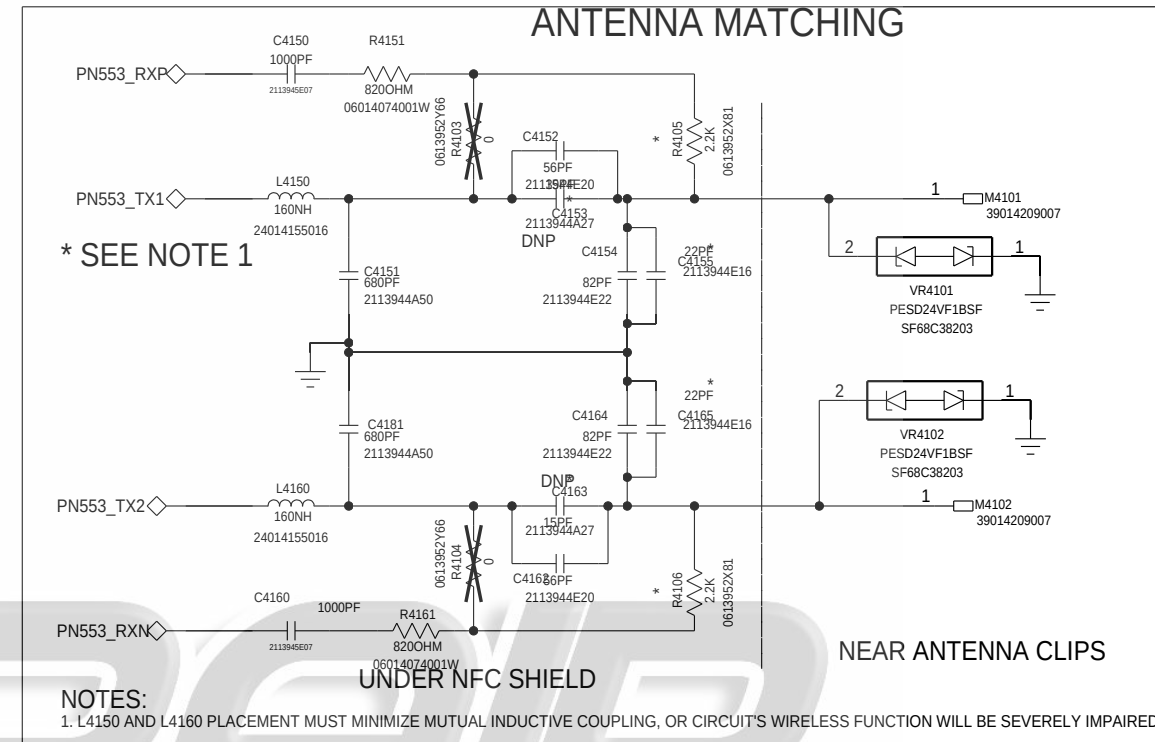
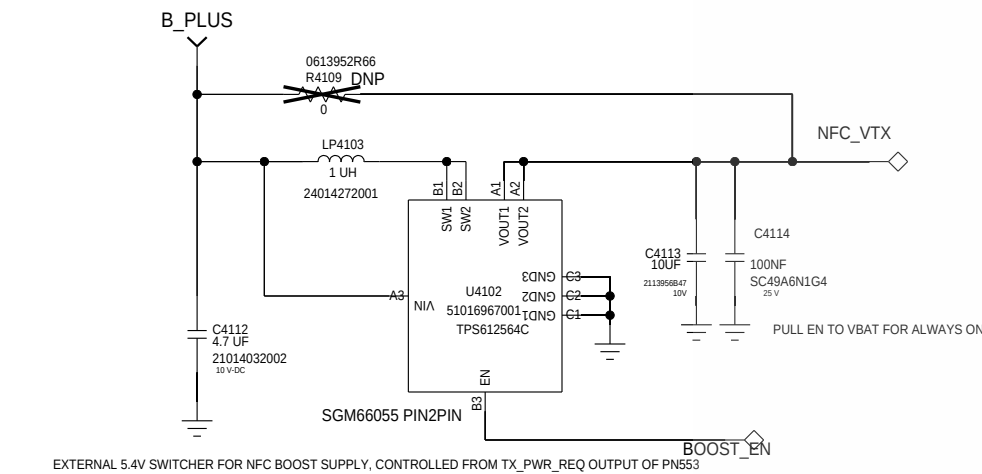
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ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FROEHLING Wed Sep 12 09:13:29 2018		connectivity	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	REV
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CAD FILE NAME:		PAGE 46 OF 66	

BASED ON 80-WL022-41 REV. A

REF: 4100-4199

HIF2 HIF1 WRITE READ

HIF2	HIF1	WRITE	READ
LOW	LOW	0X50	0X51
LOW	HIGH	0X52	0X53
HIGH	LOW	0X54	0X55
HIGH	HIGH	0X56	0X57

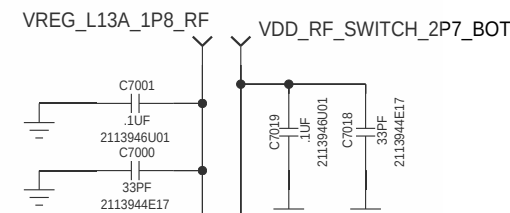
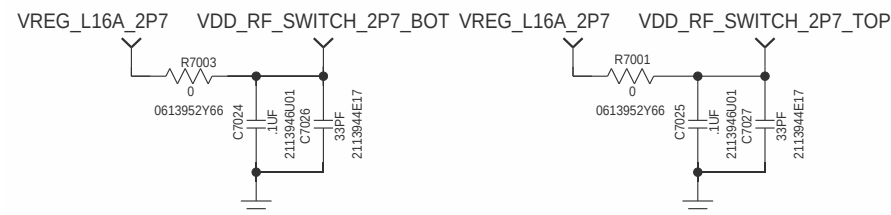


 APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA		BLOCK TITLE: connectivity	
LAST MODIFIED BY: MOTOROLA Wed Sep 12 09:13:29 2018			
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		CAD FILE NAME:	
		PAGE 47 OF 66	

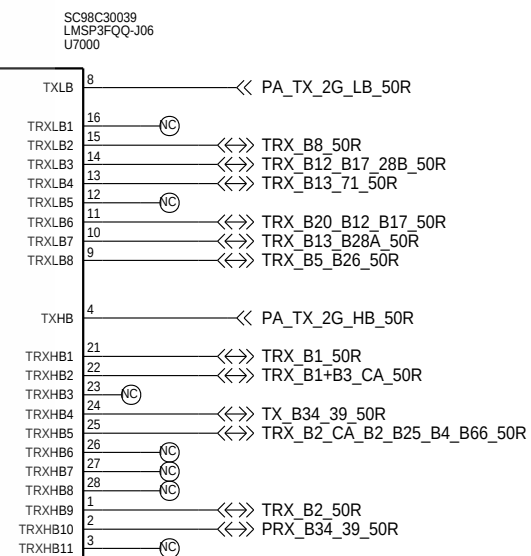
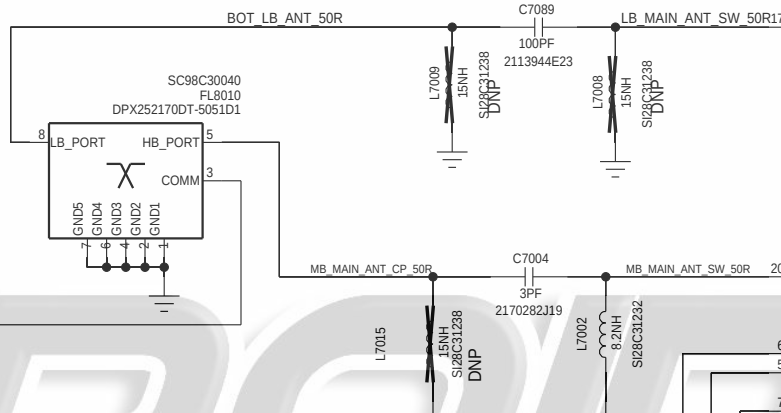
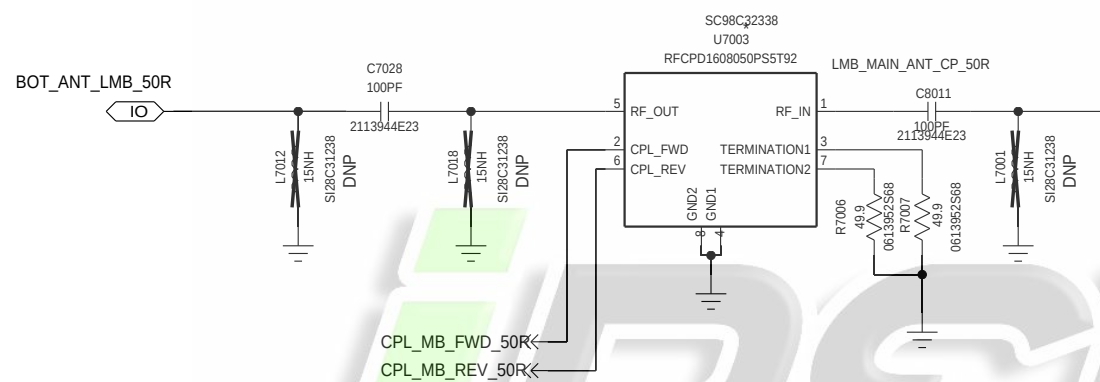
RFFE:TRX ASM

REF 7000-7099

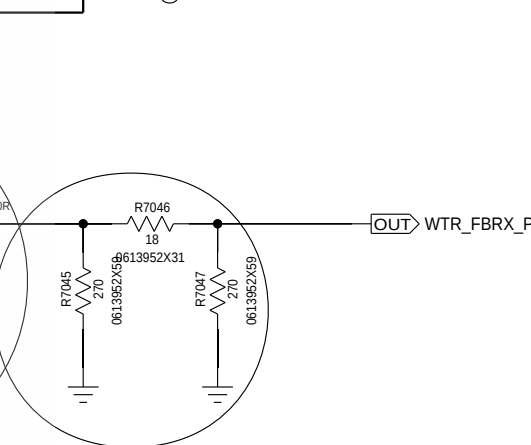
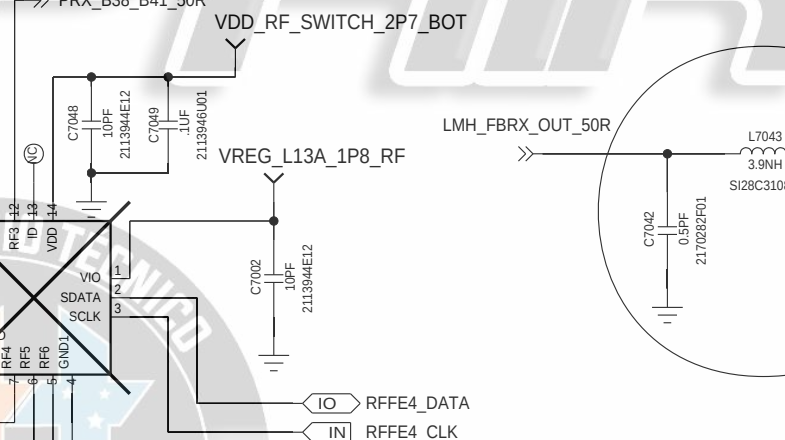
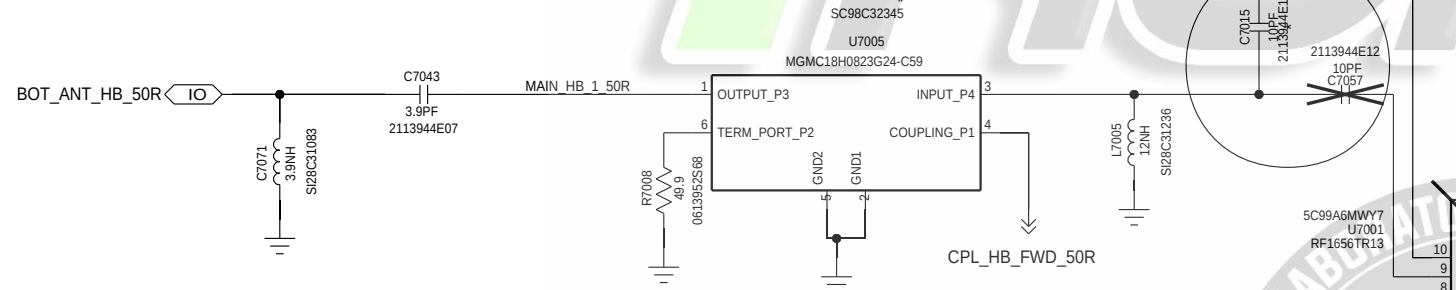
VDD_SWITCH_2P7 FOR RF SWITCH PWR SUPPLY



U7003
NA USE SC98C21210 LD118829M28DG014
NON NA SC98C32338 RFPCPD1608050PS5T92

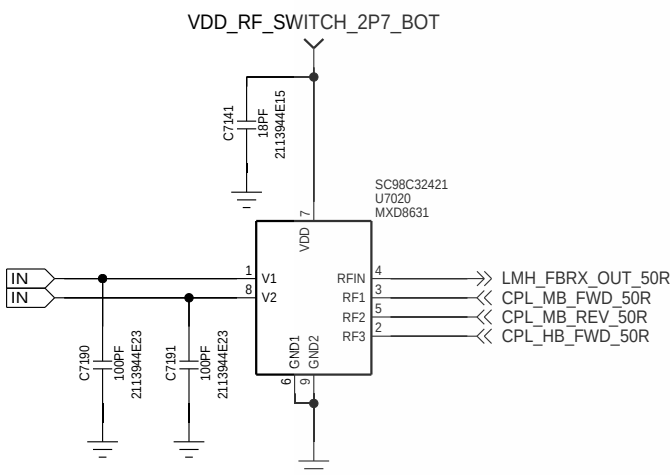


U7005
NA USE 58014033001 LD118829M25AG006
NON NA SC98C32345 MGMC18H0823G24-C59



U7001 BRAZIL+LAVERZION DNP
TMO USE 5C99A6MWY7 RF1656TR13
NON TMO SC98C37674 RFASWH656DTF03

GRFC1_GPIO93
CTL1_CPL_FBRX
CTL2_CPL_FBRX
GRFC6_GPIO98



APPROVALS

ENGINEER:

LAST MODIFIED BY:

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

rffe

PROJECT NAME

rioh.cpm

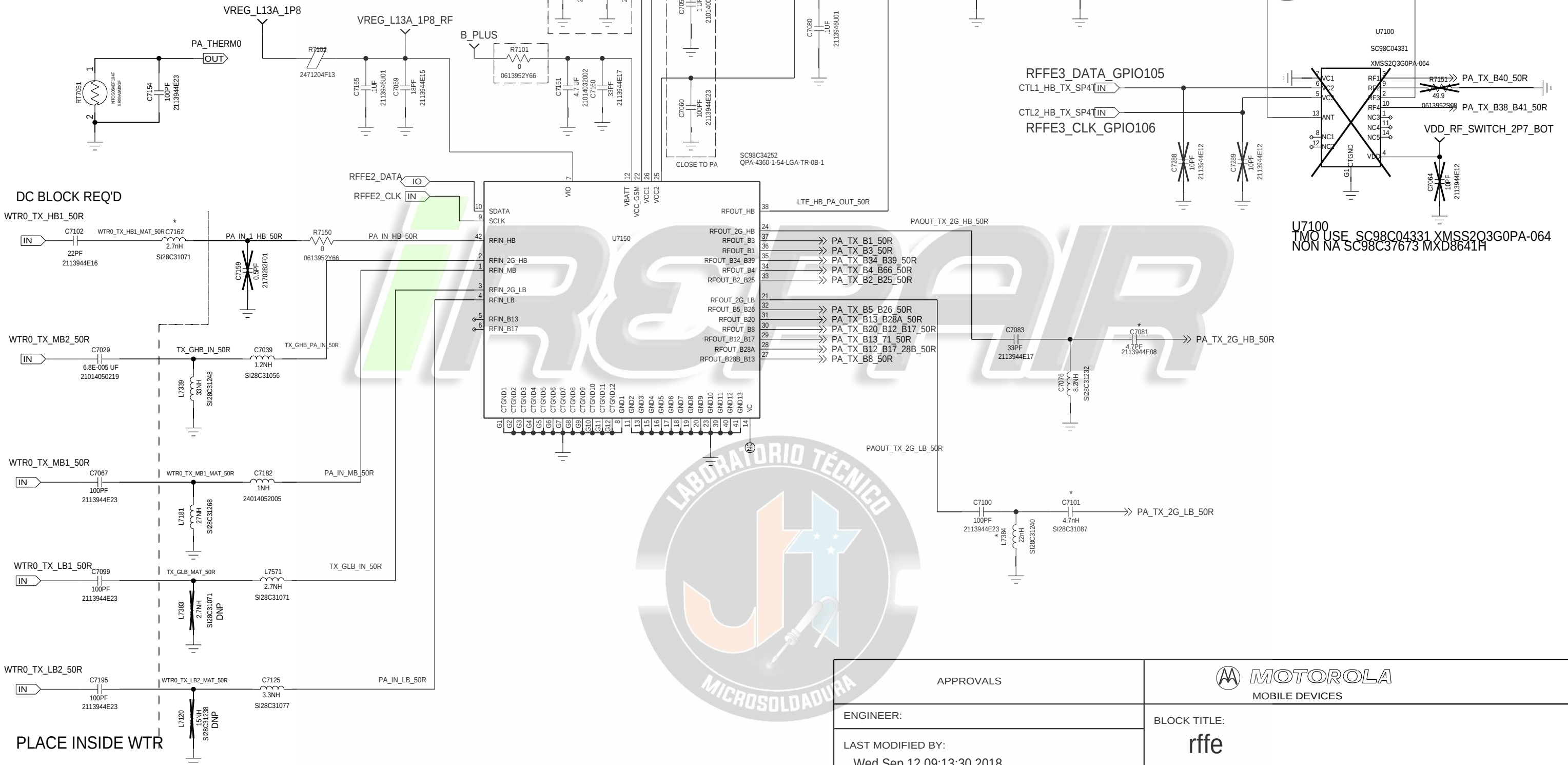
CAD FILE NAME:

REV

01

PAGE 48 OF 66

RFFE: QPA4360
REF 7150-7199

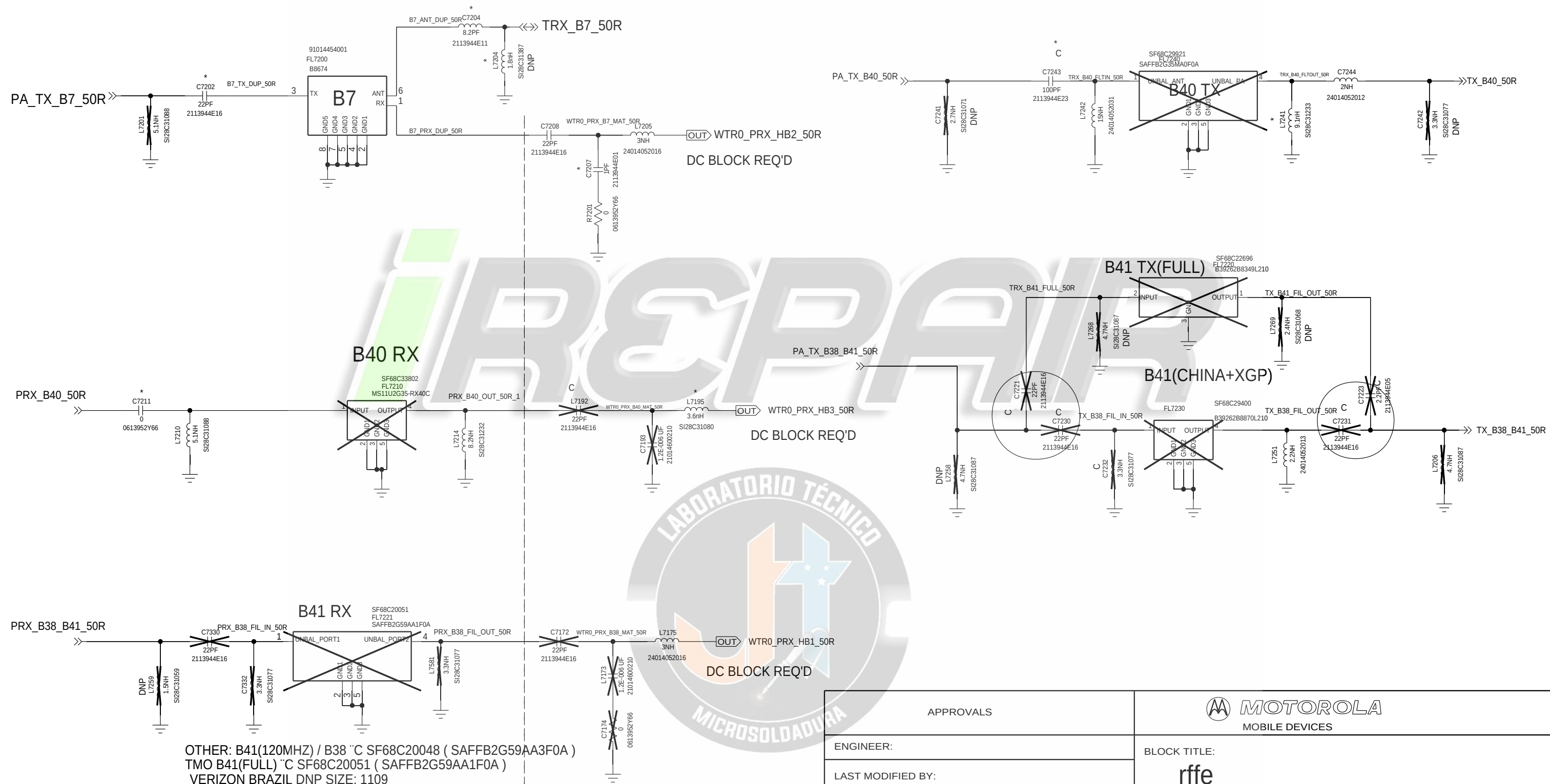


U7100
TMO USE SC98C04331 XMSS203G0PA-064
NON NA SC98C37673 MXD8641H

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: rffe		
LAST MODIFIED BY: Wed Sep 12 09:13:30 2018			
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		CAD FILE NAME:	PAGE 49 OF 66

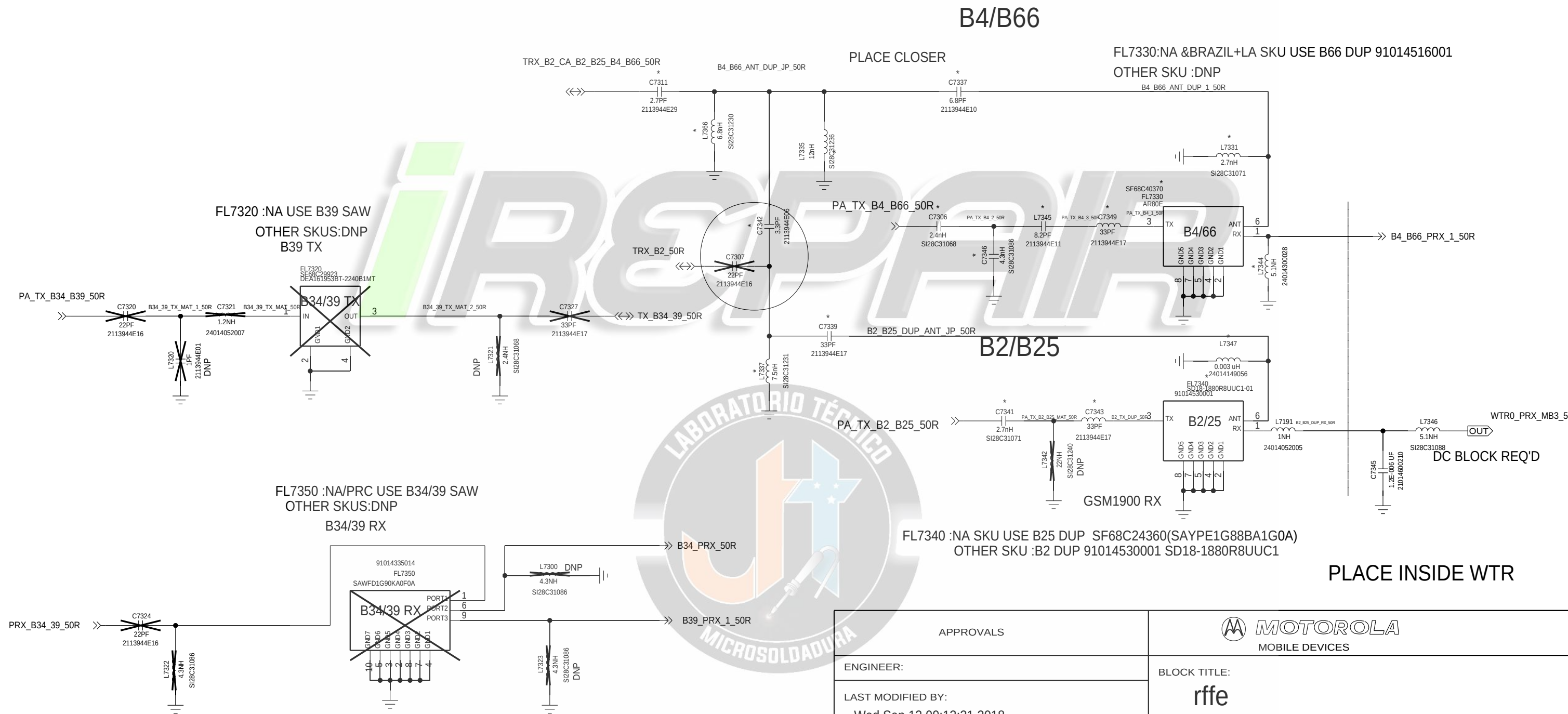
RFFE: HB_TRX_B7,38,40,41

REF 7200-7299



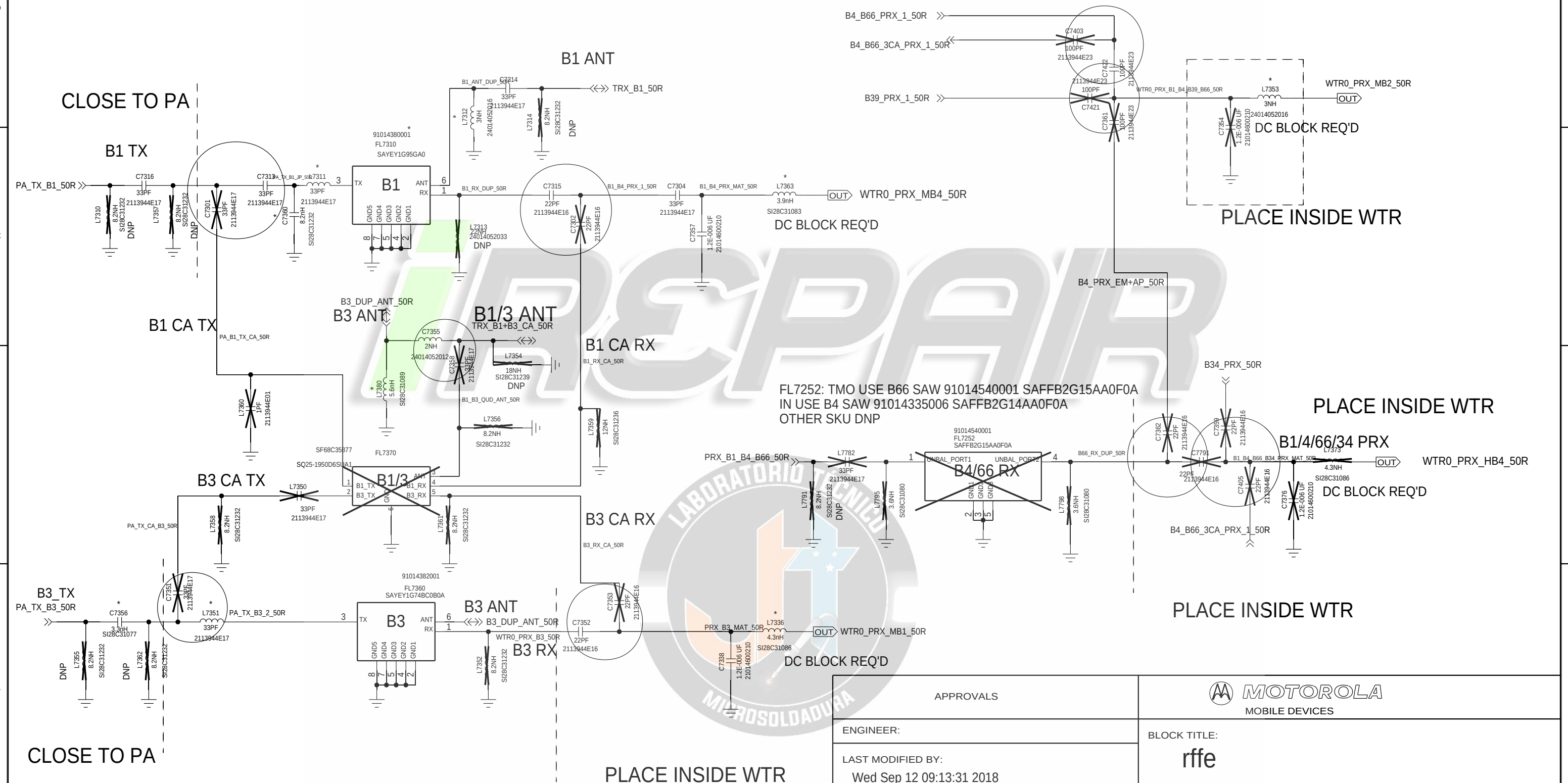
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ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		rffe	
Wed Sep 12 09:13:31 2018			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		PAGE 50 OF 66	
		REV 01	

RFFE :MB_TRX_B2,4,34,39,G1900
REF 7300-7399



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: rffe	
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		CAD FILE NAME: PAGE 51 OF 66	

RFFE: MB_TRX_B1_3_B1/3 QUAD_G1800
REF 7350-7399



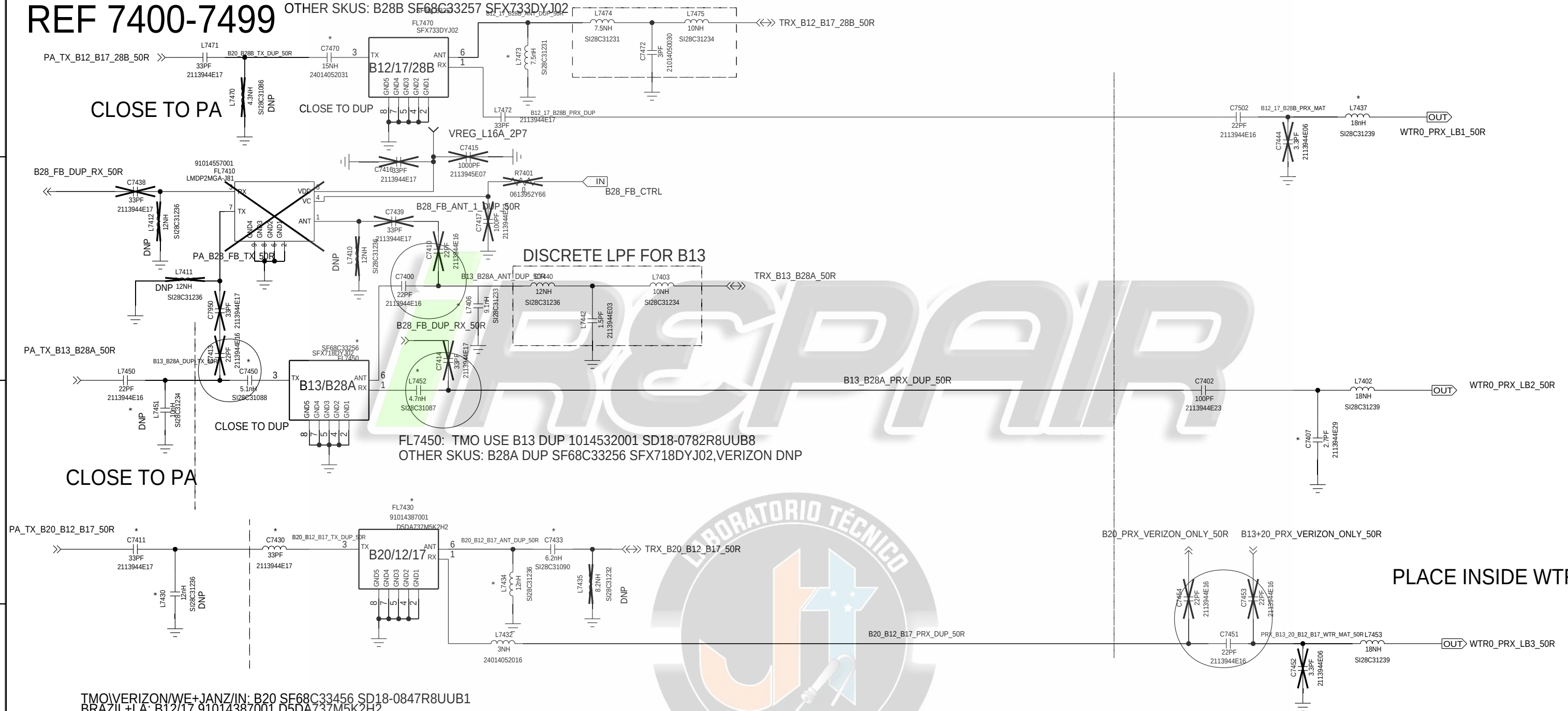
APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: Wed Sep 12 09:13:31 2018			
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		CAD FILE NAME:	PAGE 52 OF 66

REF 7400-7499

MOVERIZON B12/17 91014387001 D5DA737M5K2H2
OTHER SKUS: B28B SF68C33257 SFX733DYJ02

DISCRETE LPF

— $\langle\langle\rangle\rangle$ TRX_B12_B17_28B_50R



TMO/VERIZON/WE+JANZ/IN: B20 SF68C33456 SD18-0847R8UUB1
BRAZIL+LA: B12/17 91014387001 D5DA737M5K2H2

PLACE INSIDE WTR

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: rffe		
LAST MODIFIED BY: Wed Sep 12 09:13:31 2018			
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		CAD FILE NAME:	PAGE 53 OF 66

D

D

REF 7600-7699

D

C

B

A

D

C

B

A

RF_SWITCH
SC98C29925
U7610
RF1648BTR13-5K

U7610
NA: SP4T RF1648B
EM+AP+IN+WE+JAPAN: RF1621
BRAZIL+LA: DNP

GRFC4_GPIO96
CTL1_DRX_HB_SP3T
CTL2_DRX_HB_SP3T
GRFC5_GPIO97

NON NA: B41(120MHZ) / B38 "C SF68C20048 (SAFFB2G59AA3F0A)
TMO B41(FULL) "C SF68C20051 (SAFFB2G59AA1F0A)
VERIZON\BRAZIL+LA DNP SIZE: 1109

B40
SF68C33802
FL7620
MS11U2G35-RX40C

B7
FL7660
SF68C33455

U7660
NA(SP4T): RF1648B SC98C29925
OTHER SKUS: SP3T RF1621

W_GRFC_1
CTL2_DRX_LB_SP4T
CTL1_DRX_LB_SP4T
W_GRFC_0

APPROVALS

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MOBILE DEVICES

BLOCK TITLE:
rfe

PROJECT NAME
rio.h.cpm

CAD FILE NAME:

REV
01

REF 7700-7749 B34/39 RX

B34/39 RX

TMO\VERZION\PRC ONLY

DRX B3/B2 FLOUT 50R

TMO\BRAZIL+LA USE B3 FLITER 91014308001 SAFFB1G84AB0F0A
OTHER SKUS USE B2 FLITER 91014335005 SAFFB1G96AB0F0A

ALL SKUS DNP

~~B66 RX~~

DRX CA B1/B3/B2/B4/B25/B66 50R>>

FL7700 :TMO USE B34+B39 DUAL SAW 91014335014 SAWFD1G90KA0F0A
FL7700 :BRAZIL+LA USE B25+B66 DUAL SAW 91014518001 SAWFD1G96AC2F0A
FL7700 :OTHER SKUS USE B1+B3 DUAL SAW 91014424001 SAWFD1G84AA0F0A

B25+B66 OR B1+B3 OR B34+B39

B66/B4 OR B34

B25/B2 OR B39

PLACE INSIDE WTR



MOTOROLA
MOBILE DEVICES

J7701 NA - 09014068001W(MM8030-2610RJ3)
OTHER - SCA8C29399(818011998)

U7712 TMO/VERIZON/PRC(SP3T): RF1621 51014606014
NON NA(SPDT): RF1621A 51016674001 C7734

GRFC2 GPIO94

IN CTL1 DRX MB SP3T

GRFC3 GPIO95

APPROVALS

ENGINEER:

LAST MODIFIED BY:

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BLOCK TITLE:

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PROJECT NAME

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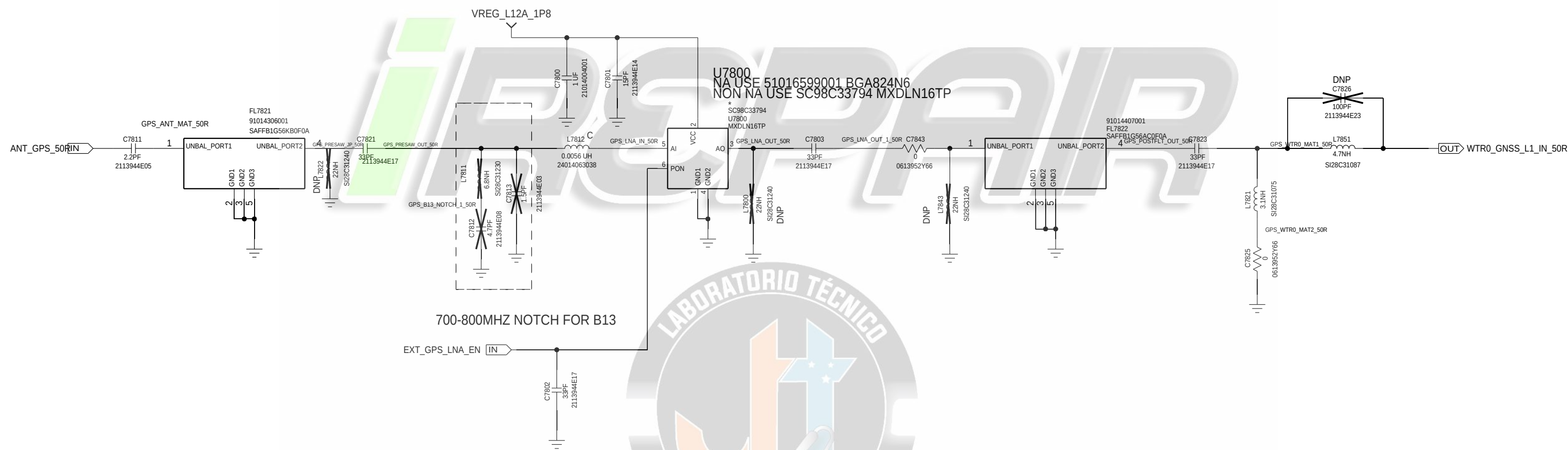
REV
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PAGE 56 OF 66

REF 7750-7799

PAGE 57 OF 66

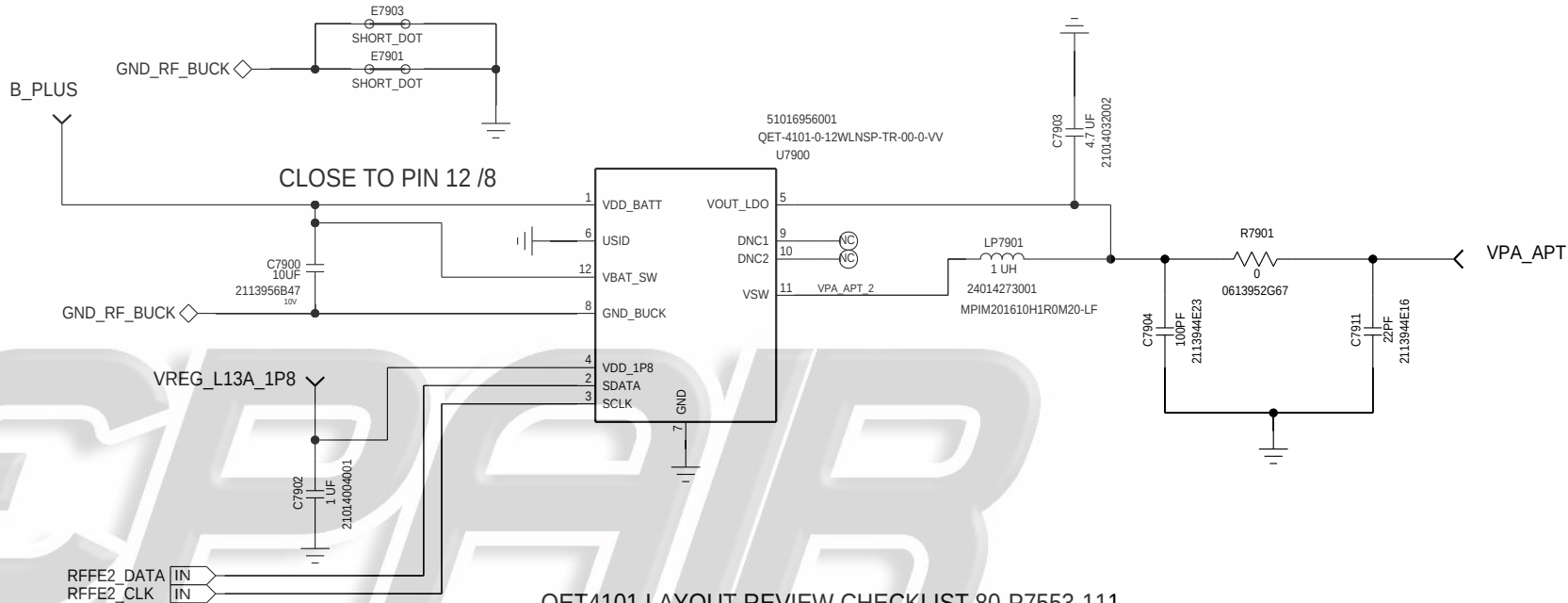
RFFE: GPS
REF7800-7899



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		rffe	
Wed Sep 12 09:13:30 2018			
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		PAGE 58 OF 66	

RFFE: QFE4101&TI DCDC
REF7900-7949

APT DCDC INDUCTOR		
1ST	MPIM201610H1R0M20-LF	DCR = 58MOHM
	MICROGATE 24014273001	ISAT = 3.1A MAX, IRMS = 2.2A MAX
2ND	CIGT201610UM1R0MNE	DCR = 48MOHM
	SECOM SI28C38192	ISAT = 3.1A MAX, IRMS = 2.8A MAX

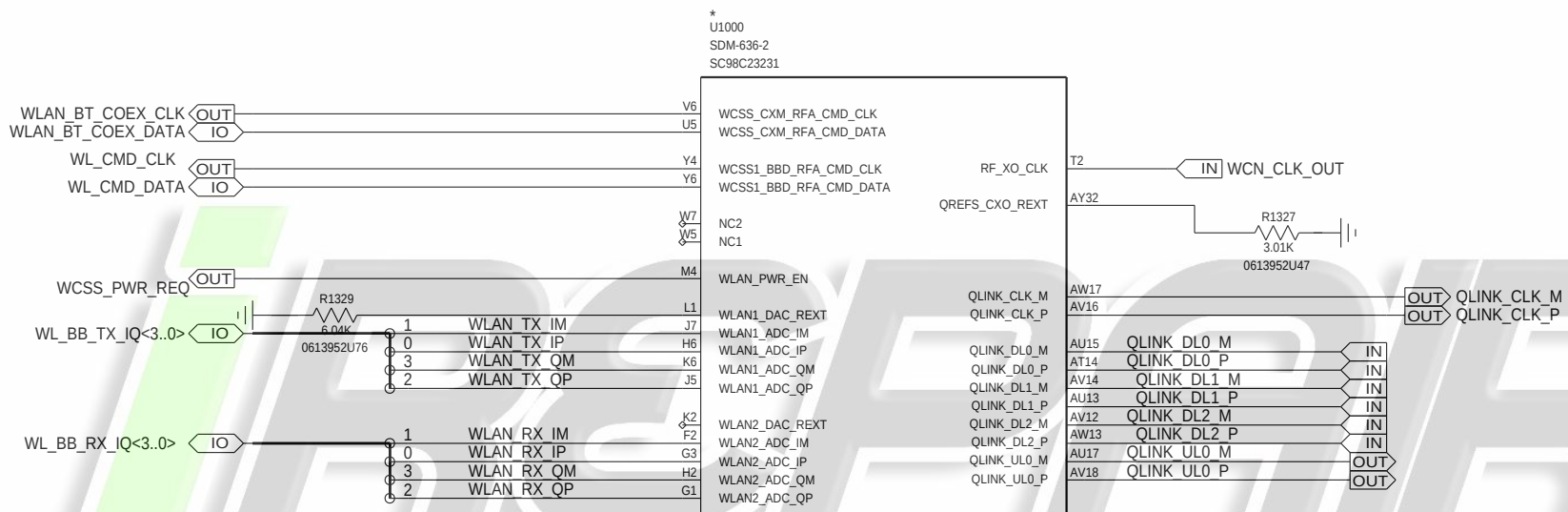


DO NOT VIA DIRECTLY TO GROUND PLANE. KEEP ISOLATED ON SURFACE LAYER

NOTE1: NO VIAS ON PIN 8 (GND_SW) AND PIN 12 (VBAT_SW). NO VIAS ON C7900 PADS. USE SHORT AND WIDE TRACES CONNECTING FROM CHIP PINS TO C7900. EXTEND THE TRACE (≥ 1MM), THEN VIAS DOWN AND CONNECT TO THE GROUND AND POWER PLANE/FILL ON THE NEXT LAYER. KEEP THE NEXT LAYER AS CONTINUOUS GROUND UNDERNEATH THE AREA FROM CHIP PINS TO THE C7900 CONNECTION.

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		rffe	
Wed Sep 12 09:13:30 2018			
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		CAD FILE NAME:	
		PAGE 59 OF 66	

MSM: RF INTERFACE
REF 1200-1299

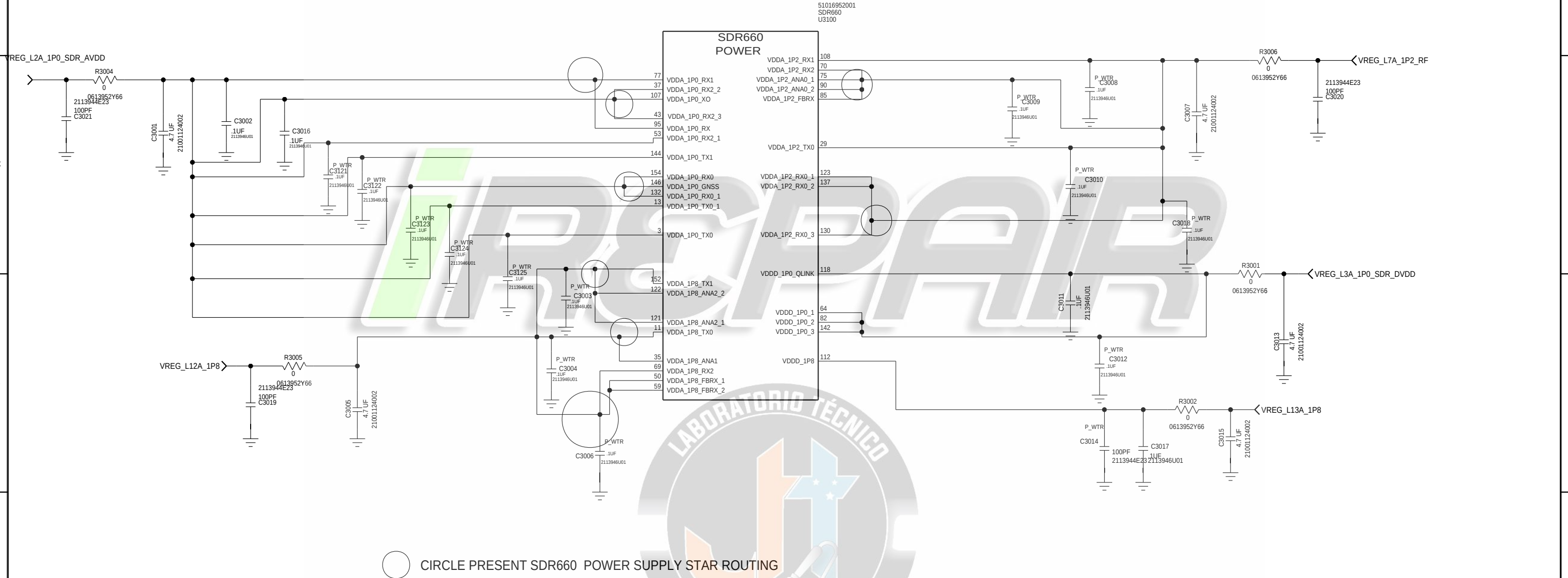


NEED TO APPLY LC FILTER FOR REDUCE B40 RX SPURS(QC 80-NP237-56)
LC FILTER COULD BE PLACED ANYWHERE BETWEEN WTR BBRX_IQ AND MSM BBRX_IQ PINS
BUT THE GND FOR SHUNT C SHOULD NOT BE SHARED WITH EITHER WTR OR MSM GND ON ANY LAYER.
A DIRECT VIA FROM SHUNT C TO MAIN GND IS RECOMMENDED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Wed Sep 12 09:13:28 2018		block_mdm9x25m_pm8019	
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		CAD FILE NAME: PAGE 60 OF 66	

SDR660: POWER

REF 3000-3099

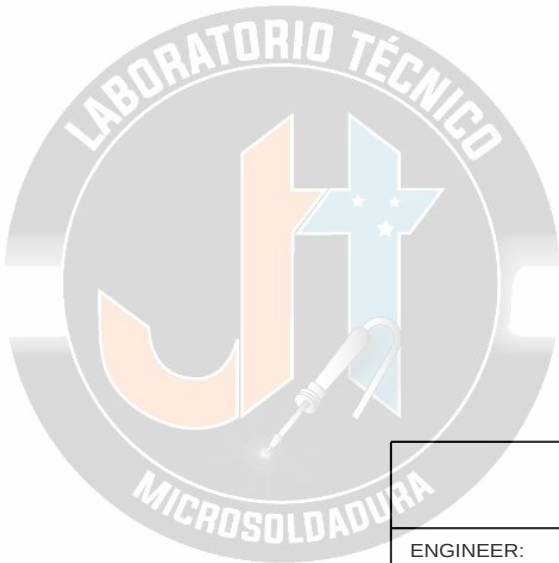
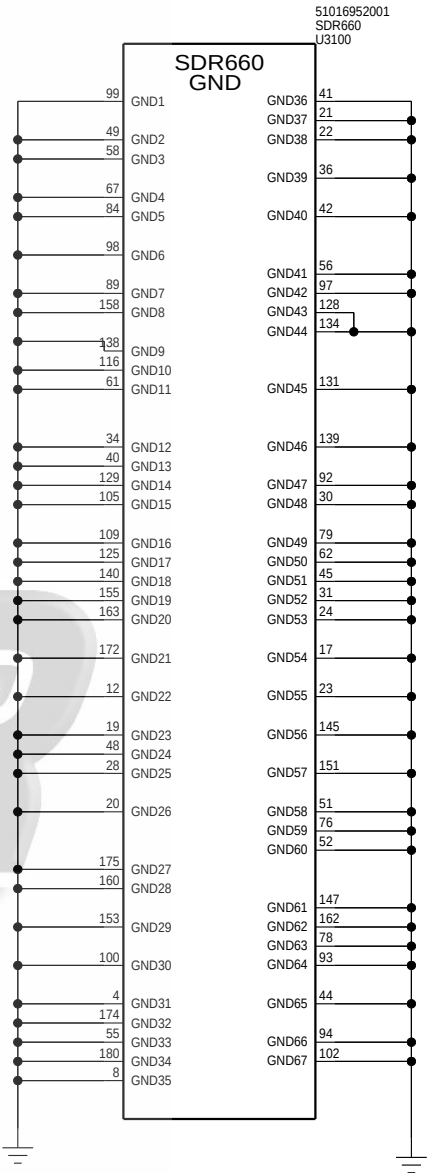
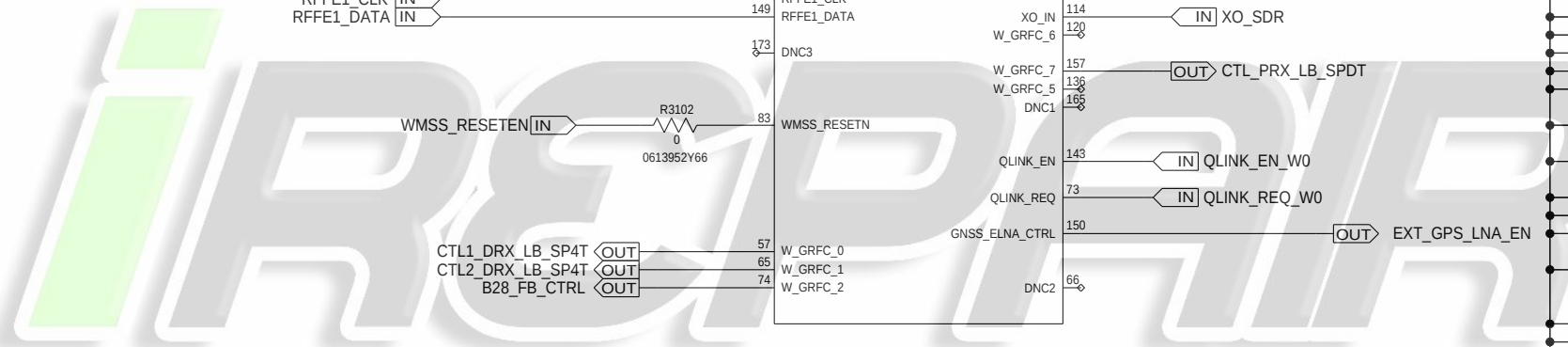


CIRCLE PRESENT SDR660 POWER SUPPLY STAR ROUTING

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Wed Sep 12 09:13:29 2018		blk_sdr660	
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		CAD FILE NAME:	
		PAGE 61 OF 66	

SDR660: TX/PWR/GROUND

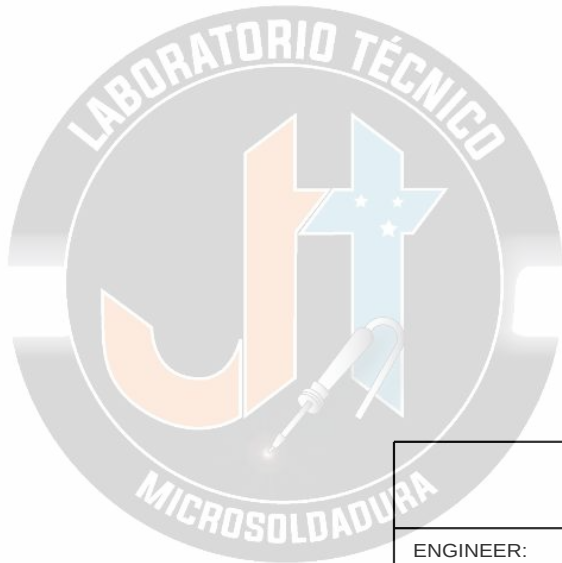
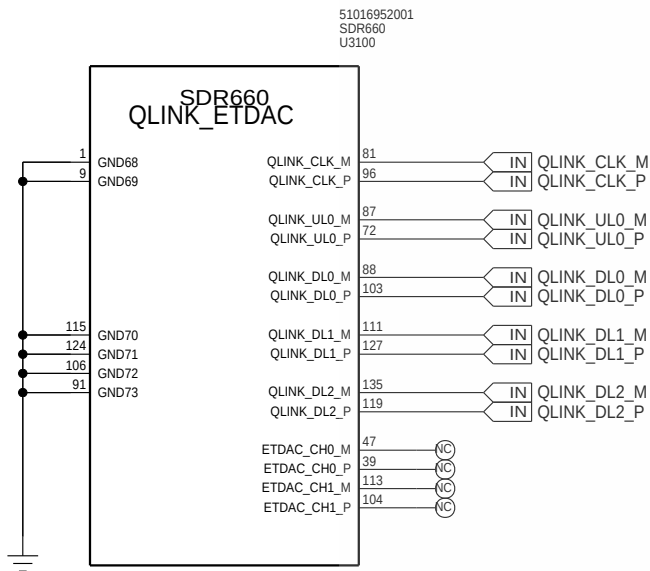
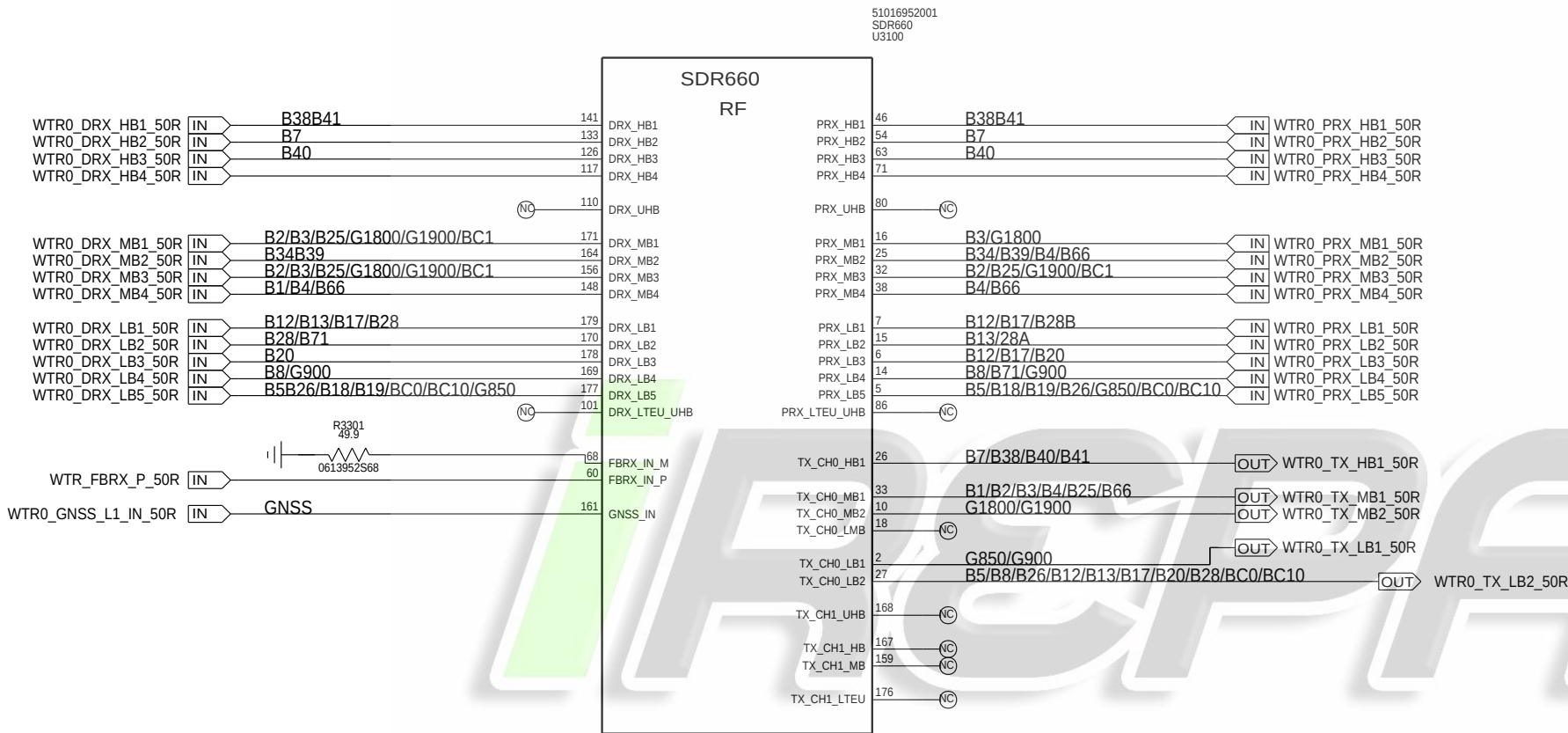
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APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Wed Sep 12 09:13:29 2018		blk_sdr660	
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		CAD FILE NAME:	
		PAGE 62 OF 66	

SDR660: PRX/DRX/GPS

REF 3200-3299

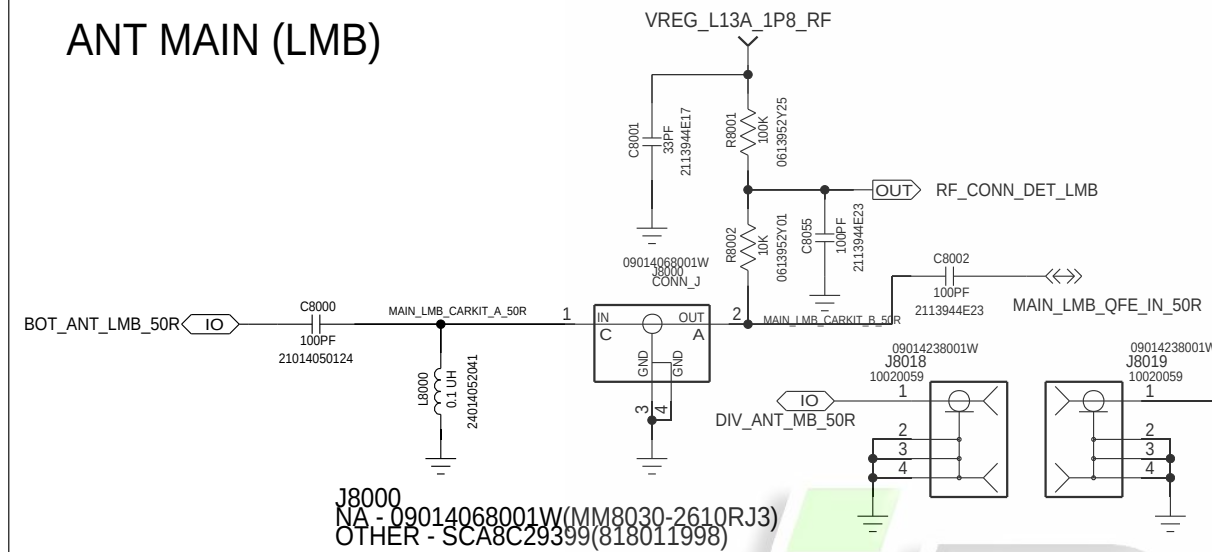


APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE: blk_sdr660	
LAST MODIFIED BY: Wed Sep 12 09:13:29 2018			
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		REV 01	
		CAD FILE NAME:	
		PAGE 63 OF 66	

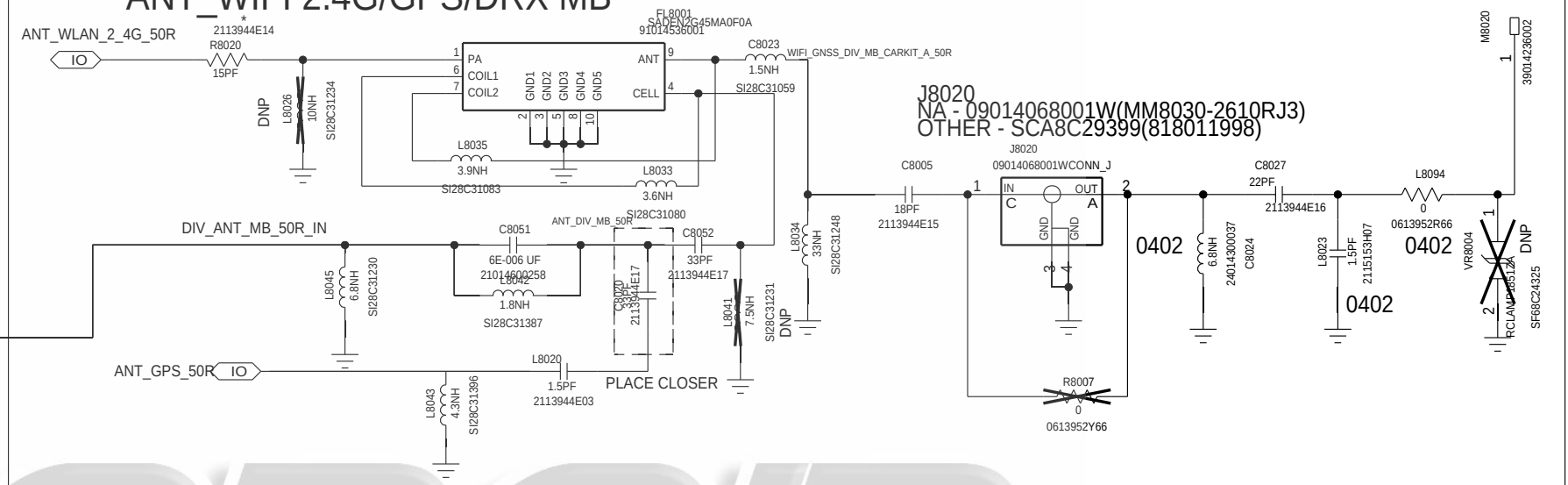
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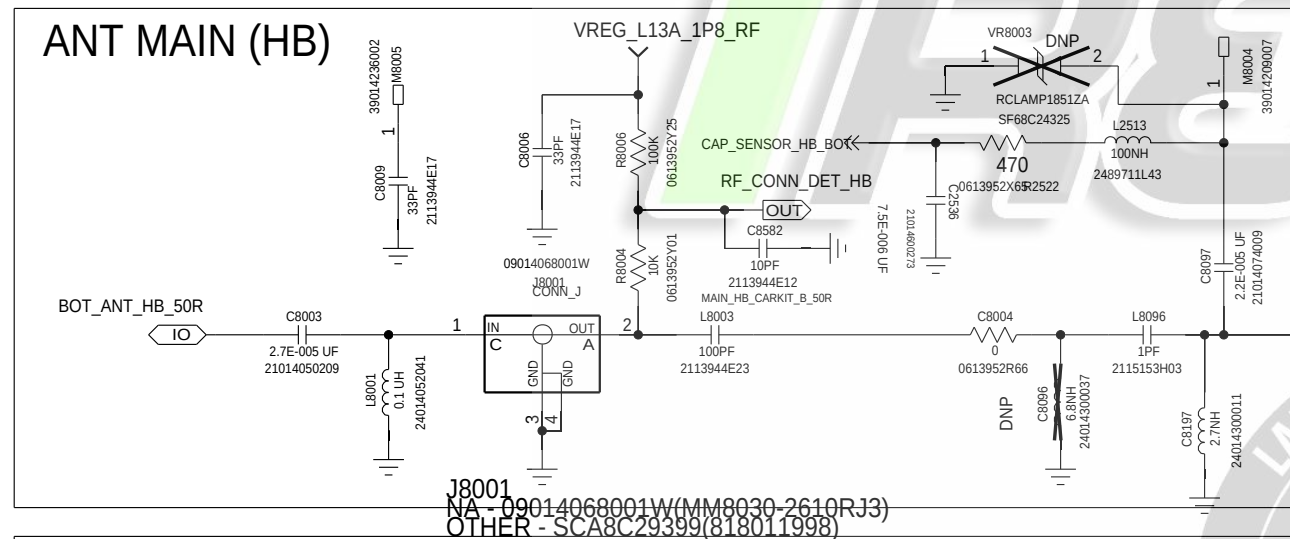
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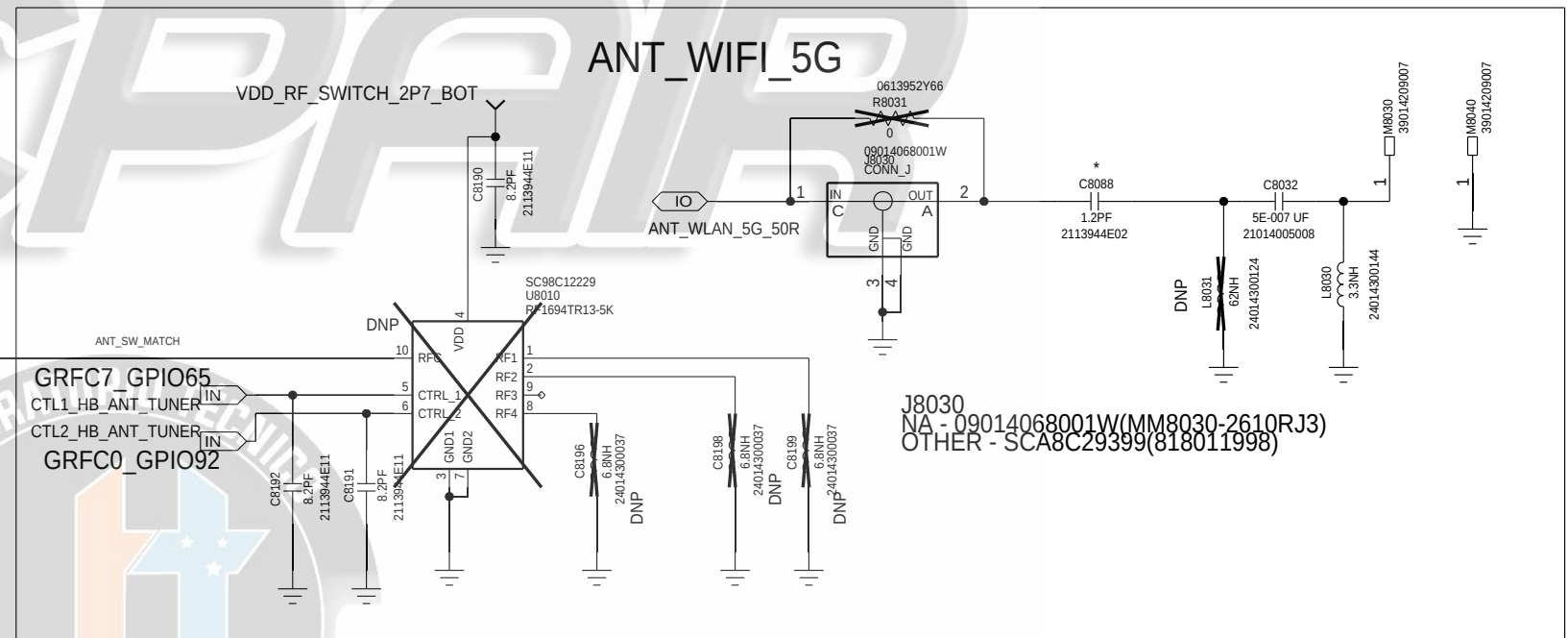
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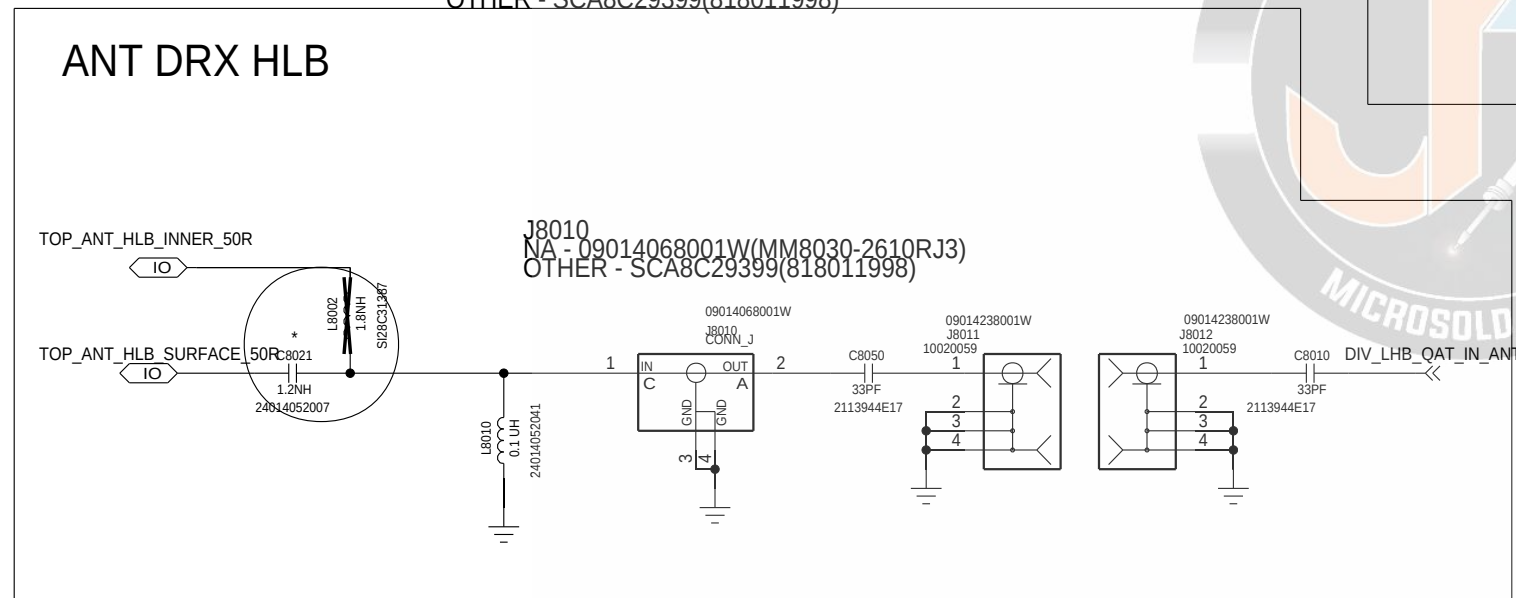
ANT MAIN (HB)



ANT_WIFI_5G



ANT DRX HLB



APPROVALS

ENGINEER:

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BLOCK TITLE:

antennas

PROJECT NAME

rio.h.cpm

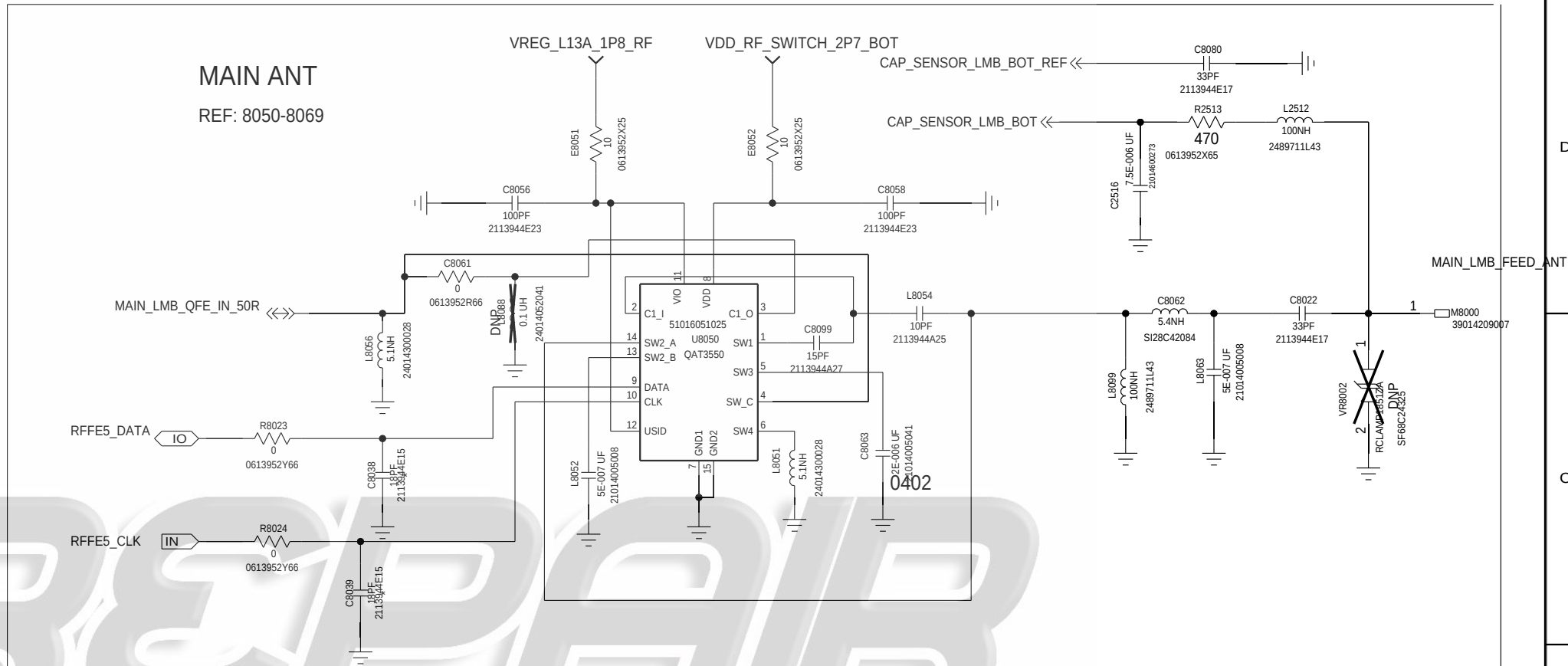
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REV

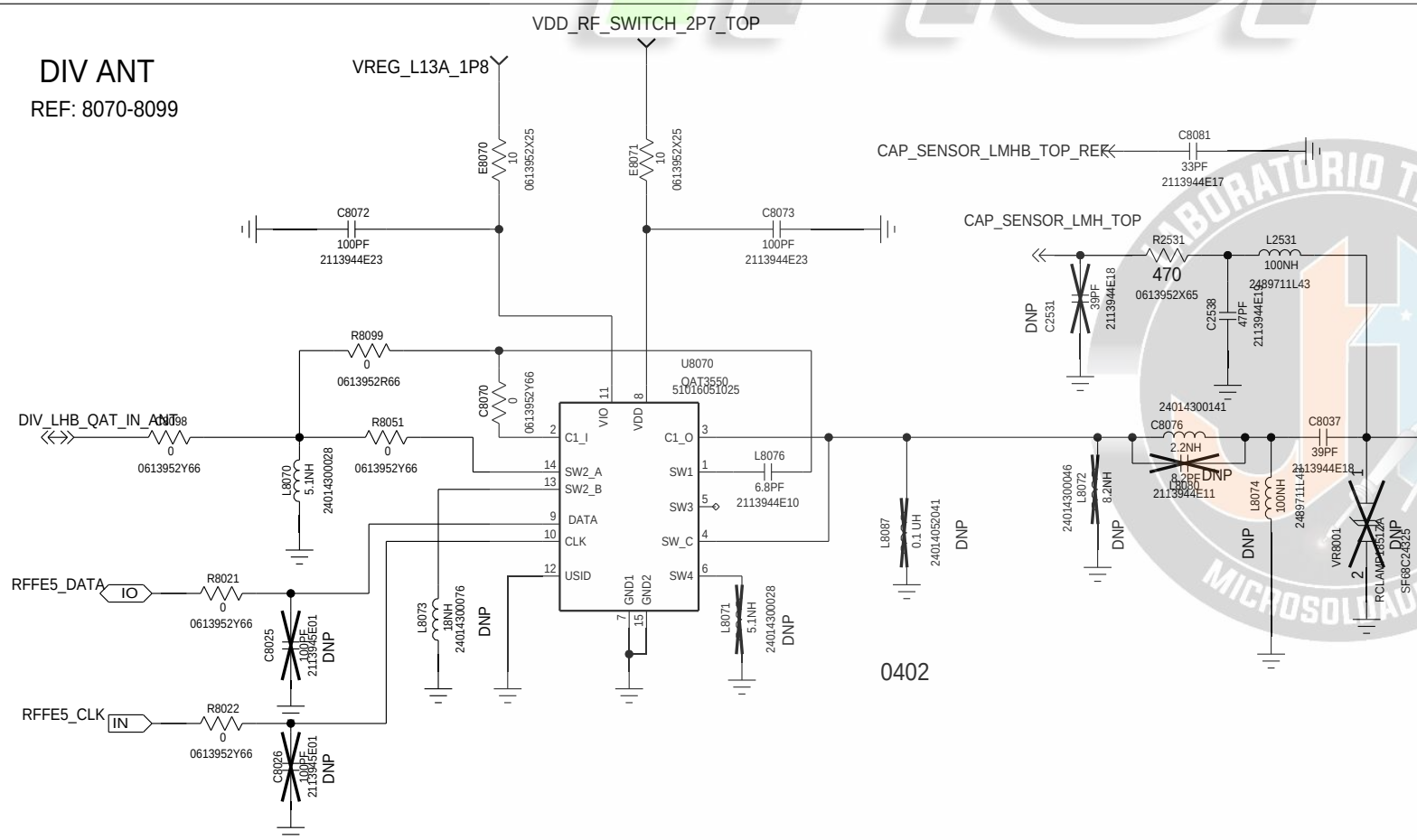
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REF 8050-8099



DIV ANT REF: 8070-8099



APPROVALS

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BLOCK TITLE:

antennas

PROJECT NAME

rioh.cpm

REV

01

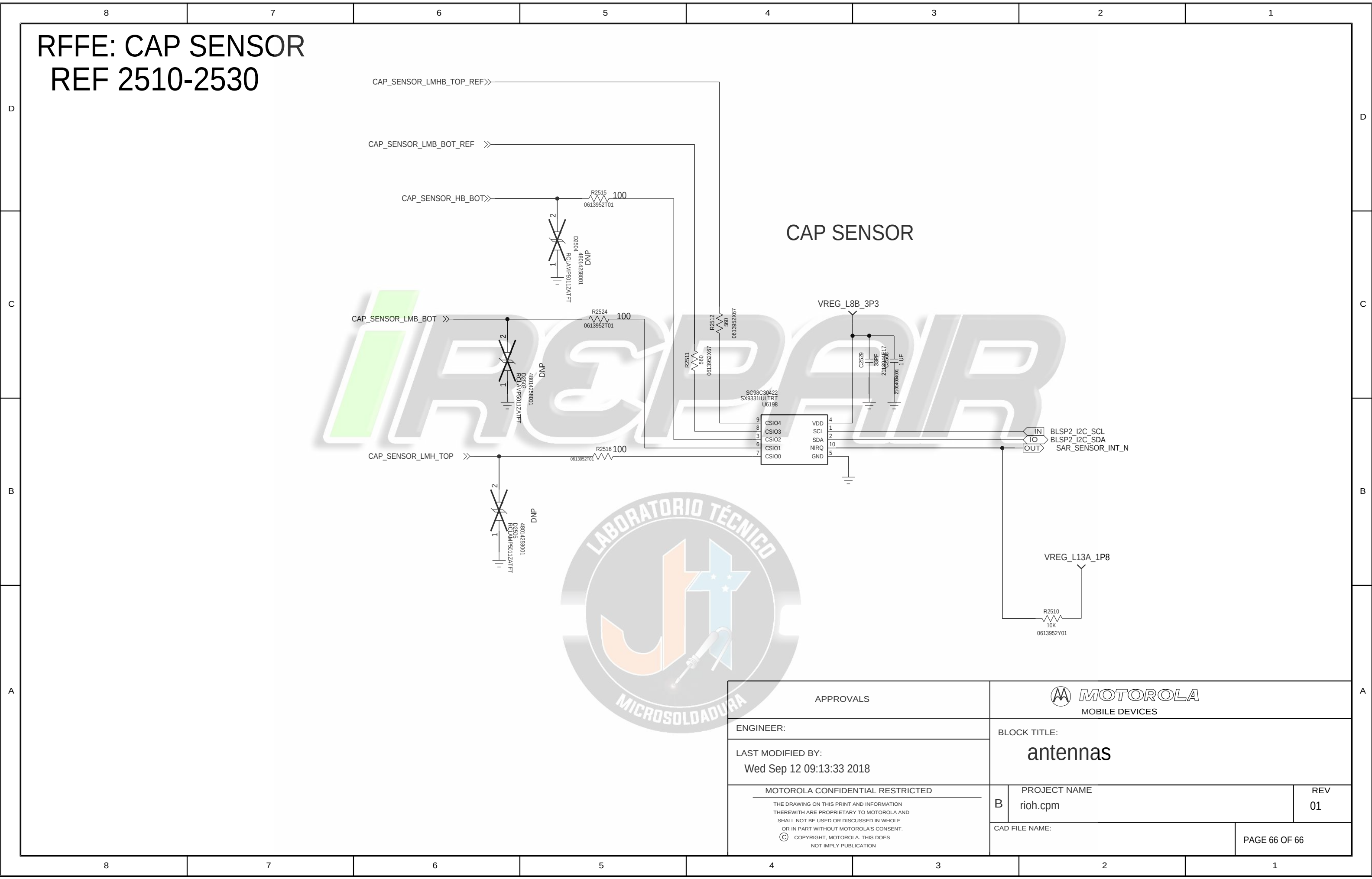
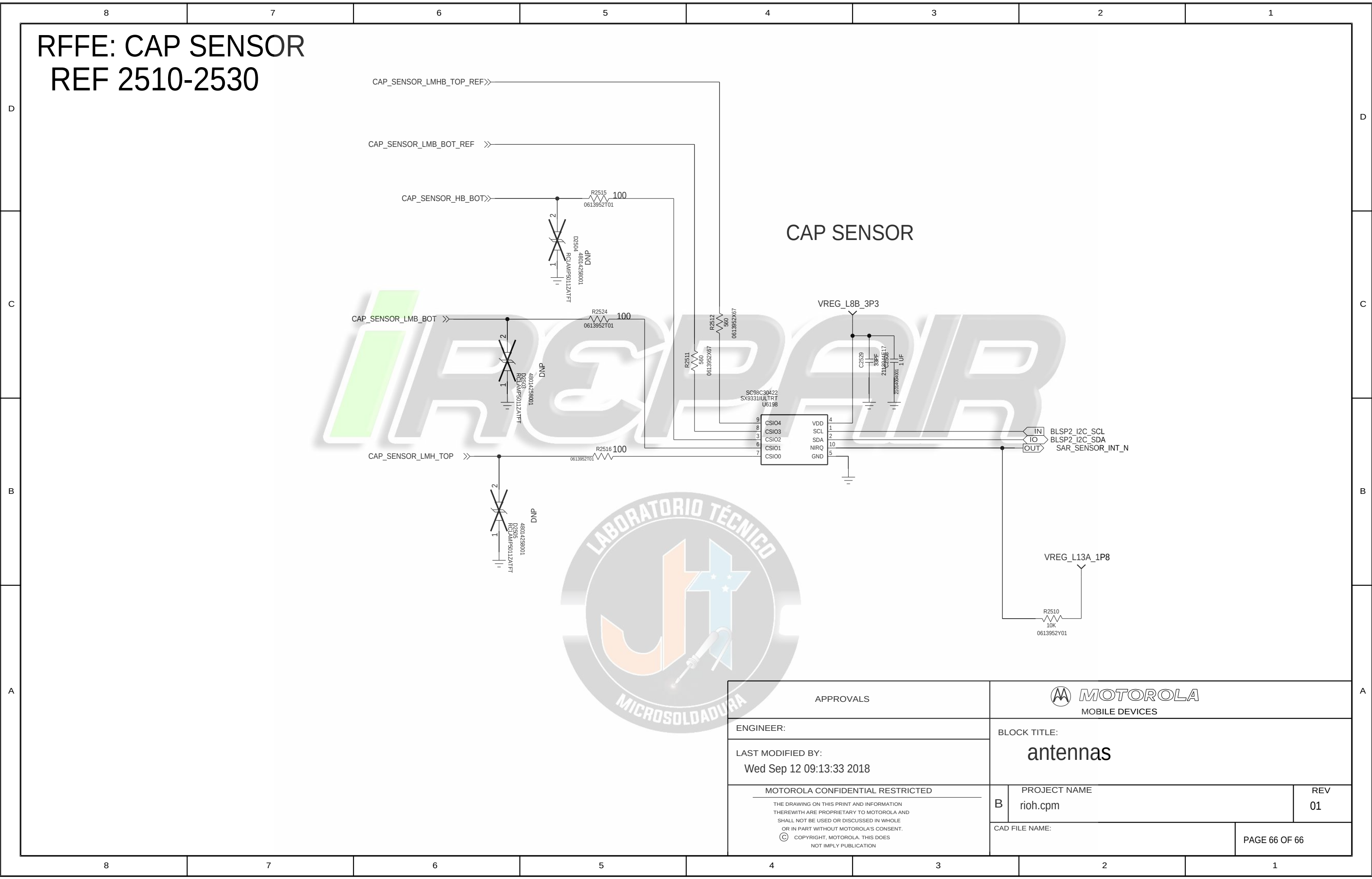
CAD FILE NAME:

PAGE 65 OF 66

8 7 6 5 4 3 2 1

RFFE: CAP SENSOR

REF 2510-2530



8 7 6 5 4 3 2 1

RFFE: CAP SENSOR

REF 2510-2530

CAP_SENSOR_LMHB_TOP_REF

CAP_SENSOR_LMB_BOT_REF

CAP_SENSOR_HB_BOT

CAP_SENSOR_LMB_BOT

CAP_SENSOR_LMH_TOP

VREG_L8B_3P3

VREG_L13A_1P8

PCB

LABORATORIO TÉCNICO
MICRO SOLDADURA

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: antennas		
LAST MODIFIED BY: Wed Sep 12 09:13:33 2018	PROJECT NAME rio.h.cpm		
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8 7 6 5 4 3 2 1